# EVERY SINGLE-CENTRED BLACK HOLE INDEX OF THE $\mathbb{Z}_2$ CHL MODEL IS POSITIVE

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Abstract. Every single-centred black hole index of the $\mathbb{Z}_2$ CHL model is positive. The proof rests on an exact Rademacher expansion at level two.

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## 1. Introduction

A quarter-BPS dyon of $\mathcal{N} = 4$ string theory in four dimensions carries an electric charge $Q$ and a magnetic charge $P$, and the protected index that counts such states is the helicity trace $B_6$. When the charges admit a single-centred black hole whose only hair are the zero modes of the broken supersymmetries, the twelve fermion zero modes fix the sign of the trace, and the horizon degrees of freedom contribute $-B_6 > 0$. Sen turned this observation into a conjecture about Fourier coefficients of Siegel modular forms [29]: the index $d^*(Q, P)$ of the single-centred black hole, extracted from the inverse of the dyon partition function with the contour of the attractor region, is positive.[1]

For type II string theory on $K3 \times T^2$, where the partition function is the Igusa cusp form $\Phi_{10}$ [16, 30, 19], the conjecture is settled. Bringmann and Murthy proved it for the two lowest magnetic charges $P^2 = 2, 4$ by writing the finite parts of the first two Fourier–Jacobi coefficients of $1/\Phi_{10}$ through class numbers [5]. Rosselló then treated all charges, by writing the single-centred index as a mixed Rademacher series and showing that one of its terms dominates the others; the existence of such series on $SL_2(\mathbb{Z})$ is due to Bringmann and Manschot [6], and Ferrari and Reys used them for these black holes [18]. The CHL models [9] are the $\mathbb{Z}_N$ orbifolds of type IIB on $K3 \times T^2$ in which the $\mathbb{Z}_N$ acts on $K3$ by a symplectic automorphism and on $T^2$ by a shift; their partition functions $\widetilde{\Phi}_k$ are Siegel modular forms of weight $k$ for subgroups of $Sp(2, \mathbb{Z})$ [23, 13, 15, 14].[2] Bhand, Sen and Singh showed that the Fourier–Jacobi coefficients of $1/\widetilde{\Phi}_k$ are meromorphic Jacobi forms, split them into an Appell–Lerch polar part and a mock Jacobi finite part, and identified the single-centred index with the Fourier coefficients of the finite part on an explicit set of charges [8, Theorem 5.4]. Cardoso, Nampuri and Rosselló computed the polar coefficients of these finite parts through continued fractions and wrote the immortal index as a Rademacher series, taking the existence of such an expansion at level $N$ as an assumption [10].

[1] Sen checked the sign on the charges of his Tables 1 to 5, for the heterotic string on $T^6$ and for the CHL models of orders $2, 3, 5, 7$, and stated the conjecture for all charges in the cones (3.7)–(3.12) of [29]. The entropy function and attractor background behind the statement are reviewed in [28].

[2] For $N = 2$ the model is the heterotic string on $T^6$ obtained from the $E_8 \times E_8$ string by exchanging the two $E_8$ factors together with a half shift on a circle [9]; for the ten-dimensional $E_8 \times E_8$ theory and its relation to M-theory see [26]. Background on the involutions of $K3$ surfaces and on Calabi–Yau threefolds fibred by $K3$ surfaces, which underlie the dual type II description, is in [7, 3].

This paper settles the $\mathbb{Z}_2$ CHL model. It has $k = 6$, duality group $\Gamma_1(2)$, and charges normalised so that $m = Q^2$, $n = P^2/2$ and $\ell = Q \cdot P$ are integers; the discriminant is $D = Q^2P^2 - (Q \cdot P)^2 = 2mn - \ell^2$.

**Theorem A.** *In the $\mathbb{Z}_2$ CHL model the single-centred index $d^*(Q, P)$ is positive for every charge with $D > 0$.*

The proof has two halves. The first is an exact formula for the coefficients of the finite parts.

**Theorem B.** *For every $m \geq 1$ the coefficients $c_m^F(n, \ell)$ of the finite part of the $m$-th Fourier–Jacobi coefficient of $1/\widetilde{\Phi}_6$ with $D = 2mn - \ell^2 > 0$ are given by an absolutely convergent vector-valued Rademacher series over the three cosets of $\Gamma^0(2)$ in $SL_2(\mathbb{Z})$, with Bessel terms $I_{15/2}$ from the principal parts and correction terms $I_8$ and $I_{17/2}$ from the shadow. Its principal parts at the cusps $0$ and $\infty$ are explicit sums over Farey walls.*

The precise statement is Theorem 6.2. The wall sums at the cusp 0 have the shape of the continued-fraction formula of Cardoso, Nampuri and Rosselló; we prove them, and their counterpart at the cusp $\infty$, by following the Fourier coefficients of $1/\widetilde{\Phi}_6$ across the walls of its poles, which are the Farey geodesics with an even denominator (Figure 2), and of a second product at the cusp 0 across all Farey geodesics. The expansion itself comes from the circle method of Rademacher and Zuckerman [25, 34], applied to the theta coefficients of the finite parts once the Appell–Lerch sum has been completed in the manner of Zwegers [35]; the completion is proved directly, by a Mordell integral, and the transformation law in all three frames follows from the product formula alone, without any statement about Siegel modular forms.

The second half is the sign. Write $x = \sqrt{D}/m$ and $\theta = \pi\ell/m$, and let $\mathcal{E}(D, m)$ be the contribution of the most polar term at the cusp 0, an explicit multiple of a Bessel function $I_{15/2}$. Every charge has, in its duality orbit or after the Fricke exchange of $Q^2$ and $P^2/2$, a representative with $m \leq n$ and $0 \leq \ell \leq m$ (Proposition 4.6); then $x \geq 1$. The terms of the Rademacher series attached to the leading family of Farey walls, those with the matrix $\left(\begin{smallmatrix} 1 & 0 \\ 1 & 1 \end{smallmatrix}\right)$, add up, after division by $\mathcal{E}$, to an approximation of

$$|P(z)|^2, \qquad P(z) = \prod_{n \geq 1} (1 - z^n)^{-8} (1 - z^{2n})^{-8}, \qquad z = e^{-\pi x + i\theta},$$

the squared modulus of the generating function of the degeneracies of the electric half-BPS states of the model (Theorem 11.1). The function $|P|^2$ stays above 0.538 on the disc $|z| \leq e^{-\pi}$, its minimum being attained at $z = -e^{-\pi}$ (Proposition 11.3, Figure 3). Everything else, namely the error of the approximation, the remaining walls, the other two cosets, the terms with $\gamma \geq 2$ and the shadow corrections, is bounded by 0.347 for $Q^2 \leq 100$ by a certificate in interval arithmetic and by 0.3382 for $Q^2 \geq 101$ by explicit inequalities (Theorems 14.1 and 14.2). The remaining charges have $D \leq 800$, and for them the index is computed exactly from the product formula (Theorem 4.7).[3]

On the first row the theorem also has a short proof by closed formulas. Following Bringmann and Murthy, an identity between Jacobi forms puts $\varphi_1$ into closed form through Hurwitz class numbers and the theta series of $D_4$, and the sign of every coefficient can be read off (Theorem 3.3).

The proof draws throughout on the work of Sen on the positivity conjecture and the attractor mechanism, of Bhand, Sen and Singh on the mock modularity of the CHL partition functions, of

[3]For $SL_2(\mathbb{Z})$ one dominant term suffices. At level two the dominant contribution is a whole family, and the family resums to an eta quotient; the bound $|P|^2 \geq 0.538$ is sharp in the limit (Remark 14.4).

Cardoso, Nampuri and Rosselló on the decay walls, of Bringmann and Manschot and of Ferrari and Reys on mixed Rademacher series, on the circle method of Rademacher and Zuckerman, on the completions of Zwegers and on the theory of Jacobi forms of Eichler and Zagier [17].

Section 2 recalls the partition function and the decomposition of Bhand, Sen and Singh, and Section 3 treats the first row. Section 4 proves the reduction and the bounded-discriminant theorem, and Section 5 describes the Farey walls. Section 6 states the Rademacher expansion, and Sections 7 to 10 prove it: the polar data at the two cusps, the transformation law and the circle method. Section 11 proves the eta-quotient picture, Section 12 collects the facts on Bessel functions, Section 13 bounds every remaining term, and Section 14 proves Theorem A. Section 15 compares the values with the product formula and with Sen's table, and Section 16 describes the extension to the other CHL models. The appendices describe the exact computation and the interval certificates and list the margins.

## 2. The partition function and the single-centred index

2.1. **Notation.** Write $\mathbf{e}(t) = e^{2\pi i t}$. The variables of the Siegel upper half-space are $(\rho, \sigma, v)$; put $x = \mathbf{e}(\sigma/2)$, $p = x^2 = \mathbf{e}(\sigma)$ and $y = \mathbf{e}(v)$. For a function of $\sigma$ alone we use the expansion in $x$. The Jacobi theta functions are

$$\theta_1(\tau,z) = -i\sum_{r\in\mathbb{Z}+\frac12}(-1)^{r-\frac12}q^{r^2/2}\zeta^r, \qquad \theta_2(\tau,z) = \sum_{r\in\mathbb{Z}+\frac12}q^{r^2/2}\zeta^r,$$
$$\theta_3(\tau,z) = \sum_{n\in\mathbb{Z}}q^{n^2/2}\zeta^n, \qquad \theta_4(\tau,z) = \sum_{n\in\mathbb{Z}}(-1)^n q^{n^2/2}\zeta^n,$$

with $q = \mathbf{e}(\tau)$, $\zeta = \mathbf{e}(z)$, and $\theta_k(\tau) = \theta_k(\tau,0)$. Let $E(\tau) = 1 - 24\sum\sigma_1(n)q^n$, $E_4 = 1 + 240\sum\sigma_3(n)q^n$, $E_6 = 1 - 504\sum\sigma_5(n)q^n$, and let

$$\mathfrak{E}(\tau) = 2E(2\tau) - E(\tau) = 1 + 24\sum_{n\ge1}\sigma_1^{\mathrm{odd}}(n)q^n, \qquad \sigma_1^{\mathrm{odd}}(n) = \sum_{d\mid n,\ d\ \mathrm{odd}} d,$$

be the Eisenstein series of weight two for $\Gamma_0(2)$. The weak Jacobi forms of weight $-2$ and $0$ and index one are

$$\phi_{-2,1}(\tau,z) = -\frac{\theta_1(\tau,z)^2}{\eta(\tau)^6} = \zeta{-}2{+}\zeta^{-1}{+}O(q), \qquad \phi_{0,1}(\tau,z) = 4\sum_{k=2}^{4}\frac{\theta_k(\tau,z)^2}{\theta_k(\tau)^2} = \zeta{+}10{+}\zeta^{-1}{+}O(q)$$

[17, §9]; we write $A = \phi_{-2,1}(\sigma,v)$ and $B = \phi_{0,1}(\sigma,v)$. The Hurwitz class number $H(D)$ is extended by $H(0) = -\frac{1}{12}$ and $H(D) = 0$ unless $D$ is a nonnegative integer with $D \equiv 0,3 \pmod 4$. Put $g(\tau) = \eta(\tau)^8\eta(2\tau)^8$ and

$$\frac{1}{g(\sigma/2)} = \sum_{j\ge0}\beta(j)\,x^{j-1}, \qquad \sum_{j\ge0}\beta(j)x^j = \prod_{r\ge1}(1-x^r)^{-e_r}, \quad e_r = \begin{cases} 8, & r \text{ odd},\\ 16, & r \text{ even}, \end{cases} \tag{1}$$

so that every $\beta(j)$ is a positive integer.

2.2. **The product formula.** The twisted elliptic genera of $K3$ for the $\mathbb{Z}_2$ symplectic involution are [13]

$$F^{(0,0)} = \phi_{0,1}, \qquad F^{(0,1)} = \tfrac13\phi_{0,1} + \tfrac23\phi_{-2,1}\,\mathfrak{E}(\tau),$$
$$F^{(1,0)} = \tfrac13\phi_{0,1} - \tfrac13\phi_{-2,1}\,\mathfrak{E}(\tfrac{\tau}{2}), \qquad F^{(1,1)} = \tfrac13\phi_{0,1} - \tfrac13\phi_{-2,1}\,\mathfrak{E}(\tfrac{\tau+1}{2}), \tag{2}$$

each a weak Jacobi form of weight 0 and index one for $\Gamma(2)$, and $SL_2(\mathbb{Z})$ permutes them.[4] Write $F^{(r,s)}(\tau,z)=\sum_{b\in\{0,1\}}\sum_{j\in2\mathbb{Z}+b}\sum_{n\in\mathbb{Z}/2}c_b^{(r,s)}(4n-j^2)q^n\zeta^j$. The dyon partition function of the model is [23, 13, 14]

(3)
$$\widetilde{\Phi}_6(\rho,\sigma,v)=\mathbf{e}\big(\rho+\tfrac{\sigma}{2}+v\big)\prod_{b=0}^{1}\prod_{r=0}^{1}\prod_{\substack{k'\in\mathbb{Z}+r/2,\ l\in\mathbb{Z},\ j\in2\mathbb{Z}+b\\ k',l\ge0;\ j<0 \text{ if } k'=l=0}}\big(1-\mathbf{e}(k'\sigma+l\rho+jv)\big)^{\sum_{s=0}^{1}(-1)^{sl}c_b^{(r,s)}(4k'l-j^2)},$$

and the index of the charges $(Q,P)$ is $(-1)^{Q\cdot P+1}$ times the coefficient of $\mathbf{e}(m\rho)x^ny^\ell$ with

$$m=Q^2,\qquad n=P^2/2,\qquad \ell=Q\cdot P, \tag{4}$$

in the expansion of $1/\widetilde{\Phi}_6$ on the appropriate contour [16, 23, 27].[5]

Expand $1/\widetilde{\Phi}_6=\sum_{m\ge-1}\varphi_m(\sigma,v)\,\mathbf{e}(m\rho)$. The factors with $l=0$ in (3) have the exponents $\sum_s c_b^{(r,s)}(-j^2)$, equal to $12,2,8,0$ for $(r,|j|)=(0,0),(0,1),(1,0),(1,1)$ and to 0 for $|j|\ge2$; together with the prefactor they give

$$\varphi_{-1}=\frac{1}{g(\sigma/2)\,A}, \tag{5}$$

since $\phi_{-2,1}=-\theta_1^2/\eta^6$. Taking the logarithm of the factors with $l\ge1$ and collecting the powers of $\mathbf{e}(\rho)$,

$$\begin{aligned}\frac{1}{\widetilde{\Phi}_6}&=\varphi_{-1}\,\mathbf{e}(-\rho)\exp\Big(\sum_{l\ge1}L_l(\sigma,v)\,\mathbf{e}(l\rho)\Big),\\ L_l&=\sum_{td=l}\frac{1}{2dt}\sum_{r,s\in\{0,1\}}(-1)^{sd}\sum_{u=0}^{2d-1}(-1)^{ru}F^{(r,s)}\Big(\frac{t\sigma+u}{d},\,tv\Big).\end{aligned} \tag{6}$$

The sum over $u$ keeps exactly the integral powers of $x$, so each $L_l$ is a series in $x$ whose coefficients are Laurent polynomials in $y$; in particular $L_l$ is entire in $v$. Hence

$$g(\tfrac{\sigma}{2})\,A\,\varphi_1=L_2+\tfrac12L_1^2,\qquad g(\tfrac{\sigma}{2})\,A\,\varphi_2=L_3+L_1L_2+\tfrac16L_1^3. \tag{7}$$

**Lemma 2.1.** *The functions $L_1$, $L_2$, $L_3$ are invariant under $\Gamma(2)$, $\Gamma(4)$, $\Gamma(6)$ respectively, acting on $\sigma$, and $L_l$ has the elliptic property of index $l$ in $v$. Every polynomial in $L_1,L_2,L_3$ that is homogeneous of total index $I$ is a Jacobi form of weight 0 and index $I$ for $\Gamma(12)$, even in $v$ and weak at every cusp.*

*Proof.* Let $\gamma=\left(\begin{smallmatrix}a&b\\c&e\end{smallmatrix}\right)$ and $M=\left(\begin{smallmatrix}t&u\\0&d\end{smallmatrix}\right)$. Then

$$M\gamma M^{-1}=\begin{pmatrix}a+uc/t & \big(tb+u(e-a)-u^2c/t\big)/d\\ dc/t & e-uc/t\end{pmatrix}.$$

For $(t,d)=(1,1)$ and $\gamma\in\Gamma(2)$, for $(t,d)=(1,2),(2,1)$ and $\gamma\in\Gamma(4)$, and for $(t,d)=(1,3),(3,1)$ and $\gamma\in\Gamma(6)$, this matrix lies in $\Gamma(2)$, under which every $F^{(r,s)}$ is invariant. This gives the invariance, and $\Gamma(12)$ is contained in all three groups. Under $v\mapsto v+\lambda\sigma+\mu$ the variable $tv$ of the term $(t,d,u)$ moves by $\lambda d\,\tau'+(t\mu-\lambda u)$ with $\tau'=(t\sigma+u)/d$, and the elliptic property of index one of $F^{(r,s)}$ produces the factor $\mathbf{e}(-\lambda^2d\tau'-2\lambda d\,tv)=\mathbf{e}(-\lambda^2l\sigma-2\lambda lv)$, which is the elliptic property of index $l$. Evenness in $v$ is inherited from the genera. For $\gamma\in SL_2(\mathbb{Z})$

[4] In the normalisation of [13], $F^{(r,s)}=\frac{1}{N}\operatorname{Tr}_{RR,g^r}\big(g^s(-1)^{F_L+F_R}q^{L_0}\bar q^{\bar L_0}y^J\big)$, so that $F^{(0,0)}$ is half the elliptic genus $2\phi_{0,1}$ of $K3$; the constant terms then give the exponents $12,2,8,0$ used below.

[5] The exponent of $\mathbf{e}(\rho)$ is $NQ^2/2=Q^2$, and the power of $x=\mathbf{e}(\sigma/2)$ is $P^2/2$, as in [8, (5.15)].

one has $M\gamma = \gamma' M'$ with $\gamma' \in SL_2(\mathbb{Z})$ and $M'$ upper triangular of the same determinant, and $SL_2(\mathbb{Z})$ permutes the $F^{(r,s)}$, none of which contains a negative power of $q$; so every term is weak at every cusp. □

2.3. **Polar part, finite part and the single-centred index.** By [8, Theorem 5.2], $\varphi_m$ is a meromorphic Jacobi form of weight $-6$ and index $m$ for $\Gamma_1(2)$ with double poles at $v \in \mathbb{Z}\sigma + \mathbb{Z}$. Its polar part is [8, (5.45)]

$$\varphi_m^P = \frac{\hat f(m)}{g(\sigma/2)}\,\mathcal{A}_{2,m}, \qquad \mathcal{A}_{2,m}(\sigma,v) = \sum_{s\in\mathbb{Z}} \frac{p^{ms^2+s} y^{2ms+1}}{(1-p^s y)^2}, \tag{8}$$

where $\hat f(m)$ is the coefficient of $q^m$ in $1/\big(\eta(\tau)^8\eta(2\tau)^8\big)$, so that $\hat f(-1),\hat f(0),\hat f(1),\hat f(2),\hat f(3) = 1, 8, 52, 256, 1122$. The finite part $\varphi_m^F = \varphi_m - \varphi_m^P = \sum c_m^F(n,\ell) x^n y^\ell$ is a mock Jacobi form.

Let $\mathcal{A}$ be the set of charges of [8, (5.46)–(5.47)]: with $N = 2$ and the notation (4), the charges with $m, n > 0$ and

$$D := Q^2P^2 - (Q\cdot P)^2 = 2mn - \ell^2 > 0, \tag{9}$$

such that

$$\min_{m_1\in 2\mathbb{Z},\ s\in\mathbb{Z},\ m_1 | s(s+1)} \Big[\big(m_1\ell + 2ms + m\big)^2 + m_1^2 D\Big] \ \geq\ m^2. \tag{10}$$

By [8, Theorem 5.4(a)], for every charge in $\mathcal{A}$,

$$d^*(Q,P) = (-1)^{\ell+1} c_m^F(n,\ell). \tag{11}$$

The single-centred index is invariant under the action [8, (5.17)–(5.19)] of $\Gamma_1(2)$ on the charges.

**Lemma 2.2.** *Every charge with $Q^2 \in \{1,2\}$, $P^2 > 0$ and $D > 0$ lies in $\mathcal{A}$.*

*Proof.* If $m_1 = 0$, the condition $m_1 \mid s(s+1)$ reads $s(s+1) = 0$, and the bracket is $m^2(2s+1)^2 = m^2$. If $m_1 \neq 0$, then $m_1^2 \geq 4 \geq m^2$ and $D \geq 1$, so the bracket is at least $m^2$. □

2.4. **A positivity lemma.** The last step of Section 3 divides a series of known sign pattern by $g$.

**Lemma 2.3.** *Let $f = -1 + \sum_{n\geq 1} a_n x^n$ with $a_n > K$ for every $n$, and let $0 < e_r \leq K$ for every $r \geq 1$. Then $f\prod_{r\geq 1}(1-x^r)^{-e_r}$ has positive coefficients at every $x^n$ with $n \geq 1$.*

*Proof.* Let $f_0 = f$ and $f_s = f_{s-1}(1-x^s)^{-e_s} = -1 + \sum_n b_n^{(s)} x^n$. We show by induction on $s$ that $b_n^{(s)} > 0$ for $n \geq 1$ and $b_n^{(s)} > K$ for $n > s$; since $f_s$ and the full product agree through $x^s$, this proves the lemma. The case $s = 0$ is the hypothesis. Write $e = e_s$ and $b = b^{(s-1)}$. If $s \nmid n$, then $b_n^{(s)} = \sum_k \binom{e+k-1}{k} b_{n-sk} \geq b_n$, all terms being positive. If $n = sm$, then

$$b_{sm}^{(s)} = \sum_{k=0}^{m-1} \binom{e+k-1}{k} b_{s(m-k)} - \binom{e+m-1}{m} > \binom{e+m-1}{m-1}\Big(K - \frac{e}{m}\Big),$$

since every $b_{s(m-k)}$ exceeds $K$ and $\sum_{k=0}^{m-1}\binom{e+k-1}{k} = \binom{e+m-1}{m-1}$. The right side is $K - e \geq 0$ for $m = 1$, and at least $(e+1)(K - \frac{e}{2}) \geq (e+1)\frac{K}{2} \geq K$ for $m \geq 2$. □

We also use the elementary bound $H(D) \leq D$ for $D \geq 3$: a reduced form $(a,b,c)$ of discriminant $-D$ has $|b| \leq a \leq \sqrt{D/3}$ and is determined by $(a,b)$, so $H(D) \leq \sqrt{D/3}\,(2\sqrt{D/3}+1) \leq D$.

## 3. The first row: $Q^2 = 1$

Let $\mathcal{H}(\sigma, v) = \sum_{n,\ell} H(4n-\ell^2)p^n y^\ell$ be the generating function of the Hurwitz class numbers, $\mathcal{B} = \mathcal{A}_{2,1}$, $r_3(D)$ the number of representations of $D$ as a sum of three squares,

$$M(\tau) = \frac{E_4(\tau) - E_4(2\tau)}{240} = \sum_{n\geq 1} \sigma_3^{\mathrm{odd}}(n) q^n, \qquad \sigma_3^{\mathrm{odd}}(n) = \sum_{d\mid n,\ n/d \text{ odd}} d^3,$$

$$\Theta(\sigma, v) = \sum_{e,\ell} r_3(2e-\ell^2)\, x^e y^\ell.$$

Then $\Theta = \frac{1}{2}\big[\theta_3(\frac{\sigma}{2}, v)\theta_3(\frac{\sigma}{2})^3 + \theta_4(\frac{\sigma}{2}, v)\theta_4(\frac{\sigma}{2})^3\big]$ is the theta series of the lattice $D_4$, with the Jacobi variable on one coordinate vector, at $\tau = \sigma/2$.[6] Write $\phi_{-2,1}(\tau, z) = \sum \alpha(4a - r^2) q^a \zeta^r$, so that $\alpha(-1) = 1$ and $\alpha(0) = -2$.

**Proposition 3.1.** *In the $\mathbb{Z}_2$ CHL model*

$$g(\tfrac{\sigma}{2})\,\varphi_1 = 52\,\mathcal{B} + 48\,M(\tfrac{\sigma}{2})\,A - \tfrac{20}{3}\,\Theta - 104\,\mathcal{H}. \tag{12}$$

*Consequently $c_1^F(n,\ell)$ depends only on $D = 2n - \ell^2$, and*

$$\begin{aligned} c_1^F(n,\ell) &= \mathcal{C}(D) := \sum_{j\geq 0} \beta(j)\,\mathcal{N}(D + 2 - 2j), \\ \mathcal{N}(D) &= 48 \sum_{t\geq 1} \sigma_3^{\mathrm{odd}}(t)\,\alpha(D-2t) - \tfrac{20}{3}\, r_3(D) - 104\, H(D). \end{aligned} \tag{13}$$

*Proof. Step 1.* We claim that

$$L_2 + \tfrac{1}{2}L_1^2 = \tfrac{13}{36}\,B^2 - \tfrac{5}{9}\,\mathfrak{E}(\tfrac{\sigma}{2})\,AB + \Big(\tfrac{4}{45}E_4(\tfrac{\sigma}{2}) + \tfrac{19}{180}E_4(\sigma)\Big)A^2. \tag{14}$$

Every term is a Jacobi form of weight 0 and index two for $\Gamma(4)$, even and holomorphic in $v$ and weak at every cusp (Lemma 2.1; $\mathfrak{E}(\sigma/2)$ and $E_4(\sigma/2)$ are modular for $\Gamma^0(2) \supset \Gamma(4)$). Let $G$ be the difference of the two sides and $G = \chi_0(\sigma) + \chi_2(\sigma)v^2 + \chi_4(\sigma)v^4 + \cdots$. If $\chi_0 = \cdots = \chi_{\nu-2} = 0$, the lowest order in $v$ of the transformation law shows that $\chi_\nu$ is a modular form of weight $\nu$ for $\Gamma(4)$ [17, §3], holomorphic at the cusps because $G$ is weak. Since $[PSL_2(\mathbb{Z}) : \bar{\Gamma}(4)] = 24$ and the width of the cusp $\infty$ is four, Sturm's theorem [31] shows that such a form vanishes once its coefficients of $x^0, \ldots, x^\nu$ do. The coefficients of $G$, computed from (6) and (2), vanish through $x^{48}$, so $\chi_0 = \chi_2 = \chi_4 = 0$. For each $\sigma$ the entire function $G(\sigma, \cdot)$ then vanishes to order at least six at $v = 0$, while a nonzero function with the elliptic property of index two has exactly four zeros in a period parallelogram [17, Theorem 1.2]. Hence $G = 0$.

*Step 2.* Bringmann and Murthy [5, Section 3] proved

$$\frac{B^2}{A} = 144\,\mathcal{B} + E_4(\sigma)\,A - 288\,\mathcal{H}, \tag{15}$$

and the identity

$$\mathfrak{E}(\tfrac{\sigma}{2})\,B = \tfrac{1}{5}\big(6E_4(\sigma) - E_4(\tfrac{\sigma}{2})\big)A + 12\,\Theta \tag{16}$$

holds because both sides are holomorphic Jacobi forms of weight two and index one for $\Gamma^0(2)$ [17, §7]: the argument of Step 1, with $[PSL_2(\mathbb{Z}) : \bar{\Gamma}^0(2)] = 3$, width two, Sturm bound $x^{\lfloor \nu/4 \rfloor}$ and two zeros in a period parallelogram, reduces it to the coefficients through $x^{48}$, which agree.

[6] A vector of $D_4$ is an integral vector of even coordinate sum; the term $\frac{1}{2}[\cdots]$ selects these, and its norm is $2e$ when the first coordinate is $\ell$ and the others contribute $2e - \ell^2$.

*Step 3.* Divide (14) by $A$ and insert (15) and (16). The coefficient of $\mathcal{B}$ is $\frac{13}{36}\cdot 144 = 52$, that of $\mathcal{H}$ is $-\frac{13}{36}\cdot 288 = -104$, that of $\Theta$ is $-\frac{5}{9}\cdot 12 = -\frac{20}{3}$, and that of $A$ is

$$\Big(\tfrac{13}{36} - \tfrac{2}{3} + \tfrac{19}{180}\Big)E_4(\sigma) + \Big(\tfrac{1}{9} + \tfrac{4}{45}\Big)E_4(\tfrac{\sigma}{2}) = \tfrac{1}{5}\big(E_4(\tfrac{\sigma}{2}) - E_4(\sigma)\big) = 48\,M(\tfrac{\sigma}{2}).$$

With (7) this is (12). The term $52\mathcal{B}/g(\sigma/2)$ is the polar part (8), and the remaining numerator has the coefficient $\mathcal{N}(2e-\ell^2)$ at $x^e y^\ell$; multiplying by (1) gives (13). □

**Lemma 3.2.** (i) $\alpha(D) > 0$ *for* $D \equiv 3 \pmod 4$, $\alpha(D) < 0$ *for* $D \equiv 0 \pmod 4$ *with* $D \geq 0$, *and* $\alpha(D) = 0$ *otherwise.*

(ii) $\mathcal{N}(D) = 0$ *for* $D < 0$, $\mathcal{N}(0) = 2$, $\mathcal{N}(D) > 0$ *for odd* $D \geq 1$, *and* $-\mathcal{N}(2t) \geq 96\,\sigma_3^{\rm odd}(t)$ *for* $t \geq 1$.

*Proof.* (i) With $\vartheta_{1,\mu}(\tau,z) = \sum_{r\equiv\mu\,(2)} q^{r^2/4}\zeta^r$ one has $\phi_{-2,1} = \big[\vartheta_{1,0}(\tau,0)\vartheta_{1,1}(\tau,z) - \vartheta_{1,1}(\tau,0)\vartheta_{1,0}(\tau,z)\big]/\eta(\tau)^6$ [5, (4.1)]. So $\sum_{D\equiv 3\,(4)}\alpha(D)q^{(D+1)/4} = \vartheta_{1,0}(\tau,0)\prod(1-q^n)^{-6}$ and $\sum_{D\equiv 0\,(4)}\alpha(D)q^{D/4} = -q^{-1/4}\vartheta_{1,1}(\tau,0)\prod(1-q^n)^{-6}$, and both right sides have positive coefficients at every exponent $\geq 0$.

(ii) For $D<0$ every argument $D-2t$ is at most $-3$, and $r_3(D) = H(D) = 0$. At $D=0$, $\mathcal{N}(0) = -\frac{20}{3} + \frac{104}{12} = 2$. For $D = 2t$ every $\alpha(2t-2t')$ is $\leq 0$ by (i), the term $t' = t$ is $\alpha(0) = -2$, and $r_3, H \geq 0$; this gives $-\mathcal{N}(2t) \geq 96\,\sigma_3^{\rm odd}(t)$. For odd $D$ every $\alpha(D-2t)$ is $\geq 0$, and the term $t = (D+1)/2$ is $\alpha(-1)\sigma_3^{\rm odd}(t) \geq t^3$, so the sum in $\mathcal{N}(D)$ is at least $6(D+1)^3$. Further $r_3(D) \leq (2\sqrt{D}+1)\max_{k\leq D} r_2(k) \leq (2\sqrt{D}+1)\,8\sqrt{D} \leq 24D$, from $r_2(k) \leq 4d(k) \leq 8\sqrt{k}$, and $H(D) \leq D$. Hence $\mathcal{N}(D) \geq 6(D+1)^3 - 264D > 0$ for odd $D \geq 7$, while $\mathcal{N}(1), \mathcal{N}(3), \mathcal{N}(5) = 8,\ 296,\ 1568$. □

**Theorem 3.3.** *In the* $\mathbb{Z}_2$ *CHL model,* $d^*(Q,P) > 0$ *for every charge with* $Q^2 = 1$, $P^2 > 0$ *and* $P^2 > (Q\cdot P)^2$, *and for every charge in the* $\Gamma_1(2)$ *orbit of such a charge.*

*Proof.* By Lemma 2.2, (11) and Proposition 3.1, $d^* = (-1)^{\ell+1}\mathcal{C}(D)$ with $D = 2n - \ell^2 \equiv \ell \pmod 2$. If $D$ is odd, every argument of $\mathcal{N}$ in (13) is odd, so every term is nonnegative by Lemma 3.2, and the term $j = 0$ is $\mathcal{N}(D+2) > 0$. If $D = 2u$ is even, $-\mathcal{C}(2u)$ is the coefficient of $x^{u+1}$ in $f\prod_r (1-x^r)^{-e_r}$ with $f = \sum_{t\geq 0}\big(-\mathcal{N}(2t)\big)x^t$. By Lemma 3.2, $f/2 = -1 + \sum_{t\geq 1} a_t x^t$ with $a_t \geq 48$, and Lemma 2.3 with $K = 16$ gives $-\mathcal{C}(2u) > 0$. The orbits follow from the $\Gamma_1(2)$ invariance of $d^*$. □

## 4. Reduction to a fundamental set of charges

From now on $m$ is arbitrary. This section shows that the sign of $d^*$ has to be checked only on charges with $m \leq n$ and $0 \leq \ell \leq m$, and proves Theorem 4.7.

4.1. **The Fricke symmetry of the product formula.** The duality group $\Gamma_1(2)$ acts on the charges by $(Q,P) \mapsto (aQ+bP,\, cQ+dP)$ with $\left(\begin{smallmatrix} a & b \\ c & d\end{smallmatrix}\right) \in \Gamma_1(2)$, that is, $c$ even and $a, d$ odd [8, (5.17)–(5.19)]. It does not exchange $Q^2$ with $P^2/2$. The product formula has, however, an extra symmetry that does.

**Lemma 4.1.** *Let* $E(A,B,j) = \sum_{s=0}^{1}(-1)^{sB} c_{j \bmod 2}^{(A \bmod 2,\, s)}(2AB - j^2)$ *be the exponent of the factor* $1 - x^A \mathbf{e}(B\rho) y^j$ *in* (3), *where* $x = \mathbf{e}(\sigma/2)$ *and* $A = 2k'$. *Then* $E(A,B,j) = E(B,A,j)$ *for all* $A, B \geq 0$ *and* $j$. *Consequently*

$$\widetilde{\Phi}_6(\rho,\sigma,v) = \widetilde{\Phi}_6\big(\tfrac{\sigma}{2},\, 2\rho,\, v\big) \tag{17}$$

*as formal products.*

*Proof.* If $A$ and $B$ have the same parity, the two sides are the same expression. Let $A$ be even and $B$ odd; then $2AB - j^2 \equiv -j^2 \pmod 4$ and $AB/2 \in \mathbb{Z}$, and

$$E(A,B,j) = c^{(0,0)}(N) - c^{(0,1)}(N), \qquad E(B,A,j) = c^{(1,0)}(N) + c^{(1,1)}(N), \qquad N = 2AB - j^2,$$

both read off at an integral power of $q$. By (2),

$$F^{(0,0)} - F^{(0,1)} = \tfrac{2}{3}\phi_{0,1} - \tfrac{2}{3}\phi_{-2,1}\,\mathfrak{E}(\tau), \qquad F^{(1,0)} + F^{(1,1)} = \tfrac{2}{3}\phi_{0,1} - \tfrac{1}{3}\phi_{-2,1}\big(\mathfrak{E}(\tfrac{\tau}{2}) + \mathfrak{E}(\tfrac{\tau+1}{2})\big).$$

Since $\mathfrak{E}(\tau) = 1 + 24\sum \sigma_1^{\mathrm{odd}}(n)q^n$ and $\sigma_1^{\mathrm{odd}}(2k) = \sigma_1^{\mathrm{odd}}(k)$,

$$\mathfrak{E}(\tfrac{\tau}{2}) + \mathfrak{E}(\tfrac{\tau+1}{2}) = 2 + 48\sum_{k\geq 1}\sigma_1^{\mathrm{odd}}(2k)q^k = 2\mathfrak{E}(\tau),$$

so $F^{(0,0)} - F^{(0,1)} = F^{(1,0)} + F^{(1,1)}$ identically, and the exponents agree. The prefactor $\mathbf{e}(\rho + \frac{\sigma}{2} + v)$ and the range $A, B \geq 0$, $j < 0$ if $A = B = 0$, of the product are symmetric under $A \leftrightarrow B$, which gives (17). □

**Remark 4.2.** The substitution $(\rho, \sigma) \mapsto (\sigma/2, 2\rho)$ is the action on the Siegel upper half-space of the Fricke involution of $\Gamma_0(2)$; Persson and Volpato showed that the CHL models with a geometric symmetry are self-dual under the corresponding Fricke S-duality [24], and (17) is its shadow on the partition function. We use only the elementary statement of Lemma 4.1.

4.2. **A chamber shared by both orientations.** For a charge $(m, n, \ell)$ and a point $(\rho_2, \sigma_2, v_2)$ of imaginary parts, let

$$c(m,n,\ell;\rho_2,\sigma_2,v_2) = \int \frac{\mathbf{e}(-m\rho - n\frac{\sigma}{2} - \ell v)}{\widetilde{\Phi}_6(\rho,\sigma,v)}\, d\rho_1\, d\sigma_1\, dv_1,$$

the integral running over a period cell of the real parts. Sen showed [27, §4] that this number is locally constant in the imaginary parts and jumps only across the walls of marginal stability, which for the $\mathbb{Z}_2$ CHL model are the hyperplanes

$$j\,v_2 + n_1\sigma_2 - m_1\rho_2 = 0, \qquad j \text{ odd}, \quad m_1 \in \mathbb{Z}, \quad n_1 \in \tfrac{1}{2}\mathbb{Z}, \quad m_1 n_1 = \tfrac{1-j^2}{4}, \tag{18}$$

coming from the double zeros of $\widetilde{\Phi}_6$ with vanishing coefficient of $\rho\sigma - v^2$.

**Lemma 4.3.** *Let* $\mathcal{R} = \{\rho_2, \sigma_2 > 0,\ 0 < -v_2 < \frac{1}{2}\min(\rho_2, \frac{\sigma_2}{2})\}$.

(i) $\mathcal{R}$ *meets no wall* (18)*, so* $c_{\mathcal{R}}(m,n,\ell) = c(m,n,\ell;\rho_2,\sigma_2,v_2)$ *does not depend on the point of* $\mathcal{R}$.
(ii) $c_{\mathcal{R}}(m,n,\ell) = c_{\mathcal{R}}(n,m,\ell)$.
(iii) *If* $D = 2mn - \ell^2 > 0$ *and* $0 \leq \ell \leq 2m$*, then* $c_m^F(n,\ell) = c_{\mathcal{R}}(m,n,\ell)$.

*Proof.* (i) For $j = \pm 1$ one of $m_1, n_1$ vanishes and the wall is $|v_2| = |n_1|\sigma_2$ with $|n_1| \geq \frac{1}{2}$, or $|v_2| = |m_1|\rho_2$ with $|m_1| \geq 1$, or $v_2 = 0$; none meets $\mathcal{R}$. For $|j| \geq 3$ the product $m_1 n_1$ is negative, so $|n_1\sigma_2 - m_1\rho_2| = |n_1|\sigma_2 + |m_1|\rho_2 \geq 2\sqrt{|m_1 n_1|\sigma_2\rho_2} = \sqrt{j^2-1}\,\sqrt{\sigma_2\rho_2}$, while on $\mathcal{R}$

$$|j\,v_2| < \tfrac{|j|}{2}\min(\rho_2, \tfrac{\sigma_2}{2}) \leq \tfrac{|j|}{2\sqrt{2}}\sqrt{\rho_2\sigma_2} < \sqrt{j^2-1}\,\sqrt{\rho_2\sigma_2}.$$

(ii) The substitution of Lemma 4.1 maps $\mathcal{R}$ to itself and exchanges the roles of $m$ and $n$.

(iii) Let $\rho_2 \to \infty$ inside $\mathcal{R}$ with $\sigma_2$ and $v_2$ fixed. The integral over $\rho_1$ picks out $\varphi_m$, and $c_{\mathcal{R}}(m,n,\ell)$ is the coefficient of $x^n y^\ell$ of $\varphi_m$ expanded in the strip $-\sigma_2 < v_2 < 0$, where $|p| < |y|^{-1} < 1$. Write $\varphi_m = \varphi_m^P + \varphi_m^F$ with (8). The finite part is holomorphic in $v$, so its expansion does not depend on the strip. In the polar part, expand $\mathcal{A}_{2,m}$ in the strip: the term $s = 0$ is $y(1-y)^{-2} = \sum_{k\geq 1} k\,y^{-k}$, the terms $s \geq 1$ contain only powers $y^{2ms+1+k}$ with $k \geq 0$, and the terms $s \leq -1$, rewritten through $(1-p^s y)^{-2} = p^{-2s}y^{-2}(1-p^{-s}y^{-1})^{-2}$, only powers $y^{2ms-1-k}$

with $k \geq 0$. None of these exponents lies in $[0, 2m]$, and the factor $1/g(\sigma/2)$ does not change the power of $y$. Hence the polar part contributes nothing at $y^\ell$ with $0 \leq \ell \leq 2m$. □

4.3. **Least representatives.**

**Lemma 4.4.** *Let $(m, n, \ell)$ be a charge with $m \geq 1$, $|\ell| \leq m$, $|\ell| \leq n$ and $D = 2mn - \ell^2 > 0$. Then it lies in the set $\mathcal{A}$ of* (10).

*Proof.* For $m_1 = 0$ the bracket in (10) is $m^2(2s+1)^2 \geq m^2$. Let $m_1 \neq 0$, so $|m_1| \geq 2$, and let $d$ be the distance from $m_1\ell$ to the nearest odd multiple of $m$; the bracket is at least $d^2 + m_1^2 D$. If $|m_1\ell| \leq m$, then $d \geq m - |m_1\ell|$ and

$$d^2 + m_1^2 D \;\geq\; m^2 - 2m|m_1\ell| + m_1^2\ell^2 + m_1^2(2mn - \ell^2) = m^2 + 2m|m_1|\big(|m_1|n - |\ell|\big) \;\geq\; m^2.$$

If $|m_1\ell| > m$, then $D \geq 2m|\ell| - \ell^2 \geq m|\ell|$ and $m_1^2 D \geq m|m_1| \cdot |m_1\ell| > 2m^2$. □

**Definition 4.5.** A charge $(m, n, \ell)$ is *reduced* if $1 \leq m \leq n$, $0 \leq \ell \leq m$ and $D = 2mn - \ell^2 > 0$. For a reduced charge put $x = \sqrt{D}/m$ and $\theta = \pi\ell/m$. Since $D \geq 2m^2 - m^2$, every reduced charge has $x \geq 1$.

**Proposition 4.6.** *For every charge $(Q, P)$ with $D > 0$ there is a reduced charge $(m', n', \ell')$ with the same discriminant and $\ell' \equiv Q \cdot P \pmod 2$ such that*

$$d^*(Q, P) = (-1)^{\ell'+1} c^F_{m'}(n', \ell') = (-1)^{\ell'+1} c_{\mathcal{R}}(m', n', \ell').$$

*Proof.* In the orbit of $(Q, P)$ choose a charge with least $m = Q^2$; since $D > 0$ the form $f(a, b) = (aQ + bP)^2 = a^2 m + 2ab\ell + 2b^2 n$ is positive definite, so the least value exists. The matrices $\left(\begin{smallmatrix} 1 & 0 \\ c & 1 \end{smallmatrix}\right)$ with $c$ even fix $Q$ and send $\ell$ to $\ell + cm$; so we may assume $|\ell| \leq m$. The matrices $\left(\begin{smallmatrix} 1 & \pm 1 \\ 0 & 1 \end{smallmatrix}\right)$ send $m$ to $m \pm 2\ell + 2n$, and minimality gives $n \geq |\ell|$. By Lemma 4.4 the charge lies in $\mathcal{A}$, so $d^* = (-1)^{\ell+1} c^F_m(n, \ell)$ by (11). The finite part is a mock Jacobi form, even in $v$, so $c^F_m(n, \ell) = c^F_m(n, |\ell|)$. If $m \leq n$, the charge $(m, n, |\ell|)$ is reduced. If $n < m$, then $0 \leq |\ell| \leq n$ and Lemma 4.3 gives $c^F_m(n, |\ell|) = c_{\mathcal{R}}(m, n, |\ell|) = c_{\mathcal{R}}(n, m, |\ell|) = c^F_n(m, |\ell|)$, and $(n, m, |\ell|)$ is reduced. The shifts change $\ell$ by even multiples of $m$ and the sign change does not alter the parity, so $\ell' \equiv Q \cdot P \pmod 2$. □

From now on we write $d^*(m, n, \ell) = (-1)^{\ell+1} c^F_m(n, \ell)$ for a reduced charge, and the conjecture is the statement $d^*(m, n, \ell) > 0$ on reduced charges.

4.4. **Bounded discriminant.**

**Theorem 4.7.** *For every charge of the $\mathbb{Z}_2$ CHL model with $0 < D \leq 800$ one has $d^* > 0$.*

*Proof.* A reduced charge with $D \leq 800$ has $m \leq \sqrt{D} \leq \lfloor\sqrt{800}\rfloor$ and $n = (D + \ell^2)/2m \leq (D + m^2)/2m$, so there are finitely many. For each $\Gamma_1(2)$ orbit with $D \leq 800$ the program of Appendix A finds a charge of least $Q^2$ satisfying (10) with $0 \leq \ell < 2m$, and computes $c_{\mathcal{R}}(m, n, \ell)$ exactly from the product formula, in the cheaper of the two orientations allowed by Lemma 4.3. For $D \leq 800$ there are 22972 orbits; in every one of them $(-1)^{\ell+1} c_{\mathcal{R}}(m, n, \ell) > 0$. By Lemma 4.3(iii) and (11) these numbers are the single-centred indices.[7] □

## 5. Farey walls

The polar coefficients that enter the Rademacher series are sums over matrices of $SL_2(\mathbb{Z})$ attached to Farey intervals, the continued-fraction walls of [10, 12]. This section collects the elementary facts about them that the estimates need.

[7] Every orbit is found with all its representatives inside the computed box, and the program checks that they give the same value; at $Q^2 = 1$ it reproduces the closed formula of Proposition 3.1.

5.1. **The walls of a polar term.** Fix $m \geq 1$, $\nu \in \frac{1}{2}\mathbb{Z}$ and $\tilde{\ell} \in \mathbb{Z}$ with $0 \leq \tilde{\ell} < 2m$ and

$$\tilde{D} = 4m\nu - \tilde{\ell}^2 < 0, \qquad t = -\tilde{D}, \qquad \omega_0 = \frac{\tilde{\ell}}{2m}, \qquad \delta = \frac{\sqrt{t}}{2m}.$$

Let $\mathcal{W}(m, \nu, \tilde{\ell})$ be the set of $G = \left(\begin{smallmatrix} p & q \\ r & s \end{smallmatrix}\right) \in SL_2(\mathbb{Z})$ with $r, s \geq 1$ and $q/s \leq \omega_0 < p/r$; for each such $G$ put

$$m_G = r^2\nu + p^2 m - pr\tilde{\ell}, \qquad n_G = s^2\nu + q^2 m - qs\tilde{\ell}, \qquad \ell_G = -2rs\nu - 2pqm + (ps+qr)\tilde{\ell}. \tag{19}$$

These are the coefficients of the quadratic form $\phi(X, Y) = \nu X^2 - \tilde{\ell}XY + mY^2$ transformed by $G$: $m_G = \phi(r, p)$, $n_G = \phi(s, q)$, and $\ell_G$ is the cross term up to sign.

**Lemma 5.1.** *Let $G \in \mathcal{W}(m, \nu, \tilde{\ell})$ and $U = p/r - \omega_0$, $W = \omega_0 - q/s$, so that $U > 0$, $W \geq 0$ and $U + W = 1/rs$.*

(i) $\ell_G^2 - 4m_G n_G = t$.

(ii) $m_G = mr^2(U^2 - \delta^2)$, $n_G = ms^2(W^2 - \delta^2)$, $\ell_G = 2mrs(UW + \delta^2) > 0$, *and* $m = m_G s^2 + n_G r^2 + \ell_G rs$.

(iii) $E_G := m - m_G - n_G - \ell_G = m\big[1 - (rU + sW)^2 + \delta^2(r - s)^2\big] \geq 0$.

(iv) *If $m_G, n_G \geq -1$, put $a = m_G$, $b = n_G$, $S = a + b + 2$ and $t_0 = m^2 + 4m$. Then*

$$t_0 - t = S(2m - S) + E_G^2 + 2E_G\ell_G + 4(a+1)(b+1), \tag{20}$$

*so $t \leq t_0$; moreover $t_0 - t \leq 2mS$ when $r = s = 1$.*

(v) *If $m_G, n_G \geq -1$, then $a \leq m/s^2$, $b \leq m/r^2$ and $r, s \leq 2m$.*

(vi) *If $m_G, n_G \geq -1$ and $rs \geq 2$, then $\delta \leq \frac{1}{4} + \frac{5}{2m}$, that is $\sqrt{t} \leq \frac{m}{2} + 5$.*

(vii) *If $r = s = 1$, then $G = \left(\begin{smallmatrix} 1 & 0 \\ 1 & 1 \end{smallmatrix}\right)$, $a = \nu + m - \tilde{\ell}$, $b = \nu$, $\ell_G = m - a - b$, $E_G = 0$ and $\tilde{\ell} = m + b - a$. Conversely every pair $(a, b)$ of integers with $a, b \geq -1$, $k := m - a - b \geq 1$, $t = k^2 - 4ab > 0$ and $-m \leq b - a < m$ arises from exactly one polar term with $\nu = b \in \mathbb{Z}$ in this way.*

*Proof.* (i) $\phi \circ G$ has the same discriminant as $\phi$. (ii) $\phi(r, p) = r^2\big(\nu - \tilde{\ell}u + mu^2\big)$ with $u = p/r$, and $\nu - \tilde{\ell}u + mu^2 = m\big((u - \omega_0)^2 - \delta^2\big)$ because $\tilde{\ell}^2 - 4m\nu = 4m^2\delta^2$; similarly for $n_G$. For $\ell_G$ write $\ell_G = 2mrs\big[(p/r - \omega_0)(\omega_0 - q/s) + \delta^2\big]$, which is (19) after expanding and using $ps - qr = 1$. The last identity follows from $\phi(r, p)s^2 + \phi(s, q)r^2 + \ell_G rs = m(ps - qr)^2$. (iii) Insert (ii) in the definition and use $U + W = 1/rs$; then $rU + sW \leq \max(r, s)(U + W) = 1/\min(r, s) \leq 1$. (iv) By (i), $t = \ell_G^2 - 4ab$ with $\ell_G = m - a - b - E_G = m + 2 - S - E_G$; expand $t_0 - t = m^2 + 4m - (m + 2 - S - E_G)^2 + 4ab$ and use $4ab = 4(a+1)(b+1) - 4S + 4$. For $r = s = 1$, $E_G = 0$ and $4(a+1)(b+1) \leq S^2$. (v) $a = mr^2(U^2 - \delta^2) \leq mr^2U^2 \leq m/s^2$, and $b \leq m/r^2$ in the same way. Since $t \geq 1$, $\delta \geq 1/2m$. From $a \geq -1$, $\delta^2 \leq U^2 + 1/(mr^2)$, and from $b \geq -1$, $\delta^2 \leq W^2 + 1/(ms^2)$. One of $U, W$ is at most $1/(2rs)$; if it is $U$, the first inequality gives $r^2 \leq 4m^2(\frac{1}{4s^2} + \frac{1}{m}) \leq m^2 + 4m$, so $r \leq m + 1$, and the second with $W \leq 1/rs$ gives $s^2 \leq 4m^2(1 + \frac{1}{m})$, so $s \leq 2m$. The other case is symmetric. (vi) If $\delta \leq \frac{1}{4}$ there is nothing to prove. Otherwise, from $a \geq -1$, $\delta - U \leq 1/(mr^2(\delta + U)) \leq 4/(mr^2)$, and from $b \geq -1$, $\delta - W \leq 4/(ms^2)$. Adding, $2\delta \leq \frac{1}{rs} + \frac{4}{m}\big(\frac{1}{r^2} + \frac{1}{s^2}\big) \leq \frac{1}{2} + \frac{5}{m}$ when $rs \geq 2$. (vii) $p - q = 1$ and $q \leq \omega_0 < q + 1$ with $0 \leq \omega_0 < 1$ force $q = 0$, $p = 1$. The formulas follow from (19). Given $(a, b)$, put $\nu = b$ and $\tilde{\ell} = m + b - a \in [0, 2m)$; then $\tilde{D} = 4mb - (m + b - a)^2 = -(k^2 - 4ab)$. □

**Remark 5.2.** Lemma 5.1(iii) is the reason why the leading family dominates: by (20) the walls with $r = s = 1$ are exactly those for which $t_0 - t$ is as small as the pair $(a, b)$ allows, and by (vi) every other wall has $\sqrt{t}$ at most about half of $\sqrt{t_0}$. Figure 1 shows the geometry.

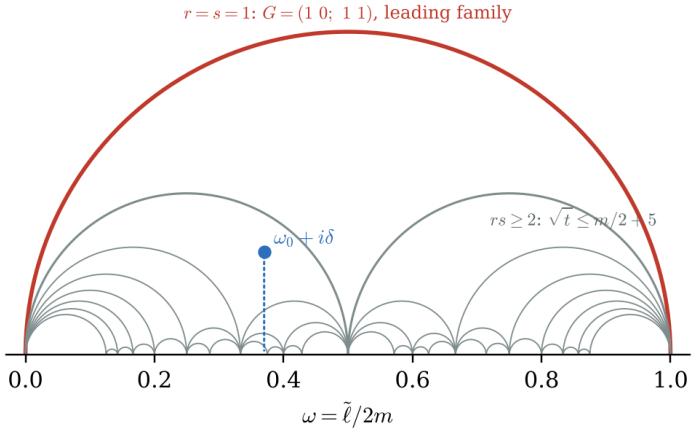



FIGURE 1. The Farey walls of a polar term at the cusp 0, drawn as geodesics from $q/s$ to $p/r$ in the upper half-plane of $\omega = \tilde{\ell}/2m$. A matrix $G = \left(\begin{smallmatrix} p & q \\ r & s \end{smallmatrix}\right)$ contributes to the polar coefficient at $(\tilde{\ell}, \nu)$ when its geodesic passes over the point $\omega_0 + i\delta$ with $\omega_0 = \tilde{\ell}/2m$ and $\delta = \sqrt{-\tilde{D}}/2m$ (Lemma 5.1). The outer red geodesic is the leading family $r = s = 1$; every other wall lies below height $\frac{1}{4} + \frac{5}{2m}$.

5.2. **Polar coefficients.** Let $d_1(k)$, $k \geq -1$, be the coefficients of

$$\frac{1}{\eta(\tau)^8\eta(2\tau)^8} = \sum_{k\geq -1} d_1(k)\, q^k = q^{-1} + 8 + 52q + 256q^2 + 1122q^3 + \cdots , \tag{21}$$

so that $d_1(k) = \hat{f}(k)$ in the notation of (8), and put $d_1(k) = 0$ unless $k \geq -1$ is an integer. For a polar term $(\nu, \tilde{\ell})$ define

$$a_S(\nu, \tilde{\ell}) = \frac{1}{16} \sum_{G\in\mathcal{W}(m,\nu,\tilde{\ell})} \ell_G\, d_1(m_G)\, d_1(n_G) \qquad (\nu \in \mathbb{Z}), \tag{22}$$

$$a_I(\nu, \tilde{\ell}) = \sum_{G\in\mathcal{W}(m,\nu,\tilde{\ell})} \ell_G \Big( [r \text{ even}]\, d_1(m_G)\, d_1(2n_G) + [s \text{ even}]\, d_1(2m_G)\, d_1(n_G) \Big) \qquad (\nu \in \tfrac{1}{2}\mathbb{Z}), \tag{23}$$

and $a_T(\nu, \tilde{\ell}) = (-1)^{2\nu} a_I(\nu, \tilde{\ell})$. The sums are finite by Lemma 5.1(v). Formula (22) is the continued-fraction formula of [10, (2.43)] for $1/\Phi_{10}$ with the coefficients of $1/\eta^{24}$ replaced by $d_1$ and with no congruence condition on $G$, and (23) is its parity-restricted form [10, (4.21)].[8]

## 6. The level-two Rademacher expansion

This section states the exact formula for the coefficients of the finite part on which the proof of Theorem A rests: a vector-valued Rademacher series over the three cosets of $\Gamma^0(2)$, whose principal parts at the two cusps are the wall sums of Section 5 and whose correction terms come from the shadow. The next four sections prove it.

[8] In (23) the condition that $r$ or $s$ be even selects the walls that are images of the basic wall under $\Gamma_0(2)$ or its conjugate; the factor $d_1(2n_G)$ vanishes unless $2n_G$ is an integer.

6.1. **Conventions, frames and Weil sums.** For $w \in \mathbb{C} \setminus (-\infty, 0]$ and $s \in \mathbb{R}$ we put $w^s = \exp(s \operatorname{Log} w)$ with the principal logarithm, $-\pi < \operatorname{Arg} w < \pi$. All powers below are of numbers in $\mathbb{H}$, in the lower half-plane, or in the right half-plane, so they are always defined, and $w^s w^{s'} = w^{s+s'}$.

Let $\Gamma^0(2)$ be the group of matrices of $SL_2(\mathbb{Z})$ with even upper right entry. It has index three, and

$$R_I = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}, \qquad R_T = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}, \qquad R_S = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$$

represent its right cosets. For $M \in SL_2(\mathbb{Z})$ the *frame* of $M$ is the $F \in \{I, T, S\}$ with $MR_F^{-1} \in \Gamma^0(2)$. The frames $I$ and $T$, which differ by a half period, belong to the cusp $\infty$, and the frame $S$ to the cusp 0.[9]

Fix $m \geq 1$. For $\gamma \geq 1$ let $\mathcal{C}_\gamma$ be the set of pairs $(\gamma, \delta)$ with $0 \leq -\delta \leq 2\gamma - 1$ and $\gcd(\gamma, \delta) = 1$. For such a pair let $\alpha \in \{0, \dots, \gamma - 1\}$ be the inverse of $\delta$ modulo $\gamma$ ($\alpha = 0$ if $\gamma = 1$), $\beta = (\alpha\delta - 1)/\gamma$, and

$$M_{\gamma,\delta} = \begin{pmatrix} \alpha & \beta \\ \gamma & \delta \end{pmatrix}, \qquad N_{\gamma,\delta} = M_{\gamma,\delta}^{-1} = \begin{pmatrix} \delta & -\beta \\ -\gamma & \alpha \end{pmatrix}, \qquad N^*_{\gamma,\delta} = \begin{pmatrix} \delta & \beta \\ \gamma & \alpha \end{pmatrix},$$

all in $SL_2(\mathbb{Z})$, with $j_{\gamma,\delta}(\tau) = -\gamma\tau + \alpha$ the automorphy factor of $N_{\gamma,\delta}$, and let $F(\gamma, \delta)$ be the frame of $N_{\gamma,\delta}$. For $\tilde{\ell}, \ell \in \mathbb{Z}$ put

$$W_{\gamma,\delta}(\tilde{\ell}, \ell) = \frac{1}{\sqrt{2m\gamma i}} \sum_{T=0}^{\gamma-1} \mathbf{e}\Big(\frac{\alpha x^2 - 2\ell x + \delta\ell^2}{4m\gamma}\Big), \qquad x = \tilde{\ell} - 2mT. \tag{24}$$

This is the matrix coefficient of the Weil representation attached to the lattice $\sqrt{2m}\,\mathbb{Z}$, evaluated at $N_{\gamma,\delta}$ [10, (2.41)]; it depends on $\tilde{\ell}$ only modulo $2m$ (Lemma 10.3).

For $\ell \in \mathbb{Z}$ let $\vartheta_\ell(\tau) = \sum_{r \equiv \ell\,(2m)} \mathbf{e}(r^2\tau/4m)$ be the unary theta series of weight $\frac{1}{2}$ and index $m$, so that $\vartheta_\ell(\tau) = \vartheta_{m,\ell}(\tau, 0)$ for the theta functions $\vartheta_{m,\ell}(\tau, z) = \sum_{r \equiv \ell\,(2m)} \mathbf{e}(r^2\tau/4m + rz)$ of [17, §5]. Its coefficients are real, so

$$\vartheta_\ell(-\bar{w}) = \overline{\vartheta_\ell(w)} \qquad (w \in \mathbb{H}). \tag{25}$$

With $N = N_{\gamma,\delta}$, $N^* = N^*_{\gamma,\delta}$ and $j = j_{\gamma,\delta}$, for $w, \tau \in \mathbb{H}$,

$$N(-\bar{w}) = -\overline{N^* w}, \qquad N^* w + N\tau = \frac{w + \tau}{(\gamma w + \alpha)\, j(\tau)}. \tag{26}$$

The first identity is immediate; for the second, the numerator $(\delta w + \beta)(-\gamma\tau + \alpha) + (\delta\tau - \beta)(\gamma w + \alpha)$ collapses to $(\alpha\delta - \beta\gamma)(w + \tau) = w + \tau$.

**Lemma 6.1.** *$|W_{1,\delta}| = (2m)^{-1/2}$, and $|W_{\gamma,\delta}(\tilde{\ell}, \ell)| \leq \sqrt{\gcd(m, \gamma)/m}$ for every $\gamma \geq 1$.*

*Proof.* For $\gamma = 1$ the sum has one term. In general, $\alpha x^2/(4m\gamma) = \alpha\tilde{\ell}^2/(4m\gamma) - \alpha\tilde{\ell}T/\gamma + \alpha m T^2/\gamma$, so up to a factor of modulus one the sum is $\sum_{T \bmod \gamma} \mathbf{e}\big((\alpha m T^2 + bT)/\gamma\big)$ for an integer $b$. A quadratic Gauss sum modulo $\gamma$ with leading coefficient $\alpha m$, $\gcd(\alpha, \gamma) = 1$, has modulus at most $\sqrt{2\gcd(m, \gamma)\gamma}$ [2, Chapter 1]. Dividing by $\sqrt{2m\gamma}$ gives the claim. □

[9] The duality group $\Gamma_1(2)$ acts on the charges; the finite part $\varphi^F_m(\sigma, v)$ is a function of $\sigma$ on which the relevant group, after the substitution $x = \mathbf{e}(\sigma/2)$, is $\Gamma^0(2)$. The cusp $\infty$ has width two, which is why it carries two frames.

6.2. **Polar data and shadows.** For $m \geq 1$ the polar terms in each frame are the pairs $(\tilde\nu, \tilde\ell)$ with $0 \leq \tilde\ell < 2m$ and $\tilde D = 4m\tilde\nu - \tilde\ell^2 < 0$, where $\tilde\nu \in \mathbb{Z}$ in the frame $S$ and $\tilde\nu \in \frac12\mathbb{Z}$ in the frames $I$ and $T$. Their coefficients are the wall sums of Section 5:

$$a_S(\tilde\nu,\tilde\ell) \text{ from (22)}, \qquad a_I(\tilde\nu,\tilde\ell) \text{ from (23)}, \qquad a_T(\tilde\nu,\tilde\ell) = (-1)^{2\tilde\nu} a_I(\tilde\nu,\tilde\ell). \tag{27}$$

By Lemma 5.1(iv) a wall with $m_G, n_G \geq -1$ has $t \leq t_0 = m^2 + 4m$, so only finitely many polar terms have a nonzero coefficient. That these numbers are the principal parts of $\varphi_m^F$ is the content of two theorems proved below:

(P$\infty$) the principal part of $\varphi_m^F$ at the cusp $\infty$ is given by (23) (Theorem 7.1);
(P0) the principal part of the finite part at the cusp 0, in the frame $S$, is given by (22) (Theorem 8.1).

The shadow of the finite part is governed by the coefficient $\hat f(m) = d_1(m)$ of the polar part (8). In the three frames it enters with the constants and orders

$$(c_S,\mu_S) = \big(\tfrac{1}{16}\hat f(m),\, 1\big), \qquad (c_I,\mu_I) = \big(\hat f(m),\, \tfrac12\big), \qquad (c_T,\mu_T) = \big(-\hat f(m),\, \tfrac12\big). \tag{28}$$

The number $\mu_F$ is the order of the pole of $1/g(\sigma/2)$ in the local parameter of the frame: $1/g$ has the term $x^{-1} = \mathbf{e}(-\sigma/2)$ at $\infty$, and its transform under $\sigma \mapsto -1/\sigma$ starts with $\mathbf{e}(-\sigma)/16$, by $\eta(-1/\tau) = \sqrt{-i\tau}\,\eta(\tau)$.[10] These are the leading coefficients of the functions $G_F$ of Proposition 9.3 (Remark 9.4). For $\gamma \geq 1$ and $g \in \mathbb{Z}$ put

$$f_{\gamma,g}(u) = \begin{cases} \dfrac{\pi^2}{\sinh^2\big(\frac{\pi u}{\gamma} - \frac{i\pi g}{2m\gamma}\big)}, & 2m\gamma \nmid g, \\[2ex] \dfrac{\pi^2}{\sinh^2(\pi u/\gamma)} - \dfrac{\gamma^2}{u^2}, & 2m\gamma \mid g, \end{cases} \qquad u \in \mathbb{R},$$

the second extended by $f_{\gamma,g}(0) = -\pi^2/3$. Since $\sinh(a-ib) = 0$ for real $a, b$ only when $a = 0$ and $b \in \pi\mathbb{Z}$, $f_{\gamma,g}$ is continuous on $\mathbb{R}$; it depends on $g$ only modulo $2m\gamma$.

6.3. **The expansion.** For a charge $(m,n,\ell)$ with $m \geq 1$, $0 \leq \ell \leq m$ and $D = 2mn - \ell^2 > 0$ define

$$\mathcal{M}(m,n,\ell) = \sum_{\gamma\geq1} \sum_{(\gamma,\delta)\in\mathcal{C}_\gamma} \sum_{(\tilde\nu,\tilde\ell)} a_F(\tilde\nu,\tilde\ell)\, \mathbf{e}\Big(\frac{\alpha\tilde D + \delta D}{4m\gamma}\Big) W_{\gamma,\delta}(\tilde\ell,\ell)\, \frac{2\pi}{\gamma}\Big(\frac{|\tilde D|}{D}\Big)^{\frac{15}{4}} I_{\frac{15}{2}}\Big(\frac{\pi\sqrt{|\tilde D|D}}{m\gamma}\Big), \tag{29}$$

$$\mathcal{K}_2(m,n,\ell) = -\sqrt{2m}\sum_{\gamma\geq1}\gamma^{-\frac12} \sum_{(\gamma,\delta)\in\mathcal{C}_\gamma} c_F\, \mathbf{e}\Big(\frac{-4m\mu_F\alpha + \delta D}{4m\gamma}\Big) W_{\gamma,\delta}(0,\ell)\Big(\frac{4m\mu_F}{D}\Big)^4 I_8\Big(\frac{2\pi}{\gamma}\sqrt{\frac{\mu_F D}{m}}\Big), \tag{30}$$

$$\begin{aligned}\mathcal{K}_3(m,n,\ell) = \frac{1}{2\pi}\sum_{\gamma\geq1}\gamma^{-2} \sum_{(\gamma,\delta)\in\mathcal{C}_\gamma} \sum_{g=-m\gamma}^{m\gamma-1} c_F\, \mathbf{e}\Big(\frac{-\alpha(4m\mu_F+g^2)+\delta D}{4m\gamma}\Big) W_{\gamma,\delta}(g,\ell)\Big(\frac{4m\mu_F}{D}\Big)^{\frac{17}{4}} \\ \times \int_{-\sqrt{\mu_F/m}}^{\sqrt{\mu_F/m}} f_{\gamma,g}(u)\Big(1-\frac{mu^2}{\mu_F}\Big)^{\frac{17}{4}} I_{\frac{17}{2}}\Big(\frac{2\pi}{\gamma}\sqrt{\frac{(\mu_F - mu^2)D}{m}}\Big)\, du,\end{aligned} \tag{31}$$

where in each line $F = F(\gamma,\delta)$ and $\alpha$ is attached to $(\gamma,\delta)$ as above.

[10] With $\tau = \sigma/2$, the substitution $\sigma \mapsto -1/\sigma$ is $\tau \mapsto -1/4\tau$, and $g(-1/4\tau) = \eta(-1/4\tau)^8\eta(-1/2\tau)^8 = 2^{12}\tau^8\eta(\sigma)^8\eta(2\sigma)^8$, whose expansion starts with $\mathbf{e}(\sigma)$.

**Theorem 6.2** (Rademacher expansion)**.** *For every charge $(m,n,\ell)$ with $m\geq 1$, $0\leq\ell\leq m$ and $D=2mn-\ell^2>0$, the three series* (29)–(31) *converge absolutely, and*

$$(-1)^{\ell+1}c_m^F(n,\ell)=\tfrac{1}{2}(-1)^{\ell}\sqrt{i}\,\big(\mathcal{M}+\mathcal{K}_2+\mathcal{K}_3\big)(m,n,\ell). \tag{32}$$

This is Theorem B of the introduction. The proof occupies Sections 7 to 10. Section 7 proves (P$\infty$) and Section 8 proves (P0), by following the Fourier coefficients of $1/\widetilde{\Phi}_6$ and of a second product $1/\Phi_6^S$ across the Farey walls. Section 9 proves the theta decomposition of the finite parts in the three frames, the completion of the Appell–Lerch sum by a Mordell integral and the resulting transformation law. Section 10 runs the circle method on Farey arcs and ends with the proof of Theorem 6.2.[11]

**Lemma 6.3.** *The three series* (29)–(31) *converge absolutely.*

*Proof.* There are at most $2\gamma$ pairs in $\mathcal{C}_\gamma$, the polar terms are finitely many, and $|W|\leq 1$ by Lemma 6.1. Since $I_\nu(z)/z^\nu$ is increasing, $I_\nu(z/\gamma)\leq\gamma^{-\nu}I_\nu(z)$ for $\gamma\geq 1$. So the $\gamma$-th term of (29) is $O(\gamma\cdot\gamma^{-1}\cdot\gamma^{-15/2})$, that of (30) is $O(\gamma^{1-1/2-8})$. In (31), for $2m\gamma\nmid g$ one has $|\sinh(a-ib)|^2=\sinh^2 a+\sin^2 b\geq\sin^2 b$, so $|f_{\gamma,g}|\leq\pi^2\csc^2(\pi g/2m\gamma)$, and for $g=0$ the function $\pi^2/\sinh^2 y-\pi^2/y^2$ takes values in $[-\pi^2/3,0)$. With the identity $\sum_{g=1}^{N-1}\csc^2(\pi g/N)=(N^2-1)/3$ for $N=2m\gamma$,

$$\sum_{g=-m\gamma}^{m\gamma-1}\sup_u|f_{\gamma,g}(u)|\leq\frac{\pi^2(4m^2\gamma^2-1)}{3}+\frac{\pi^2}{3}=\frac{4\pi^2m^2\gamma^2}{3}, \tag{33}$$

so the $\gamma$-th term of (31) is $O(\gamma\cdot\gamma^{-2}\cdot\gamma^2\cdot\gamma^{-17/2})$. Only absolute bounds are used below, except for the terms with $\gamma=1$ in the frame $S$, which are computed explicitly in Section 11. □

**Remark 6.4.** The proof of Theorem 6.2 uses no computation; its ingredients were also checked independently. The wall sum (23) agrees with the coefficients of $\varphi_m^F$ computed exactly from the product formula on all 1498 polar terms with $m\leq 12$, and (22) agrees with the coefficients of the finite part in the frame $S$, computed exactly from the second product of Section 8, on all 954 polar terms with $m\leq 12$. The right side of (32), truncated at $\gamma\leq 8$ in $\mathcal{M}$ and at $\gamma\leq 4$ in $\mathcal{K}_2$ and $\mathcal{K}_3$, agrees with the exact coefficients for $1\leq Q^2\leq 4$, $n\in\{4,8,16,30\}$ and $0\leq\ell\leq m$: the relative difference is below $6\cdot10^{-9}$ at $n=4$, $5\cdot10^{-12}$ at $n=8$ and $5\cdot10^{-21}$ at $n=30$, and it decreases with $n$ as the omitted terms with larger $\gamma$ do. The programs are listed in Appendix B.

**Remark 6.5.** What the proof of Theorem A uses from (P0) is the structure of Lemma 5.1: the leading family of walls and the bound (vi) for the others. From (P$\infty$) it uses only that every wall in it has $r$ or $s$ even, hence $rs\geq 2$, so that Lemma 5.1(vi) applies.

## 7. The polar coefficients at the cusp $\infty$

**Theorem 7.1.** *Let $m\geq 1$, $\nu\in\frac{1}{2}\mathbb{Z}$ and $0\leq\tilde{\ell}<2m$ with $4m\nu-\tilde{\ell}^2<0$. Then*

$$c_m^F(2\nu,\tilde{\ell})=a_I(\nu,\tilde{\ell}),$$

[11] For weakly holomorphic forms at level $N$ the expansion is the circle method of Rademacher and Zuckerman [25, 34]. For mixed mock Jacobi forms on $SL_2(\mathbb{Z})$, the case of $1/\Phi_{10}$, it was proved by Bringmann and Manschot [6], with the correction terms coming from the completion of Zwegers [35], and it was used for black holes by Ferrari and Reys [18]. At level two the expansion is written down, and assumed, in the derivation of the immortal index by Cardoso, Nampuri and Rosselló [10, §4].

*with $a_I$ the wall sum* (23)*. Consequently (P$\infty$) of Section 6 holds: the principal part of $\varphi_m^F$ at the cusp $\infty$ in the frame $I$ is given by* (23)*, and in the frame $T$, which is obtained by $\sigma \mapsto \sigma+1$, that is $x \mapsto -x$, by $a_T = (-1)^{2\nu} a_I$.*

In outline: the product converges on a tube domain and defines a meromorphic function whose poles lie over walls in the space of imaginary parts; the walls are the Farey geodesics with an even denominator; across each wall the Fourier coefficients of $1/\widetilde{\Phi}_6$ jump by a product of two coefficients $d_1$ (a residue computation, transported by $\Gamma_0(2)$); the polar coefficients of $\varphi_m^F$ are the coefficients in one chamber $K_0$; and following a vertical path from $K_0$ to a chamber in which the coefficient vanishes gives the wall sum. Every statement below is proved here; from Section 2 we take the definitions (2), (3), (8), the identities (5), (6) (rearrangements of the product formula) and Lemma 2.1, and from Section 4 Lemma 4.1.

7.1. **Notation.** Write $\Omega = \left(\begin{smallmatrix} \rho & v \\ v & \sigma \end{smallmatrix}\right)$, $\Omega = X + iY$ with

$$Y = \begin{pmatrix} M_1 & M_3 \\ M_3 & M_2 \end{pmatrix} = \operatorname{Im}\Omega,$$

and let $\mathcal{P}$ be the open cone of positive definite real symmetric $2\times 2$ matrices. A *charge* is a triple $T = (m, \nu, \ell) \in \Lambda := \mathbb{Z} \times \frac{1}{2}\mathbb{Z} \times \mathbb{Z}$; it stands for the monomial $\mathbf{e}(\langle T, \Omega\rangle)$ with

$$\langle T, \Omega\rangle = m\rho + \nu\sigma + \ell v = \operatorname{tr}(S_T\Omega), \qquad S_T = \begin{pmatrix} m & \ell/2 \\ \ell/2 & \nu \end{pmatrix}, \qquad N(T) := 4\det S_T = 4m\nu - \ell^2,$$

so that $\mathbf{e}(\langle T, \Omega\rangle) = \mathbf{e}(m\rho)x^{2\nu}y^{\ell}$. In the notation of Lemma 4.1, the factor $1 - x^A\mathbf{e}(B\rho)y^j$ of (3) is $1 - \mathbf{e}(\langle g, \Omega\rangle)$ with the *factor charge* $g = (B, A/2, j)$, its exponent is $E(g) := E(A, B, j)$, and $N(g) = 2AB - j^2$. Let

$$\mathcal{I} = \{(B, A/2, j) :\ A, B \in \mathbb{Z}_{\geq 0},\ j \in \mathbb{Z},\ j < 0 \text{ if } A = B = 0\}, \qquad \varrho = (1, \tfrac{1}{2}, 1),$$

so that (3) reads $\widetilde{\Phi}_6 = \mathbf{e}(\langle \varrho, \Omega\rangle) \prod_{g\in\mathcal{I}} \big(1 - \mathbf{e}(\langle g, \Omega\rangle)\big)^{E(g)}$.

The coefficient $c_b^{(r,s)}(N)$ of (3) is the coefficient of $q^n\zeta^j$ in $F^{(r,s)}$ for any $(n, j)$ with $4n - j^2 = N$, $j \equiv b \pmod 2$: this is the usual consequence of the elliptic invariance of index one under $z \mapsto z + \lambda\tau + \mu$, $\lambda, \mu \in \mathbb{Z}$. Since the expansions of the $F^{(r,s)}$ contain no negative powers of $q$, taking $j = b \in \{0, 1\}$ gives

$$c_b^{(r,s)}(N) = 0 \quad \text{for } N < -1, \qquad \text{hence} \qquad E(g) = 0 \quad \text{if } N(g) < -1. \tag{34}$$

For $Y \in \mathcal{P}$ put

$$\tau_Y = \frac{-M_3 + i\sqrt{\det Y}}{M_2} \in \mathbb{H}, \qquad \text{so that} \qquad Y = \frac{\sqrt{\det Y}}{\operatorname{Im}\tau_Y} \begin{pmatrix} |\tau_Y|^2 & -\operatorname{Re}\tau_Y \\ -\operatorname{Re}\tau_Y & 1 \end{pmatrix}.$$

The map $Y \mapsto (\tau_Y, \det Y)$ is a bijection $\mathcal{P} \to \mathbb{H} \times \mathbb{R}_{>0}$, and for $\tau = \tau_Y = \mathsf{x} + i\mathsf{y}$,

$$\langle T, Y\rangle = \frac{\sqrt{\det Y}}{\mathsf{y}}\big(m|\tau|^2 - \ell\,\mathsf{x} + \nu\big). \tag{35}$$

For $A = \left(\begin{smallmatrix} a & b \\ c & d \end{smallmatrix}\right) \in SL_2(\mathbb{Z})$ the substitution $\Omega \mapsto A\Omega A^T$ preserves $\det Y$, and

$$\tau_{AYA^T} = M_A\,\tau_Y, \qquad M_A = \begin{pmatrix} a & -b \\ -c & d \end{pmatrix}. \tag{36}$$

Indeed $Y = \frac{\sqrt{\det Y}}{\mathsf{y}} \operatorname{Re}(zz^*)$ with $z = (-\tau, 1)^T$, and $Az = (-c\tau + d)\,(-\tau', 1)^T$ with $\tau' = M_A\tau$, $\operatorname{Im}\tau' = \mathsf{y}/|-c\tau + d|^2$. Since $A \mapsto M_A$ is conjugation by $\operatorname{diag}(1, -1)$, it maps $\Gamma_0(2)$ onto itself.

On charges, $\langle T, A\Omega A^T\rangle = \langle A^*T, \Omega\rangle$ with $S_{A^*T} = A^T S_T A$, that is

$$A^*(m,\nu,\ell) = \big(ma^2 + \ell ac + \nu c^2,\ mb^2 + \ell bd + \nu d^2,\ 2mab + \ell(ad+bc) + 2\nu cd\big), \tag{37}$$

which maps $\Lambda$ to itself when $c$ is even.

7.2. **Convergence.**

**Lemma 7.2.** *There are constants $K, \kappa > 0$ with $|c_b^{(r,s)}(N)| \le K e^{\kappa\sqrt{N+1}}$ for all $r, s, b$ and $N \ge -1$.*

*Proof.* For a series $f = \sum a(n,j) q^n \zeta^j$ with $n \in \frac14\mathbb{Z}$ write $f^+ = \sum_n \big(\sum_j |a(n,j)|\big) q^n$; then $(fg)^+ \le f^+ g^+$ coefficientwise. By (2) each $F^{(r,s)}$ is a linear combination of $\theta_k(\tau,z)^2/\theta_k(\tau)^2$ ($k = 2,3,4$), of $\theta_1(\tau,z)^2/\eta(\tau)^6$ and of its products with $\mathfrak{E}(\tau)$, $\mathfrak{E}(\tau/2)$, $\mathfrak{E}((\tau+1)/2)$. The coefficients of $\theta_k(\tau,z)^2 q^{-\epsilon_k}$ ($\epsilon_1 = \epsilon_2 = \frac14$, $\epsilon_3 = \epsilon_4 = 0$) in $q^{1/2}$ are bounded by the number of representations of an integer as a sum of two squares, hence by a polynomial; those of $\mathfrak{E}$ are $24\sigma_1^{\mathrm{odd}}(n) \le 24n^2$. The remaining factors $q^{\epsilon_k}/\theta_k(\tau)^2$ and $q^{1/4}/\eta(\tau)^6$ are eta quotients, $\theta_2 = 2\eta(2\tau)^2/\eta(\tau)$, $\theta_3 = \eta(\tau)^5/(\eta(\tau/2)^2\eta(2\tau)^2)$, $\theta_4 = \eta(\tau/2)^2/\eta(\tau)$, so their coefficients in $X = q^{1/2}$ are bounded in absolute value by those of $\prod_{i\ge1}(1-X^i)^{-E_0}$ for a fixed $E_0$. Let $p_E(k)$ be the coefficient of $X^k$ in $\prod_i(1-X^i)^{-E}$. For $0 < X < 1$, $-\log(1-X^i) = \sum_{n\ge1} X^{in}/n$ and $X^n/(1-X^n) \le X/(n(1-X))$ give $\log\prod_i(1-X^i)^{-1} \le \pi^2/(6(1-X))$, hence $p_E(k) \le X^{-k}\exp(E\pi^2/(6(1-X)))$. With $X = 1-(k+1)^{-1/2}$ and $-\log(1-u) \le 2u$ for $u \le \frac12$ this gives $p_E(k) \le \exp\big((2+E\pi^2/6)\sqrt{k+1}\big)$ for $k \ge 3$. Multiplying the finitely many majorants, the coefficient of $q^n$ in $F^{(r,s)+}$ is at most $K_1(n+1)^{B_1}e^{\kappa_1\sqrt{2n+1}}$. Finally $c_b^{(r,s)}(N)$ is the coefficient of $q^n\zeta^b$ with $n = (N+b^2)/4 \le (N+1)/4$. □

Fix $K, \kappa$ as in Lemma 7.2, put $R_0 = (\kappa/\pi)^2$ and

$$\mathcal{D} = \{Y \in \mathcal{P} :\ \det Y > R_0\}, \qquad \mathcal{T} = \{\Omega :\ \operatorname{Im}\Omega \in \mathcal{D}\}.$$

The set $\mathcal{D}$ is connected (it is $\mathbb{H}\times(R_0,\infty)$ under $Y \mapsto (\tau_Y, \det Y)$), and so is $\mathcal{T}$.

**Lemma 7.3.** *Let $g = (B,k',j) \in \mathcal{I}$ with $N(g) = -1$. Then $E(g) = 2$ if $k' \in \mathbb{Z}$ and $B$ is even, and $E(g) = 0$ otherwise. If $N(g) \ge 0$, then $S_g$ is positive semidefinite and nonzero.*

*Proof.* Let $N = -1$, so $j$ is odd. The coefficient $c_1^{(r,s)}(-1)$ is the coefficient of $q^0\zeta$ in $F^{(r,s)}$. By (2), with $\phi_{0,1} = \zeta + 10 + \zeta^{-1} + O(q)$, $\phi_{-2,1} = \zeta - 2 + \zeta^{-1} + O(q)$ and $\mathfrak{E} = 1 + O(q)$, it equals $1,1,0,0$ for $(r,s) = (0,0),(0,1),(1,0),(1,1)$. Hence $E(g) = 1 + (-1)^B$ if $r = 0$, that is $k' \in \mathbb{Z}$, and $E(g) = 0$ if $r = 1$. If $N(g) \ge 0$ then $B, k' \ge 0$ and $4Bk' \ge j^2$, so $S_g$ is positive semidefinite; it is zero only for $g = (0,0,0) \notin \mathcal{I}$. □

A factor charge $g \in \mathcal{I}$ with $N(g) = -1$ and $E(g) \ne 0$ is called a *wall charge*, and $\{Y : \langle g, Y\rangle = 0\}$ its *wall*. By (35) the wall of $g = (B,k',j)$ is the cone over the geodesic

$$B|\tau|^2 - j\,\mathsf{x} + k' = 0 \tag{38}$$

in $\mathbb{H}$, whose endpoints are the roots of $Bt^2 - jt + k'$, a polynomial of discriminant $j^2 - 4Bk' = 1$. A *Farey edge* is a pair of fractions $a/c, b/d \in \mathbb{Q}\cup\{\infty\}$ in lowest terms, $c, d \ge 0$, $\infty = 1/0$, with $|ad - bc| = 1$, viewed as the geodesic of $\mathbb{H}$ joining them; it is *even* if $cd$ is even and *odd* if $c, d$ are both odd. Two adjacent fractions cannot both have even denominators, so every Farey edge is either even or odd.

**Lemma 7.4.** *The map $g \mapsto$(38) is a bijection from the set of wall charges onto the set of even Farey edges. If the edge of $g$ joins $a/c$ and $b/d$ with $c, d \geq 1$, then $g = (cd, ab, ad + bc)$ and*

$$\langle g, Y\rangle = \frac{\sqrt{\det Y}}{\mathsf{y}}\, Q, \qquad Q = \operatorname{Re}\big[(c\tau - a)(d\bar\tau - b)\big], \qquad Q^2 + \mathsf{y}^2 = |c\tau - a|^2\, |d\tau - b|^2, \tag{39}$$

*with $\tau = \tau_Y$; $\langle g, Y\rangle > 0$ outside the half-disc bounded by the edge. The vertical edges $\{\infty, n\}$, $n \in \mathbb{Z}$, are the walls of $(0, n, 1)$ for $n \geq 1$ and $(0, -n, -1)$ for $n \leq 0$, with $\langle g, Y\rangle > 0$ on the side $\mathsf{x} < n$, respectively $\mathsf{x} > n$.*

*Proof.* Let $g = (B, k', j)$ be a wall charge, so $k' \in \mathbb{Z}$ and $B$ is even, and $\gcd(B, j, k') = 1$ because its square divides the discriminant 1. If $B > 0$, the roots $(j \pm 1)/2B$ are rational, so by Gauss's lemma $Bt^2 - jt + k' = (ct - a)(dt - b)$ with integers, and comparing coefficients $cd = B$, $ad + bc = j$, $ab = k'$ and $(ad - bc)^2 = j^2 - 4Bk' = 1$. Since $cd = B > 0$ we may take $c, d \geq 1$; then $cd = B$ is even. If $B = 0$ then $j = \pm 1$ and the root is $k'/j = jk'$; the conditions $k' \geq 0$, and $j < 0$ if $k' = 0$, give each integer exactly once, as stated. Conversely, for an even edge $\{a/c, b/d\}$ with $c, d \geq 1$ the charge $\pm(cd, ab, ad + bc)$ with the sign making $cd > 0$ has $N = 4abcd - (ad + bc)^2 = -(ad - bc)^2 = -1$, $k' = ab \in \mathbb{Z}$ and $B = cd$ even, so $E = 2$ by Lemma 7.3; it lies in $\mathcal{I}$ because $ab \geq 0$: the two endpoints lie on the same side of 0, since by Lemma 7.5(F1) below a Farey edge does not cross the edge $\{0, \infty\}$. For (39), $(c\tau - a)(d\bar\tau - b) = cd|\tau|^2 - (ad\bar\tau + bc\tau) + ab$ has real part $B|\tau|^2 - j\mathsf{x} + k'$ and imaginary part $(ad - bc)\mathsf{y} = \pm\mathsf{y}$; combine with (35). The sign statements follow from (38). □

**Lemma 7.5.** (F3) *If $a/c < e/f < b/d$ with $bc - ad = 1$, $c, d, f \geq 1$, then $f \geq c + d$.*

(F1) *Two distinct Farey edges do not cross: their intersection in $\mathbb{H}$ is empty.*

(F0) *For every compact set $\mathcal{K} \subset \mathbb{H}$ only finitely many Farey edges meet $\mathcal{K}$; more precisely an edge with $c, d \geq 1$ meeting $\{\operatorname{Im}\tau \geq \mathsf{y}_0\}$ has $cd \leq 1/(2\mathsf{y}_0)$.*

*Proof.* (F3) $e/f - a/c \geq 1/(cf)$ and $b/d - e/f \geq 1/(df)$, and their sum is $1/(cd)$, so $1/(cd) \geq (c+d)/(cdf)$. (F1) Two geodesics cross only if their endpoints interleave. A vertical edge $\{\infty, n\}$ would have $n$ strictly inside an edge $(a/c, b/d)$ with $c, d \geq 1$, so $1 \geq c + d$ by (F3), which is impossible. For two edges $(a/c, b/d)$, $(a'/c', b'/d')$ with all denominators $\geq 1$ and interleaving endpoints, one endpoint of each lies strictly inside the other; by (F3) its denominator is at least the sum of the denominators of the other edge, and the two inequalities contradict each other. (F0) The edge is a semicircle of radius $1/(2cd)$, and its centre lies within 1 of $\operatorname{Re}\tau$; vertical edges meet $\mathcal{K}$ only for finitely many $n$. □

**Proposition 7.6.** (i) *For every compact $\mathcal{K} \subset \mathcal{D}$ and every $X_0$ the set of wall charges with $\langle g, Y\rangle \leq X_0$ for some $Y \in \mathcal{K}$ is finite, and $\sum_{g \in \mathcal{I}} |E(g)|\, e^{-2\pi\langle g, Y\rangle}$, taken over the $g$ with $\langle g, Y\rangle > 0$, converges uniformly on $\mathcal{K}$.*

(ii) *The product* (3) *converges absolutely and locally uniformly on $\mathcal{T}$, and $\widetilde{\Phi}_6$ is holomorphic on $\mathcal{T}$.*

(iii) *The zeros of $\widetilde{\Phi}_6$ in $\mathcal{T}$ are the points with $\langle g, \Omega\rangle \in \mathbb{Z}$ for a wall charge $g$; there $\operatorname{Im}\Omega$ lies on the wall of $g$, and the zero is a double zero of the factor $(1 - \mathbf{e}(\langle g, \Omega\rangle))^2$, all other factors being nonzero. Hence $\mathcal{F} := 1/\widetilde{\Phi}_6$ is meromorphic on $\mathcal{T}$ and holomorphic on the tube over the complement of the walls.*

*Proof.* (i) Factors with $N(g) < -1$ have $E(g) = 0$ by (34). For $N(g) \geq 0$, $S_g$ is positive semidefinite (Lemma 7.3), and for positive semidefinite $S$ and $Y \in \mathcal{P}$, $\operatorname{tr}(SY) \geq 2\sqrt{\det S \det Y}$ and $\operatorname{tr}(SY) \geq \lambda_{\min}(Y) \operatorname{tr} S$. Hence

$$2\pi\langle g, Y\rangle \geq \pi\sqrt{N(g) \det Y} + \pi\lambda_{\min}(Y)(B + k').$$

By Lemma 7.2 and $\pi\sqrt{\det Y} \geq \kappa$ on $\mathcal{D}$, $|E(g)|e^{-2\pi\langle g,Y\rangle} \leq 4Ke^{\kappa}e^{-\pi\lambda_{\min}(Y)(B+k')}$, and for given $(B,k')$ there are at most $4(B+k')+1$ values of $j$. Since $\lambda_{\min}$ is continuous and positive, these terms are summable uniformly on $\mathcal{K}$. For a non-vertical wall charge, $Q = cd\big(|\tau-\omega|^2-(2cd)^{-2}\big) \geq -1/(4cd)$, $\omega$ the centre of the edge, so by (39) and $|c\tau-a| \geq c\mathsf{y}$, $|d\tau-b| \geq d\mathsf{y}$, the inequality $\langle g,Y\rangle \leq X$ implies $cd\,\mathsf{y}^2 \leq Z$ and $|c\mathsf{x}-a| \leq Z/(d\mathsf{y})$ with $Z = \big(\max(X\mathsf{y}/\sqrt{\det Y},\tfrac14)^2+\mathsf{y}^2\big)^{1/2}$; on $\mathcal{K}$ the quantities $\mathsf{x},\mathsf{y},\det Y$ are bounded and $\mathsf{y},\det Y$ bounded below, so the number of wall charges with $\langle g,Y\rangle \leq X$ is $O((1+X)^3)$ uniformly on $\mathcal{K}$ (vertical walls contribute $O(1+X)$). This gives finiteness and uniform convergence of the sum over wall charges. (ii) By (i), on a neighbourhood of a compact set only finitely many factors have $\langle g,Y\rangle \leq \frac{1}{2\pi}\log 2$, and the remaining ones satisfy $|\mathbf{e}(\langle g,\Omega\rangle)| \leq \frac12$ with $\sum|E(g)||\mathbf{e}(\langle g,\Omega\rangle)|$ uniformly convergent; their product converges absolutely and uniformly to a holomorphic function without zeros, and the finitely many others are entire. (iii) The infinite part has no zeros. Among the finitely many remaining factors, those with $N(g) \geq 0$ satisfy $|\mathbf{e}(\langle g,\Omega\rangle)| < 1$, so only wall charges can vanish, exactly when $\langle g,\Omega\rangle \in \mathbb{Z}$, which forces $\langle g,Y\rangle = 0$. Distinct Farey edges are disjoint in $\mathbb{H}$ (Lemma 7.5(F1)), so at such a point no other factor vanishes. □

7.3. **Chambers and Fourier coefficients.** The *walls* in $\mathbb{H}$ are the even Farey edges; by (F0) their union is closed. A *chamber* is a connected component $C$ of the complement of the walls in $\mathbb{H}$, and we put $\tilde{C} = \{Y \in \mathcal{D} : \tau_Y \in C\}$, which is connected. Let $K_0$ be the open ideal quadrilateral with vertices $0, \frac12, 1, \infty$, and $K_0'$ the one with vertices $-1, -\frac12, 0, \infty$.

**Lemma 7.7.** *Let $C$ be a chamber.*

(i) *For each wall charge $g$ the sign of $\langle g,Y\rangle$ is constant on $\tilde{C}$, and the set $\mathcal{I}_C^-$ of wall charges with $\langle g,Y\rangle < 0$ on $\tilde{C}$ is finite.*

(ii) *On the tube over $\tilde{C}$,*

$$\mathcal{F}(\Omega) = \mathbf{e}\Big(-\Big\langle \varrho + 2\sum_{g\in\mathcal{I}_C^-} g,\ \Omega\Big\rangle\Big)\prod_{g\in\mathcal{I}_C}\big(1-\mathbf{e}(\langle g,\Omega\rangle)\big)^{-E(g)}, \qquad \mathcal{I}_C = (\mathcal{I}\setminus\mathcal{I}_C^-)\cup(-\mathcal{I}_C^-),$$

*with $E(-g) := E(g)$, and $\langle g,Y\rangle > 0$ for every $g \in \mathcal{I}_C$ with $E(g) \neq 0$ and $Y \in \tilde{C}$.*

(iii) *Expanding each factor binomially gives $\mathcal{F}(\Omega) = \sum_{T\in\Lambda} d(T;C)\,\mathbf{e}(\langle T,\Omega\rangle)$ with integers $d(T;C)$, absolutely and locally uniformly convergent on the tube over $\tilde{C}$, and for every $Y \in \tilde{C}$*

$$d(T;C) = \frac12\int_0^1\int_0^2\int_0^1 \mathcal{F}(X+iY)\,\mathbf{e}(-\langle T,X+iY\rangle)\,dX_v\,dX_\sigma\,dX_\rho. \tag{40}$$

*Proof.* (i) $\langle g,Y\rangle$ vanishes only on the wall of $g$, which does not meet $C$; the finiteness is Proposition 7.6(i) at one point. (ii) Use $(1-\mathbf{e}(w))^{-2} = \mathbf{e}(-2w)(1-\mathbf{e}(-w))^{-2}$ for $g \in \mathcal{I}_C^-$, and Lemma 7.3 for $N(g) \geq 0$. (iii) Each factor $(1-u)^{-E} = \sum_k \binom{-E}{k}(-u)^k$ is majorised by $(1-|u|)^{-|E|}$, and $\prod(1-|u_g|)^{-|E(g)|}$ converges locally uniformly by Proposition 7.6(i). For given $T$ only finitely many products of monomials contribute, because $\{g \in \mathcal{I}_C : 0 < \langle g,Y\rangle \leq X_0\}$ is finite for every $X_0$ (Proposition 7.6(i) for the walls, and $\lambda_{\min}(Y)(B+k') \leq \langle g,Y\rangle$ for $N(g) \geq 0$); so $d(T;C)$ is a finite sum of integers. $\mathcal{F}$ has period 1 in $\rho$ and $v$ and period 2 in $\sigma$, and (40) follows from absolute convergence. □

The edge $(0,1)$ is odd, so $K_0$ is the set of $\tau$ with $0 < \mathsf{x} < 1$ lying outside the half-discs on $(0,\frac12)$ and $(\frac12,1)$. Every wall charge is positive on $\tilde{K}_0$. For the vertical walls this is Lemma 7.4. By (F3) no Farey edge with $c,d \geq 1$ contains 0 or 1 strictly inside its interval, and only $(0,1)$ contains $\frac12$; so the half-disc of a non-vertical wall lies in $\{\mathsf{x} \leq 0\}$, in $\{\mathsf{x} \geq 1\}$, or in the closed

half-disc on $(0,\frac12)$ or on $(\frac12,1)$, and every point of $K_0$ is outside it. Hence $\mathcal{I}^-_{K_0}=\varnothing$, the expansion of Lemma 7.7 on $\tilde K_0$ is the expansion of the product (3) itself, and $K_0$ is a chamber: it is open, connected, contains no wall point, and its boundary consists of wall points.

7.4. **Symmetries.**

**Lemma 7.8.** *For $\Omega\in\mathcal{T}$:*

(i) $\widetilde\Phi_6(E\Omega E^T)=\widetilde\Phi_6(\Omega)$ *for* $E=\left(\begin{smallmatrix}1&1\\0&1\end{smallmatrix}\right)$, *that is* $\widetilde\Phi_6(\rho+2v+\sigma,\sigma,v+\sigma)=\widetilde\Phi_6(\rho,\sigma,v)$;

(ii) $\widetilde\Phi_6(\rho,\sigma,-v)=\widetilde\Phi_6(\rho,\sigma,v)$;

(iii) $\widetilde\Phi_6(\sigma/2,2\rho,v)=\widetilde\Phi_6(\rho,\sigma,v)$;

(iv) $\widetilde\Phi_6(A\Omega A^T)=\widetilde\Phi_6(\Omega)$ *for every* $A\in\Gamma_0(2)$, *and consequently* $d(T;C')=d(A^*T;C)$ *whenever* $C'=M_A(C)$.

*Proof.* All substitutions preserve $\mathcal{D}$ (they act on $Y$ by $Y\mapsto PYP^T$ with $\det P=\pm1$), so both sides are absolutely convergent products on $\mathcal{T}$ and may be compared factor by factor.

(i) $\langle g,E\Omega E^T\rangle=\langle E^*g,\Omega\rangle$ with $E^*(B,k',j)=(B,k'+B+j,j+2B)$ by (37). This map preserves $N$, $B$, $j \bmod 2$ and $2k' \bmod 2$, hence $E(E^*g)=E(g)$. Among the charges with $N\ge-1$ it maps $\mathcal{I}$ onto $(\mathcal{I}\setminus\{(0,1,1)\})\cup\{(0,-1,-1)\}$: for $B>0$, $4B(k'+B+j)=N+(j+2B)^2\ge0$ (if $j+2B=0$ then $N\equiv0 \pmod 4$, so $N\ge0$), and the same holds for $E^{-1}$; for $B=0$ only $j\in\{0,\pm1\}$ occur, $(0,k',0)$ is fixed, $(0,k',1)\mapsto(0,k'+1,1)$ and $(0,k',-1)\mapsto(0,k'-1,-1)$. Both exceptional charges have $E=2$, so the product over $\mathcal{I}$ acquires the factor $(1-\mathbf{e}(-\sigma-v))^2/(1-\mathbf{e}(\sigma+v))^2=\mathbf{e}(-2\sigma-2v)$, while $\langle\varrho,E\Omega E^T\rangle=\langle\varrho,\Omega\rangle+2\sigma+2v$.

(ii) $v\mapsto-v$ maps $(B,k',j)$ to $(B,k',-j)$, which preserves $E$ and maps $\mathcal{I}$ onto itself except that $(0,0,-1)$ becomes $(0,0,1)$; the factors with $E\ne0$ and $A=B=0$ have $j=-1$ by (34). Since $\mathbf{e}(v)(1-\mathbf{e}(-v))^2=\mathbf{e}(-v)(1-\mathbf{e}(v))^2$, the two products agree.

(iii) is Lemma 4.1: the factor sets correspond under $(A,B,j)\mapsto(B,A,j)$ with equal exponents, and the products converge absolutely on $\mathcal{T}$.

(iv) Let $V(\Omega)=W\Omega W^T$ with $W=\left(\begin{smallmatrix}0&-1/\sqrt2\\\sqrt2&0\end{smallmatrix}\right)$, so $V(\rho,\sigma,v)=(\sigma/2,2\rho,-v)$ and $\widetilde\Phi_6\circ V=\widetilde\Phi_6$ by (ii) and (iii). Since $WEW^{-1}=\left(\begin{smallmatrix}1&0\\-2&1\end{smallmatrix}\right)=:L^{-1}$, (i) and $V$-invariance give $\widetilde\Phi_6(L^{-1}\Omega L^{-T})=\widetilde\Phi_6(E(W^{-1}\Omega W^{-T})E^T)=\widetilde\Phi_6(W^{-1}\Omega W^{-T})=\widetilde\Phi_6(\Omega)$. The matrices $E$, $L$ and $-I$ generate $\Gamma_0(2)$: for $A=\left(\begin{smallmatrix}a&b\\c&d\end{smallmatrix}\right)\in\Gamma_0(2)$ with $c\ne0$, $a$ is odd; replacing $A$ by $E^kA$ makes $|a|\le|c|/2$, and then for suitable $k$ the matrix $L^kA\in\Gamma_0(2)$ has lower left entry $c+2ka$ with $|c+2ka|\le|a|<|c|$; after finitely many steps $c=0$ and $A=\pm E^k$. Finally, if $C'=M_A(C)$ and $Y\in\tilde C'$, then $A^{-1}YA^{-T}\in\tilde C$ by (36), and

$$\mathcal{F}(\Omega)=\mathcal{F}(A^{-1}\Omega A^{-T})=\sum_S d(S;C)\,\mathbf{e}(\langle S,A^{-1}\Omega A^{-T}\rangle),$$

so the coefficient of $\mathbf{e}(\langle T,\Omega\rangle)$ is $d(S;C)$ with $(A^{-1})^*S=T$, that is $S=A^*T$. □

7.5. **The chambers.** For an even or odd Farey edge $e=(a/c,b/d)$ with $c,d\ge1$ let $\Delta(e)$ be the open ideal triangle with vertices $a/c$, $b/d$ and the mediant $(a+b)/(c+d)$, that is, the set of points inside the half-disc of $e$ and outside the half-discs of its two *children* $(a/c,(a+b)/(c+d))$ and $((a+b)/(c+d),b/d)$; for $n\in\mathbb{Z}$ let $\Delta(n)$ be the open triangle with vertices $n,n+1,\infty$.

**Lemma 7.9.** (i) *The triangles $\Delta(e)$, $\Delta(n)$ are pairwise disjoint, and $\mathbb{H}$ is the union of their closures.*

(ii) *Each triangle has exactly one odd side.*

(iii) *The chambers are exactly the ideal quadrilaterals $Q(e_0)=\Delta\cup\Delta'\cup e_0$, where $e_0$ is an odd edge and $\Delta,\Delta'$ are the two triangles having $e_0$ as a side. In particular $K_0=Q((0,1))$ and $K_0'=Q((-1,0))$.*

(iv) $\Gamma_0(2)$ *acts transitively on the odd edges; hence every chamber is* $M(K_0)$ *for some* $M \in \Gamma_0(2)$*, and every chamber is* $M_A(K_0)$ *for some* $A \in \Gamma_0(2)$*.*

*Proof.* (i) Let $\tau$ lie on no Farey edge. By (F0) only finitely many edges have $\tau$ in their half-disc, and by (F1) their intervals are nested. If there is none, $n < \operatorname{Re}\tau < n+1$ for some $n$, and $\tau$ is not in the half-disc of $(n, n+1)$, so $\tau \in \Delta(n)$. Otherwise let $e$ be the innermost one; $\tau$ is not in the half-disc of a child of $e$, which would be a smaller such edge, so $\tau \in \Delta(e)$. Points on edges lie in the closure of adjacent triangles by the same argument applied to nearby points. Moreover no edge meets an open triangle: by (F1) and the argument below, an edge meeting the half-disc of $e$ is $e$, a child of $e$, or lies under a child. Hence the open triangles are the components of the complement of all Farey edges. For disjointness, suppose $\tau \in \Delta(e) \cap \Delta(e')$ with $e \neq e'$; then $\tau$ is in both half-discs, so one interval, say that of $e'$, lies strictly inside that of $e = (a/c, b/d)$ and has an endpoint strictly between $a/c$ and $b/d$. That endpoint has denominator at least $c+d$ by (F3); if the interval of $e'$ contained the mediant strictly inside, (F3) applied to $e'$ would bound the sum of its denominators by $c+d$, which is impossible. So $e'$ lies in the closed half-disc of a child of $e$, and $\tau$ lies inside that child's half-disc, a contradiction. The cases involving $\Delta(n)$ are similar and easier. (ii) The vertices of $\Delta(e)$ have denominators $c, d, c+d$, and those of $\Delta(n)$ have $1, 1, 0$; exactly one is even, since no two adjacent fractions both have even denominators. The side joining the two odd ones is odd, the other two are even. (iii) By (i), (F0) and (F1), every edge is a side of exactly two triangles, one on each side: near an interior point of the edge, each side is covered by the closure of a single triangle. By (ii) the sets $Q(e_0)$ are pairwise disjoint; each is open and connected and contains no wall point, and their union is the complement of the walls. So they are the components. The triangles $\Delta(0)$ and $\Delta((0,1))$ share the odd edge $(0,1)$, which gives $K_0$, and similarly for $K_0'$. (iv) A matrix of $SL_2(\mathbb{Z})$ maps Farey edges to Farey edges and, if it lies in $\Gamma_0(2)$, preserves the parity of denominators: $\left(\begin{smallmatrix}\alpha & \beta\\ \gamma & \delta\end{smallmatrix}\right)(a/c)$ has denominator $\gamma a + \delta c \equiv c \pmod 2$. So $\Gamma_0(2)$ permutes triangles, walls, odd edges and chambers, and maps $Q(e_0)$ to $Q(Me_0)$. Given an odd edge $(u_1/v_1, u_2/v_2)$ with $u_1v_2 - u_2v_1 = 1$, the matrix $M = \left(\begin{smallmatrix}u_1-u_2 & u_2\\ v_1-v_2 & v_2\end{smallmatrix}\right)$ lies in $\Gamma_0(2)$ and maps $1 \mapsto u_1/v_1$, $0 \mapsto u_2/v_2$. The last statement follows from (36), since $A \mapsto M_A$ is a bijection of $\Gamma_0(2)$. □

### 7.6. **The jump across a wall.**

**Lemma 7.10** (Jump lemma)**.** *For every charge* $T = (m, \nu, \ell)$*,*

$$d(T; K_0) - d(T; K_0') = -\ell\, d_1(m)\, d_1(2\nu).$$

*Proof.* The wall between $K_0$ and $K_0'$ is $W_0 = \{\operatorname{Re}\tau = 0\} = \{M_3 = 0\}$, the wall of $g_0 = (0,0,-1)$, with $\langle g_0, Y\rangle = -M_3 > 0$ on $\tilde{K}_0$. Fix $Y_0 = (M_1, M_2, 0) \in \mathcal{D}$ and a compact neighbourhood $U$ of $Y_0$ in $\mathcal{D}$, symmetric under $M_3 \mapsto -M_3$, which meets no wall other than $W_0$ (Lemma 7.5(F0),(F1)) and whose two halves $U \cap \{M_3 < 0\}$, $U \cap \{M_3 > 0\}$ lie in $\tilde{K}_0$, $\tilde{K}_0'$. On the tube over $U$ all factor charges $g \neq g_0$ with $E(g) \neq 0$ have $\langle g, Y\rangle > 0$, and since

$$\mathbf{e}(-v)\big(1 - \mathbf{e}(-v)\big)^{-2} = -\frac{1}{4\sin^2 \pi v},$$

we have $\mathcal{F} = -H/(4\sin^2\pi v)$ with

$$H(\Omega) = \mathbf{e}\big(-\rho - \tfrac{\sigma}{2}\big) \prod_{g \in \mathcal{I},\ g \neq g_0} \big(1 - \mathbf{e}(\langle g, \Omega\rangle)\big)^{-E(g)},$$

holomorphic on the tube over $U$. By Lemma 7.8(ii) $\mathcal{F}$ is even in $v$, hence so is $H$.

Take $Y_\pm = (M_1, M_2, \mp\epsilon) \in U$, so $Y_+ \in \tilde{K}_0$, $Y_- \in \tilde{K}_0'$, and apply (40). For fixed real parts of $\rho, \sigma$ the function $v \mapsto \mathcal{F}e(-\langle T, \Omega\rangle)$ is 1-periodic and meromorphic on $|\operatorname{Im} v| \le \epsilon$ with poles only at $v \in \mathbb{Z}$. The difference of the integrals over $\operatorname{Im} v = -\epsilon$ and $\operatorname{Im} v = \epsilon$ is the integral over the positively oriented boundary of a period rectangle, so

$$d(T;K_0) - d(T;K_0') = \frac{1}{2}\int_0^1\int_0^2 2\pi i\, R(\rho,\sigma)\, \mathbf{e}(-m\rho-\nu\sigma)\, dX_\sigma\, dX_\rho, \qquad R = \operatorname*{Res}_{v=0}\Big[-\frac{H\,\mathbf{e}(-\ell v)}{4\sin^2\pi v}\Big].$$

Since $1/(4\sin^2\pi v) = 1/(4\pi^2 v^2) + O(1)$ and $\partial_v H(\rho,\sigma,0) = 0$, the coefficient of $v$ in $H\mathbf{e}(-\ell v)$ is $-2\pi i\ell H(\rho,\sigma,0)$, so $R = \frac{2\pi i\ell}{4\pi^2} H(\rho,\sigma,0)$ and $2\pi i R = -\ell H(\rho,\sigma,0)$.

It remains to compute $H(\rho,\sigma,0)$. At $v=0$ the factors with given $(A,B) \neq (0,0)$ combine to $(1-x^A\mathbf{e}(B\rho))^{\sum_j E(A,B,j)}$ (finitely many $j$ by (34)), and

$$\sum_j E(A,B,j) = \sum_{s=0}^{1} (-1)^{sB}\, \big[q^{AB/2}\big] F^{(A \bmod 2,\, s)}(\tau,0).$$

By (2), $\phi_{0,1}(\tau,0) = 12$ and $\phi_{-2,1}(\tau,0) = 0$, so $F^{(0,0)}(\tau,0) = 12$ and $F^{(r,s)}(\tau,0) = 4$ otherwise, constants. Hence $\sum_j E = 0$ if $A, B \ge 1$, $\sum_j E(A,0,j) = 16$ or $8$ for $A$ even or odd, and $\sum_j E(0,B,j) = 12 + 4(-1)^B$. With $e_r$ as in (1) and $g(\tau) = \eta(\tau)^8\eta(2\tau)^8 = q\prod_n (1-q^n)^{e_n}$,

$$\begin{aligned} H(\rho,\sigma,0) &= \frac{\mathbf{e}(-\rho-\sigma/2)}{\prod_{A\ge1}(1-x^A)^{e_A}\prod_{B\ge1}(1-\mathbf{e}(B\rho))^{e_B}} \\ &= \frac{1}{g(\rho)\,g(\sigma/2)} = \sum_{k,k'\ge -1} d_1(k) d_1(k')\, \mathbf{e}(k\rho)\, x^{k'} \end{aligned}$$

by (21), absolutely convergent at $\operatorname{Im}(\rho,\sigma) = (M_1, M_2)$. The coefficient of $\mathbf{e}(m\rho)x^{2\nu}$ is $d_1(m)d_1(2\nu)$. □

For a wall that is not vertical write $G = \left(\begin{smallmatrix} p & q \\ r & s \end{smallmatrix}\right) \in SL_2(\mathbb{Z})$ with $r, s \ge 1$ for the matrix mapping $\infty \mapsto p/r$ and $0 \mapsto q/s$, so that $q/s < p/r$ are its endpoints; $G$ is unique, and exactly one of $r, s$ is even. For a charge $T = (m, \nu, \ell)$ define $m_G, n_G, \ell_G$ by (19) with $\tilde{\ell} = \ell$. The *inside* of the wall is the half-disc it bounds.

**Corollary 7.11** (Transport)**.** *Let $C_{\text{out}}$ and $C_{\text{in}}$ be the chambers adjacent to the wall of $G$ outside and inside. Then*

$$d(T;C_{\text{out}}) - d(T;C_{\text{in}}) = J_G(T) := \ell_G\Big([r \text{ even}]\, d_1(m_G)\, d_1(2n_G) + [s \text{ even}]\, d_1(2m_G)\, d_1(n_G)\Big).$$

*Proof.* If $r$ is even, $G \in \Gamma_0(2)$ maps $W_0$ onto the wall, and it maps the half-plane $\operatorname{Re}\tau > 0$, which contains $K_0$, onto the half-disc of the wall, because the boundary point 1 goes to the mediant $(p+q)/(r+s)$, which lies between $q/s$ and $p/r$; so $C_{\text{in}} = G(K_0)$, $C_{\text{out}} = G(K_0')$. Let $A = \left(\begin{smallmatrix} p & -q \\ -r & s \end{smallmatrix}\right)$, so $M_A = G$; by (37) $A^*T = (m_G, n_G, \ell_G)$, and Lemmas 7.8(iv) and 7.10 give $d(T;C_{\text{in}}) - d(T;C_{\text{out}}) = -\ell_G\, d_1(m_G)\, d_1(2n_G)$. If $s$ is even, $G' = \left(\begin{smallmatrix} q & -p \\ s & -r \end{smallmatrix}\right) \in \Gamma_0(2)$ maps $\infty \mapsto q/s$, $0 \mapsto p/r$ and 1 to $(q-p)/(s-r)$, which lies outside $[q/s, p/r]$ by (F3), its denominator $|s-r|$ being positive and smaller than $r+s$; so $C_{\text{out}} = G'(K_0)$, $C_{\text{in}} = G'(K_0')$. With $A' = \left(\begin{smallmatrix} q & p \\ -s & -r \end{smallmatrix}\right)$, $M_{A'} = G'$ and $A'^*T = (n_G, m_G, -\ell_G)$, so $d(T;C_{\text{out}}) - d(T;C_{\text{in}}) = \ell_G\, d_1(n_G)\, d_1(2m_G)$. □

Crossing an odd edge does not change the chamber, so no coefficient jumps there; correspondingly the bracket in (23) vanishes when $r$ and $s$ are odd.

7.7. **Vanishing.**

**Lemma 7.12.** (i) $d((m,\nu,\ell);K_0)=0$ *for* $m\le -2$.

(ii) *Let* $\kappa=a'/c'$ *in lowest terms with* $c'\ge 2$ *even, and let* $C$ *be the chamber containing* $\kappa+i\mathsf{y}$ *for all small* $\mathsf{y}>0$. *Then* $d((m,\nu,\ell);C)=d(T';K_0)$ *for a charge* $T'$ *whose first entry is* $V(\kappa):=\nu c'^2-\ell a'c'+ma'^2$; *hence* $d((m,\nu,\ell);C)=0$ *if* $V(\kappa)\le -2$.

*Proof.* (i) Fix $M_2>0$ and $-M_2<M_3<0$ and let $M_1\to\infty$. Then $\tau_Y$ has fixed real part in $(0,1)$ and imaginary part $\to\infty$, and $\det Y\to\infty$, so $Y\in\tilde K_0$ for $M_1$ large. In $\mathcal{F}=1/\widetilde{\Phi}_6$ the inverse of the prefactor has modulus $e^{2\pi(M_1+M_2/2+M_3)}$, the factors with $B=0$ do not depend on $M_1$ and their product is bounded on the real torus, and the factors with $B\ge 1$ have $\langle g,Y\rangle$ increasing in $M_1$, so by Proposition 7.6(i) and dominated convergence their product tends to 1 uniformly in the real parts. Hence $|\mathcal{F}|\le C_0e^{2\pi M_1}$ for large $M_1$, and (40) gives $|d(T;K_0)|\le C_1e^{2\pi(m+1)M_1}\to 0$ for $m\le -2$. (ii) Choose $b',d'$ with $a'd'-b'c'=1$; then $M=\left(\begin{smallmatrix}a' & b'\\ c' & d'\end{smallmatrix}\right)\in\Gamma_0(2)$ and $M^{-1}(\kappa+i\mathsf{y})=-d'/c'+i/(c'^2\mathsf{y})$. Since $c'\nmid d'$, for $\mathsf{y}<4/c'^2$ this point lies in the chamber $K_0+n$, $n=\lfloor -d'/c'\rfloor$, so $C=M_n(K_0)$ with $M_n=M(\left(\begin{smallmatrix}1 & n\\ 0 & 1\end{smallmatrix}\right))\in\Gamma_0(2)$, whose first column is $(a',c')^T$. Let $A$ with $M_A=M_n$; its first column is $(a',-c')^T$, and by (37) the first entry of $A^*T$ is $ma'^2-\ell a'c'+\nu c'^2$. Apply Lemma 7.8(iv) and (i). □

In terms of $\omega_0=\ell/2m$ and $t=\ell^2-4m\nu$, $V(\kappa)=c'^2m\big((\kappa-\omega_0)^2-t/4m^2\big)$.

7.8. **The polar coefficients as chamber coefficients.**

**Proposition 7.13.** *Let* $m\ge 1$, $\nu\in\frac12\mathbb{Z}$ *and* $0\le\ell\le 2m$. *Then* $c_m^F(2\nu,\ell)=d((m,\nu,\ell);K_0)$.

*Proof.* Let $\mathcal{S}=\{(\sigma,v):0<-\operatorname{Im}v<\operatorname{Im}\sigma\}$.

(a) For $(\sigma,v)\in\mathcal{S}$ and $\operatorname{Im}\rho$ large, $Y\in\tilde K_0$, as in the proof of Lemma 7.12(i). Collecting powers of $\mathbf{e}(\rho)$ in the absolutely convergent expansion of Lemma 7.7, $\mathcal{F}=\sum_{m\ge -1}\varphi_m^{K_0}(\sigma,v)\mathbf{e}(m\rho)$ by Lemma 7.12(i), where $\varphi_m^{K_0}=\sum_{\nu,\ell}d((m,\nu,\ell);K_0)x^{2\nu}y^\ell$ converges absolutely and locally uniformly on $\mathcal{S}$ (for given $(\operatorname{Im}\sigma,\operatorname{Im}v)$ take $M_1$ large).

(b) On the same region, the factors of (3) with $B=0$ and nonzero exponent have $\langle g,Y\rangle>0$ for $(\sigma,v)\in\mathcal{S}$ (they are $(0,k',0)$, $(0,k',\pm1)$ with $k'\ge 1$ integral, and $(0,0,-1)$), and together with the prefactor they give $\varphi_{-1}\mathbf{e}(-\rho)$ by (5). Taking logarithms of the factors with $B\ge 1$ is a rearrangement of an absolutely convergent double series, and gives (6). Writing $\exp\big(\sum_{l\ge1}L_lX^l\big)=\sum_{k\ge0}P_{k-1}(L_1,\dots,L_k)X^k$ and

$$\varphi_m:=\varphi_{-1}\,P_m(L_1,\dots,L_{m+1}),$$

uniqueness of Fourier coefficients in $\rho$ gives $\varphi_m^{K_0}=\varphi_m$ on $\mathcal{S}$. These $\varphi_m$ are the Fourier–Jacobi coefficients of Section 2.

(c) Each $L_l$ is holomorphic on $\mathbb{H}\times\mathbb{C}$ (every term of (6) is a weak Jacobi form evaluated at $((t\sigma+u)/d,tv)$), even in $v$, with the elliptic property of index $l$ (Lemma 2.1); $P_m$ is weighted homogeneous of weight $m+1$ when $L_l$ has weight $l$; and $\varphi_{-1}=1/(g(\sigma/2)A)$ with $A=-\theta_1(\sigma,v)^2/\eta(\sigma)^6$ of index one, whose zeros are double zeros along $v\in\mathbb{Z}\sigma+\mathbb{Z}$. So $\varphi_m$ is meromorphic on $\mathbb{H}\times\mathbb{C}$, even in $v$, has the elliptic property of index $m$, and has at most double poles, along $v\in\mathbb{Z}\sigma+\mathbb{Z}$.

(d) Principal part at $v=0$. Since $F^{(r,s)}(\tau,0)$ are the constants $12,4,4,4$, the sum over $u$ in (6) gives

$$L_l(\sigma,0)=\sum_{td=l}\frac{1}{2dt}\,2d\big(12+4(-1)^d\big),\qquad \sum_{l\ge1}L_l(\sigma,0)X^l=-\sum_{d\ge1}e_d\log(1-X^d),$$

so $\sum_k P_{k-1}(L(\sigma,0))X^k = \prod_d (1-X^d)^{-e_d} = \sum_{k\ge -1} d_1(k)X^{k+1}$ and $P_m(L(\sigma,0)) = d_1(m)$. As $\theta_1(\sigma,v) = 2\pi\eta(\sigma)^3 v + O(v^3)$, $A = -4\pi^2 v^2 + O(v^4)$, and by evenness

$$\varphi_m = -\frac{d_1(m)}{4\pi^2 g(\sigma/2)\, v^2} + O(1) \qquad (v \to 0).$$

(e) The polar part (8) has the same principal part at $v=0$: the sum $\mathcal{A}_{2,m}$ converges locally uniformly away from its poles $v \in -s\sigma + \mathbb{Z}$, only the term $s=0$ is singular at $v=0$, and $y/(1-y)^2 = -1/(4\sin^2\pi v)$. It also has the elliptic property of index $m$: under $v \mapsto v+\sigma$ the term $s$ becomes $p^{-m}y^{-2m}$ times the term $s+1$. Hence $\varphi_m^F = \varphi_m - \varphi_m^P$ has index $m$ and at most double poles along $v \in \mathbb{Z}\sigma + \mathbb{Z}$, and near $v=0$ the function $v^2\varphi_m^F$ is holomorphic with vanishing Taylor coefficients of $v^0$ and $v^1$ (both principal parts are even, with the same $v^{-2}$ coefficient). So $\varphi_m^F$ is holomorphic near $v=0$, and by the elliptic property on $\mathbb{H}\times\mathbb{C}$. Thus $c_m^F(n,\ell) = \frac{1}{2}\int_0^2 \int_0^1 \varphi_m^F\, x^{-n} y^{-\ell}\, dX_v\, dX_\sigma$, independently of $(\operatorname{Im}\sigma, \operatorname{Im} v)$.

(f) On $\mathcal{S}$ the expansion of $\mathcal{A}_{2,m}$ contains no $y^\ell$ with $0 \le \ell \le 2m$: for $s \ge 1$, $|p^s y| < 1$ and the term $s$ is $\sum_{k\ge 0}(k+1)p^{ms^2+s+sk}y^{2ms+1+k}$, with exponents of $y$ at least $2m+1$; for $s \le 0$, $|p^s y| > 1$ and the term is $\sum_{k\ge 0}(k+1)p^{ms^2+s-s(k+2)}y^{2ms-1-k}$, with exponents at most $-1$.

Taking $(\operatorname{Im}\sigma, \operatorname{Im} v) \in \mathcal{S}$ in (e), (b) and (f) give $c_m^F(2\nu,\ell) = [x^{2\nu}y^\ell]\varphi_m^{K_0} = d((m,\nu,\ell);K_0)$. □

Proposition 7.13 uses no averaging at $\ell = 0$ or $\ell = 2m$: the chamber $K_0$ is the chamber of the Fourier–Jacobi expansion in the strip $\mathcal{S}$, which corresponds to $0 < \operatorname{Re}\tau < 1$, and the polar part has no terms with $0 \le \ell \le 2m$ there.

7.9. **Proof of Theorem 7.1.** Fix a polar term $T = (m,\nu,\tilde{\ell})$, and let $\omega_0$, $t$, $\delta$ be as in Section 5; so $0 \le \omega_0 < 1$ and $\delta > 0$. Call a non-vertical wall $G$ *relevant* if $m_G \ge -1$ and $n_G \ge -1$; by Corollary 7.11 and the definition of $d_1$, only relevant walls have a nonzero jump.

**Lemma 7.14.** *Let $0 < \delta' < \delta$ and $J = [\omega_0 - \delta', \omega_0 + \delta']$. Every relevant wall whose open interval $(q/s, p/r)$ meets $J$ has $r, s \le (1/m + 2\delta' + 1)/(\delta^2 - \delta'^2)$. In particular there are finitely many such walls.*

*Proof.* Let $q/s < \kappa < p/r$ with $\kappa \in J$, put $U' = p/r - \kappa$, $W' = \kappa - q/s$, so $U', W' > 0$, $U' + W' = 1/rs$, and $h = \kappa - \omega_0$, $|h| \le \delta'$. By Lemma 5.1(ii) (whose formulas for $m_G$, $n_G$ hold for every $G$), $m_G \ge -1$ means $(U'+h)^2 \ge \delta^2 - 1/(mr^2)$. Since $|U'+h| \le 1/(rs) + \delta'$ and $rs \ge r$, $r^2 \ge r$,

$$\delta^2 - \delta'^2 \le \frac{2\delta'}{rs} + \frac{1}{r^2 s^2} + \frac{1}{mr^2} \le \frac{2\delta' + 1 + 1/m}{r},$$

which bounds $r$; $n_G \ge -1$ and $W'$ bound $s$ in the same way. For given $r, s$ the endpoint $p/r$ lies in $(\omega_0 - \delta', \omega_0 + \delta' + 1)$, which leaves finitely many $p$, and $q$ is then determined by $ps - qr = 1$. □

Let $\delta' = \delta/2$, let $\mathcal{E}$ be the finite set of endpoints greater than $\omega_0$ of the relevant walls of Lemma 7.14, and let $\beta = \min(\mathcal{E} \cup \{\omega_0 + \delta', 1\}) > \omega_0$. Choose $\kappa^* = (2j+1)/2^k \in (\omega_0, \beta)$ with $k$ so large that

$$V(\kappa^*) = 4^k m\big((\kappa^* - \omega_0)^2 - \delta^2\big) \le 4^k m(\delta'^2 - \delta^2) \le -2.$$

Consider the path $\gamma(\mathsf{y}) = \kappa^* + i\mathsf{y}$, $\mathsf{y}$ decreasing from $+\infty$ to 0 (Figure 2). For $\mathsf{y} > \frac{1}{4}$ it lies in $K_0$, because $0 < \kappa^* < 1$. By Lemma 7.5(F3), a Farey edge whose interval contains $\kappa^*$ strictly has denominators adding up to at most $2^k$, so $\gamma$ meets finitely many walls, each exactly once, transversally, passing from outside to inside; it meets no vertical wall. Distinct walls meet $\gamma$ at distinct heights by (F1). Between consecutive walls $\gamma$ stays in one chamber (it may cross

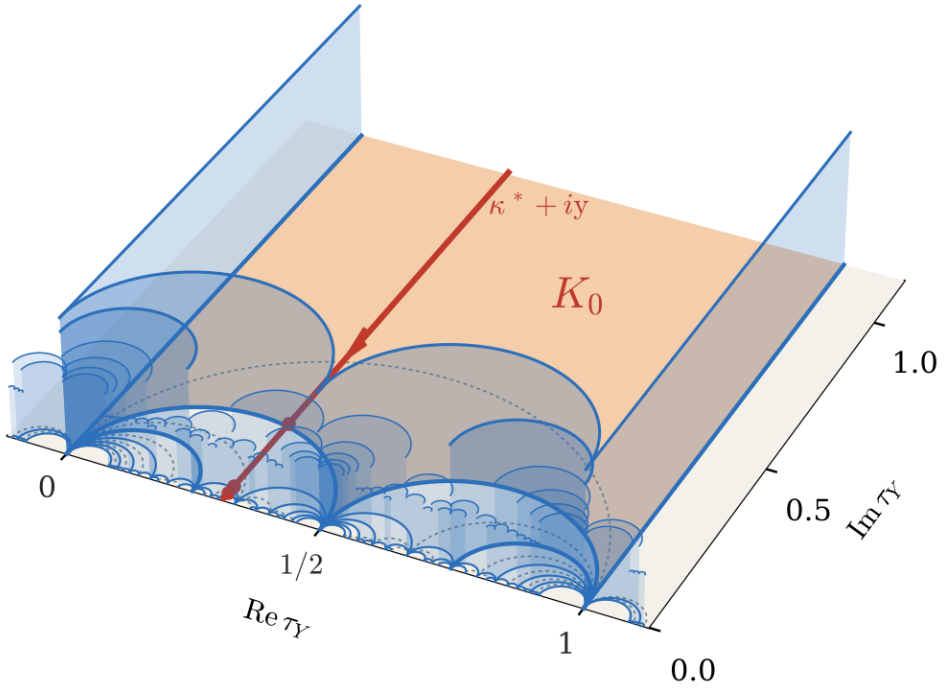



FIGURE 2. Walls and chambers in the upper half-plane of $\tau_Y$. The walls are the Farey geodesics with an even denominator, drawn as curtains; the Farey geodesics with two odd denominators (dashed) are not walls. The shaded region is the chamber $K_0$, the ideal quadrilateral with vertices $0, \frac{1}{2}, 1, \infty$, in which the polar coefficients of $\varphi_m^F$ are read off. The path $\kappa^*+iy$ of the proof of Theorem 7.1 descends from $K_0$ and crosses finitely many walls (dots), each contributing one term of the wall sum.

odd edges), and below the last one it stays in the chamber $C$ of Lemma 7.12(ii) for $\kappa = \kappa^*$, where the coefficient of $T$ vanishes since $V(\kappa^*) \le -2$. Summing Corollary 7.11 along $\gamma$,

$$d(T;K_0) = d(T;K_0) - d(T;C) = \sum_{G:\ q/s<\kappa^*<p/r} J_G(T).$$

Only relevant walls contribute. A relevant wall with $q/s < \kappa^* < p/r$ meets $J$, so $p/r \in \mathcal{E}$ or $p/r \le \omega_0$; the latter is excluded by $p/r > \kappa^* > \omega_0$, so $p/r > \omega_0$, and $q/s \le \omega_0$ because otherwise $q/s \in \mathcal{E}$ and $q/s > \kappa^*$. Conversely, a relevant wall with $q/s \le \omega_0 < p/r$ has an interval meeting $J$, so $p/r \in \mathcal{E}$, $p/r > \kappa^*$ and $q/s < \kappa^*$. Hence the sum runs over the relevant walls in $\mathcal{W}(m,\nu,\tilde\ell)$, and since $a_I(\nu,\tilde\ell) = \sum_{G\in\mathcal{W}(m,\nu,\tilde\ell)} J_G(T)$ by (23), it equals $a_I(\nu,\tilde\ell)$; the remaining $G \in \mathcal{W}(m,\nu,\tilde\ell)$ (odd edges, or not relevant) contribute 0. By Proposition 7.13, $d(T;K_0) = c_m^F(2\nu,\tilde\ell)$. □

**Remark 7.15.** The opposite endpoint convention, $q/s < \omega_0 \le p/r$, gives the same value. If $\omega_0 > 0$, choose $\kappa^*$ slightly to the left of $\omega_0$, in $(0,\omega_0)$ and closer to $\omega_0$ than every endpoint smaller than $\omega_0$ of the walls of Lemma 7.14; the argument above applies verbatim. If $\omega_0 = 0$, then $\tilde\ell = 0$, the path starts in $K_0'$, and $d(T;K_0') = d(T;K_0)$ by Lemma 7.10. In particular no averaging over the two conventions is needed when $2m \mid \tilde\ell$.

**Remark 7.16.** The proof does not use the global meromorphy of $1/\widetilde{\Phi}_6$ on the Siegel upper half-space, the identification of $\widetilde{\Phi}_6$ with a Borcherds lift or with the square of a theta-type form, nor [8, Theorem 5.2]: the meromorphy of $\varphi_m$, its poles and the holomorphy of $\varphi_m^F$ are proved in Proposition 7.13 from the product formula. The formula (8) for the polar part is taken from [8, (5.45)] as a definition.

## 8. The polar coefficients at the cusp 0

This section proves (P0) of Section 6. For a function $f$ on $\mathbb{H}\times\mathbb{C}$ and integers $k$, $m$ put

$$(f|_{k,m}S)(\sigma,v)=\sigma^{-k}\,\mathbf{e}\Big(-\frac{mv^2}{\sigma}\Big)f\Big(-\frac{1}{\sigma},\frac{v}{\sigma}\Big),$$

the action of $R_S$ in weight $k$ and index $m$. The polar part (8) satisfies $\varphi_m^P|_{-6,m}S=\frac{d_1(m)}{16}\,g(\sigma)^{-1}\,\mathcal{A}_{2,m}|_{2,m}S$, because $g(-1/2\sigma)=16\,\sigma^8 g(\sigma)$ (from $\eta(-1/\tau)=\sqrt{-i\tau}\,\eta(\tau)$). Accordingly we define the *finite part in the frame $S$* by

$$\begin{aligned}\psi_m^S&=\varphi_m|_{-6,m}S-\frac{d_1(m)}{16}\,\frac{\mathcal{A}_{2,m}(\sigma,v)}{g(\sigma)}=\varphi_m^F|_{-6,m}S+\frac{d_1(m)}{16}\,\frac{h_m(\sigma,v)}{g(\sigma)},\\ h_m&=\mathcal{A}_{2,m}|_{2,m}S-\mathcal{A}_{2,m}.\end{aligned}\tag{41}$$

It is the subtraction recorded by the constants $(c_S,\mu_S)=(d_1(m)/16,1)$ of (28).

**Theorem 8.1.** *Let $m\geq 1$.*

(i) *$\psi_m^S$ is holomorphic on $\mathbb{H}\times\mathbb{C}$, of period one in $\sigma$, and has an expansion $\psi_m^S=\sum_{\nu\geq -1}\sum_{\ell}c_m^S(\nu,\ell)\,p^\nu y^\ell$ with $16\,c_m^S(\nu,\ell)\in\mathbb{Z}$.*

(ii) *For every $\nu\in\mathbb{Z}$ and $0\leq\tilde\ell<2m$ with $4m\nu-\tilde\ell^2<0$,*

$$c_m^S(\nu,\tilde\ell)=a_S(\nu,\tilde\ell),$$

*the wall sum (22). The opposite endpoint convention $q/s<\omega_0\leq p/r$ in $\mathcal{W}(m,\nu,\tilde\ell)$ gives the same value.*

(iii) *$h_m$ is holomorphic on $\mathbb{H}\times\mathbb{C}$, even in $v$, with the elliptic property of index $m$.*

*Consequently (P0) holds: the principal part of the finite part at the cusp 0, in the frame $S$ with the subtraction (28), is given by (22).*

The term $h_m/g$ in (41) is the transformation defect of the Appell–Lerch sum; it carries no polar terms, see Remark 8.10. The proof runs parallel to Section 7, for a second product $\Phi_6^S$ whose Fourier–Jacobi coefficients are the functions $16\,\varphi_m|_{-6,m}S$: we prove this from the transformation of the twisted genera (Lemmas 8.2 to 8.3). For $1/\Phi_6^S$ every Farey edge is a wall, the chambers are the Farey triangles, and the group of symmetries is all of $SL_2(\mathbb{Z})$. With the factor $\frac{1}{16}$ from (41), the path argument of Section 7 then gives (22).

8.1. **The genera under $SL_2(\mathbb{Z})$.** Let $SL_2(\mathbb{Z})$ act on weak Jacobi forms of weight 0 and index one by $(F|\gamma)(\tau,z)=\mathbf{e}(-cz^2/(c\tau+d))F(\gamma\tau,z/(c\tau+d))$, and on $(\mathbb{Z}/2)^2$ from the right by $(r,s)\gamma=(r,s)\left(\begin{smallmatrix}a&b\\c&d\end{smallmatrix}\right)$ mod 2.

**Lemma 8.2.** *$F^{(r,s)}|\gamma=F^{(r,s)\gamma}$ for all $\gamma\in SL_2(\mathbb{Z})$ and $(r,s)\in(\mathbb{Z}/2)^2$.*

*Proof.* Both sides define right actions, so it suffices to take $\gamma=T=\left(\begin{smallmatrix}1&1\\0&1\end{smallmatrix}\right)$, for which $(r,s)T=(r,r+s)$, and $\gamma=S$, for which $(r,s)S=(s,r)$. The forms $\phi_{0,1}$ and $\phi_{-2,1}$ are invariant under $SL_2(\mathbb{Z})$ in weights 0 and $-2$ [17, §9]. For $T$ the claim is clear from (2), since $\mathfrak{E}$ has period

one. For $S$ we use the transformation $E(-1/\tau) = \tau^2 E(\tau) + 6\tau/(\pi i)$ of the quasimodular $E$. Applied at $\tau$ and at $\tau/2$ it gives

$$\tau^{-2}\mathfrak{E}(-1/\tau) = -\tfrac{1}{2}\mathfrak{E}(\tau/2), \qquad \text{and, replacing } \tau \text{ by } 2\tau, \qquad \tau^{-2}\mathfrak{E}\big(-1/(2\tau)\big) = -2\mathfrak{E}(\tau),$$

so $\mathfrak{E}$ and $\mathfrak{E}(\tau/2)$ are exchanged by $\tau \mapsto -1/\tau$ in weight two, up to the factors $-\frac{1}{2}$ and $-2$. From $\sigma_1(2n) = 3\sigma_1(n) - 2\sigma_1(n/2)$ one gets $\mathfrak{E}(\tau/2) + \mathfrak{E}((\tau+1)/2) = 2\mathfrak{E}(\tau)$, so $\mathfrak{E}((\tau+1)/2) = 2\mathfrak{E}(\tau) - \mathfrak{E}(\tau/2)$ is invariant under $\tau \mapsto -1/\tau$ in weight two. Hence $F^{(0,1)}|S = \frac{1}{3}\phi_{0,1} + \frac{2}{3}\phi_{-2,1} \cdot (-\frac{1}{2})\mathfrak{E}(\tau/2) = F^{(1,0)}$, $F^{(1,0)}|S = \frac{1}{3}\phi_{0,1} - \frac{1}{3}\phi_{-2,1} \cdot (-2)\mathfrak{E} = F^{(0,1)}$ and $F^{(1,1)}|S = F^{(1,1)}$; $F^{(0,0)} = \phi_{0,1}$ is invariant. □

For an integer matrix $M = \left(\begin{smallmatrix} a & b \\ c & d \end{smallmatrix}\right)$ with $\det M = l > 0$ and a function $F$ of index one put

$$(F|[M])(\tau, z) = \mathbf{e}\Big(-\frac{lcz^2}{c\tau + d}\Big) F\Big(M\tau, \frac{lz}{c\tau + d}\Big),$$

so that $F|[M] = F((t\tau+u)/d, tz)$ for $M = \left(\begin{smallmatrix} t & u \\ 0 & d \end{smallmatrix}\right)$. By Lemma 9.1(i) of Section 9, a statement of pure matrix algebra, $F|[\gamma M] = (F|\gamma)|[M]$ for $\gamma \in SL_2(\mathbb{Z})$, and $F|[MS] = (F|[M])|_{0,l}S$; these are the terms of the operator $V_l$ of [17, §4] written without the normalising powers of $l$.

**Lemma 8.3.** *For $l \geq 1$ let*

$$L'_l = \sum_{td=l} \frac{1}{2dt} \sum_{r,s \in \{0,1\}} (-1)^{rt} \sum_{u=0}^{2d-1} F^{(r,s)}\Big(\frac{t\sigma + u}{d}, tv\Big).$$

*Then $L_l|_{0,l}S = L'_l$, and*

$$L'_l = \sum_{t|l} \frac{1}{t} \sum_{k \geq 0} \sum_{j \in \mathbb{Z}} \big[C_0 + (-1)^t C_1\big]\big(4\tfrac{l}{t}k - j^2\big)\, p^{tk} y^{tj},$$

*where $C_0(N)$ and $C_1(N)$ are the coefficients of $q^n\zeta^j$, $4n - j^2 = N$, $n \in \mathbb{Z}$, in $F^{(0,0)} + F^{(0,1)} = \frac{4}{3}\phi_{0,1} + \frac{2}{3}\phi_{-2,1}\mathfrak{E}$ and in the integral-power part of $F^{(1,0)} + F^{(1,1)}$, which is $\frac{2}{3}\phi_{0,1} - \frac{2}{3}\phi_{-2,1}\mathfrak{E}$. In particular $C_0(N) = C_1(N) = 0$ for $N < -1$, $C_0(-1) = 2$, $C_1(-1) = 0$, $C_0(0) = 12$, $C_1(0) = 8$, and*

$$C_0(N) = c_b^{(0,0)}(N) + c_b^{(0,1)}(N), \qquad C_1(N) = c_b^{(1,0)}(N) + c_b^{(1,1)}(N), \qquad b \equiv N \pmod 2,$$

*in the notation of* (3).

*Proof.* Let $\mathcal{M}_l$ be the set of integer matrices of determinant $l$, and let $SL_2(\mathbb{Z})$ act on $\mathcal{M}_l \times (\mathbb{Z}/2)^2$ by $\gamma \cdot (M, w) = (\gamma M, w\gamma^{-1})$. With $w = (r, s)$ and $M = \left(\begin{smallmatrix} t & u \\ 0 & d \end{smallmatrix}\right)$ the sign in (6) is $(-1)^{sd+ru} = (-1)^{(wM)_2}$, the second entry of the row vector $wM$. The term $(-1)^{(wM)_2}F^w|[M]$ depends only on the orbit of $(M, w)$, by Lemma 8.2 and $F|[\gamma M] = (F|\gamma)|[M]$. The stabiliser of $M$ in $SL_2(\mathbb{Z})$ is trivial, and every orbit contains exactly one $(M, w)$ with $M$ upper triangular, $t, d > 0$, $0 \leq u < d$ (Hermite normal form). In (6), $u$ runs modulo $2d$, and $(TM, w)$ lies in the orbit of $(M, wT)$; so each orbit occurs twice, and

$$L_l = \frac{1}{l} \sum_{(M,w) \in SL_2(\mathbb{Z}) \backslash (\mathcal{M}_l \times (\mathbb{Z}/2)^2)} (-1)^{(wM)_2} F^w|[M].$$

Applying $|_{0,l}S$ and $F|[MS] = (F|[M])|_{0,l}S$, and substituting $M' = MS$, which permutes the orbits, with $(wM)_2 = (wM'S^{-1})_2 = (wM')_1$, gives the same sum with the sign $(-1)^{(wM')_1}$. For upper triangular $M'$ that sign is $(-1)^{rt}$, and returning to $u$ modulo $2d$ gives $L'_l$.

For the expansion, $F^{(r,s)} = \sum c(n, j) q^n \zeta^j$ with $n \in \frac{1}{2}\mathbb{Z}$, and $\sum_{u=0}^{2d-1} \mathbf{e}(nu/d) = 2d$ if $n \in d\mathbb{Z}$ and 0 otherwise. So the term $(t, d)$ is $\frac{1}{t}\sum_{k,j} c(dk, j) p^{tk} y^{tj}$, and only integral powers $n = dk$

of $q$ survive. Since $c(n,j)$ depends only on $4n-j^2$ and $j$ mod 2, and for integral $n$ the parity of $j$ is that of $N=4n-j^2$, the formula follows. The genera have no negative powers of $q$, which gives $C_0=C_1=0$ for $N<-1$. The values at $N=-1,0$ are read off from $\phi_{0,1}=\zeta+10+\zeta^{-1}+O(q)$, $\phi_{-2,1}=\zeta-2+\zeta^{-1}+O(q)$ and $\mathfrak{E}=1+O(q)$. For the integral-power part of $F^{(1,0)}+F^{(1,1)}$ note that the integral-power parts of $\mathfrak{E}(\tau/2)$ and $\mathfrak{E}((\tau+1)/2)$ are both $\frac12(\mathfrak{E}(\tau/2)+\mathfrak{E}((\tau+1)/2))=\mathfrak{E}(\tau)$. □

8.2. **The product at the cusp 0.** Let $\mathcal{I}^S=\{(B,k,j)\in\mathbb{Z}^3:\ B,k\ge0,\ j<0 \text{ if } B=k=0\}$, and for $g=(B,k,j)$ put $N(g)=4Bk-j^2$ and $\mathbf{e}(\langle g,\Omega\rangle)=\mathbf{e}(B\rho+k\sigma+jv)$. Define

$$\Phi_6^S(\Omega)=\mathbf{e}(\rho+\sigma+v)\prod_{g\in\mathcal{I}^S}\big(1-\mathbf{e}(\langle g,\Omega\rangle)\big)^{C_0(N(g))}\big(1+\mathbf{e}(\langle g,\Omega\rangle)\big)^{C_1(N(g))}. \tag{42}$$

Charges are now triples $T=(m,\nu,\ell)\in\mathbb{Z}^3$, and all the notation of Section 7 ($Y$, $\tau_Y$, (35), (36), (37), $\mathcal{D}$, the tube $\mathcal{T}$) is used unchanged.

**Proposition 8.4.** (i) *The product* (42) *converges absolutely and locally uniformly on* $\mathcal{T}$. *Its zeros there are the double zeros of the factors* $(1-\mathbf{e}(\langle g,\Omega\rangle))^2$ *with* $N(g)=-1$, *at the points where* $\langle g,\Omega\rangle\in\mathbb{Z}$. *The walls of* $1/\Phi_6^S$ *are the cones over all Farey edges:* $g\mapsto$(38) *is a bijection from* $\{g\in\mathcal{I}^S:\ N(g)=-1\}$ *onto the set of Farey edges, each with exponent* $C_0(-1)=2$.

(ii) *For* $(\sigma,v)$ *in the strip* $\mathcal{S}=\{0<-\operatorname{Im}v<\operatorname{Im}\sigma\}$ *and* $\operatorname{Im}\rho$ *large,* $\tau_Y$ *lies in the Farey triangle* $\Delta(0)$ *with vertices* $0,1,\infty$, *and*

$$\frac{1}{\Phi_6^S}=\sum_{m\ge-1}\varphi'_m(\sigma,v)\,\mathbf{e}(m\rho),\qquad \varphi'_m=\frac{P_m(L'_1,\dots,L'_{m+1})}{g(\sigma)\,A},$$

*with* $P_m$ *as in the proof of Proposition 7.13.*

(iii) $\varphi'_m=16\,\varphi_m|_{-6,m}S$ *as meromorphic functions on* $\mathbb{H}\times\mathbb{C}$.

*Proof.* (i) By Lemma 8.3 the exponents are sums of coefficients of the genera, so they satisfy the bound of Lemma 7.2, and they vanish for $N<-1$. For $N(g)\ge0$ the matrix $S_g$ is positive semidefinite and nonzero, and $|\mathbf{e}(\langle g,\Omega\rangle)|<1$ on $\mathcal{T}$, so neither $1-\mathbf{e}$ nor $1+\mathbf{e}$ vanishes. The proof of Proposition 7.6 then applies verbatim. A charge with $N(g)=-1$ has exponents $(2,0)$; by the first part of the proof of Lemma 7.4, without the parity condition, these charges correspond bijectively to the Farey edges, with $g=(cd,ab,ad+bc)$ for the edge joining $a/c$ and $b/d$, $c,d\ge1$, and $g=(0,n,1)$, $(0,-n,-1)$ for the vertical edges.

(ii) For such $\Omega$, $\operatorname{Re}\tau_Y=-M_3/M_2\in(0,1)$ and $\operatorname{Im}\tau_Y\to\infty$ as $M_1\to\infty$, so $\tau_Y\in\Delta(0)$. The factors with $B=0$ are $(1-p^k)^{12}(1+p^k)^8=(1-p^k)^4(1-p^{2k})^8$ for $k\ge1$ and $(1-p^ky)^2$, $(1-p^ky^{-1})^2$ for $k\ge1$, together with $(1-y^{-1})^2$; by the triple product for $\theta_1$ and $g(\sigma)=p\prod(1-p^n)^8(1-p^{2n})^8$, together with the prefactor they give $\mathbf{e}(\rho)\,g(\sigma)\,A$. They have $\langle g,Y\rangle>0$ on $\mathcal{S}$. For $B\ge1$ take logarithms, $\log(1-X)^{C_0}(1+X)^{C_1}=-\sum_{t\ge1}\frac1t\big(C_0+(-1)^tC_1\big)X^t$; collecting the powers of $\mathbf{e}(\rho)$, the exponent of $\mathbf{e}(l\rho)$ is $L'_l$ by Lemma 8.3. This is a rearrangement of an absolutely convergent series, as in Proposition 7.13(b).

(iii) By Lemma 8.3, $L'_l=L_l|_{0,l}S$. The function $L_l$ is even in $v$ (Lemma 2.1), so $L_l|_{0,l}S^2=L_l$, and $P_m(L')=P_m(L)|_{0,m+1}S$ because the slash is multiplicative in weight and index (Lemma 9.1(ii)). Moreover $\phi_{-2,1}|_{-2,1}S=\phi_{-2,1}$, and $g(-1/2\sigma)=16\sigma^8g(\sigma)$ gives $g(\sigma/2)^{-1}|_{-8,0}S=\frac{1}{16}g(\sigma)^{-1}$. With (5) and $\varphi_m=\varphi_{-1}P_m(L)$, $\varphi_m|_{-6,m}S=\big(\varphi_{-1}|_{-6,-1}S\big)\big(P_m(L)|_{0,m+1}S\big)=\frac{1}{16\,g(\sigma)A}\,P_m(L')$.

□

8.3. **Chambers, symmetries and the jump.** Since every Farey edge is a wall, the chambers of $1/\Phi_6^S$ (the components of the complement of the walls in $\mathbb{H}$) are the open Farey triangles $\Delta(e)$, $\Delta(n)$ of Section 7, by Lemma 7.9(i). Lemma 7.7 holds for $1/\Phi_6^S$ with the same proof, the factor $(1+\mathbf{e}(\langle g,\Omega\rangle))^{-C_1}$ occurring only for $N(g)\geq 0$; we write $d^S(T;C)$ for the coefficient of $\mathbf{e}(\langle T,\Omega\rangle)$, $T\in\mathbb{Z}^3$, in the chamber $C$, so that

$$d^S(T;C)=\int_0^1\int_0^1\int_0^1 \frac{\mathbf{e}(-\langle T,X+iY\rangle)}{\Phi_6^S(X+iY)}\,dX_v\,dX_\sigma\,dX_\rho \qquad (Y\in\tilde{C}).$$

All wall charges are positive on $\tilde{\Delta}(0)$: by (F3) no Farey edge with $c,d\geq 1$ contains 0 or 1 strictly inside its interval, so its half-disc lies in $\{\mathsf{x}\leq 0\}$, in $\{\mathsf{x}\geq 1\}$ or is that of $(0,1)$; and the vertical walls are positive there by Lemma 7.4. Hence the expansion in $\Delta(0)$ is the expansion of the product (42) itself.

**Lemma 8.5.** *$\Phi_6^S(A\Omega A^T)=\Phi_6^S(\Omega)$ for every $A\in SL_2(\mathbb{Z})$ (on $\mathcal{T}$), and $d^S(T;M_A(C))=d^S(A^*T;C)$ for every chamber $C$. The group $SL_2(\mathbb{Z})$ acts transitively on the Farey triangles, so every chamber is $M_A(\Delta(0))$ for some $A\in SL_2(\mathbb{Z})$.*

*Proof.* For $E=\left(\begin{smallmatrix}1&1\\0&1\end{smallmatrix}\right)$ the proof of Lemma 7.8(i) applies: $E^*(B,k,j)=(B,k+B+j,j+2B)$ preserves $N$, on which alone the exponents depend, and with $\varrho^S=(1,1,1)$ one has $E^*\varrho^S=\varrho^S+(0,2,2)$, compensated by the two exceptional factors as before. Next, (42) is symmetric under $(\rho,\sigma,v)\mapsto(\sigma,\rho,v)$, since $(B,k,j)\mapsto(k,B,j)$ preserves $\mathcal{I}^S$ and $N$; and it is even in $v$ by the argument of Lemma 7.8(ii). Together, $\Phi_6^S$ is invariant under $\Omega\mapsto J\Omega J^T$ with $J=\left(\begin{smallmatrix}0&-1\\1&0\end{smallmatrix}\right)$, and $E$, $J$ generate $SL_2(\mathbb{Z})$. The statement on coefficients is proved as in Lemma 7.8(iv). Finally $SL_2(\mathbb{Z})$ maps Farey edges to Farey edges and triangles to triangles, and it is transitive on triangles: the matrix $\left(\begin{smallmatrix}b&a\\d&c\end{smallmatrix}\right)$ maps $0,\infty$ to $a/c,b/d$ when $bc-ad=1$, hence maps $\Delta(0)$ or $\Delta(-1)$ to a triangle with side $(a/c,b/d)$, and $\Delta(-1)=T^{-1}\Delta(0)$. □

**Lemma 8.6** (Jump lemma at the cusp 0)**.** *For every $T=(m,\nu,\ell)\in\mathbb{Z}^3$, $d^S(T;\Delta(0))-d^S(T;\Delta(-1))=-\ell\,d_1(m)\,d_1(\nu)$.*

*Proof.* The proof of Lemma 7.10 applies with $g_0=(0,0,-1)$ and $1/\Phi_6^S=-H^S/(4\sin^2\pi v)$, $H^S$ even and holomorphic near $M_3=0$; it remains to compute $H^S(\rho,\sigma,0)$. At $v=0$ the factors with given $(B,k)\neq(0,0)$ combine to $(1-X)^{\sum_j C_0}(1+X)^{\sum_j C_1}$ with $X=\mathbf{e}(B\rho+k\sigma)$, and $\sum_j C_0(4Bk-j^2)$, $\sum_j C_1(4Bk-j^2)$ are the coefficients of $q^{Bk}$ in $(F^{(0,0)}+F^{(0,1)})(\tau,0)=16$ and in the integral part of $(F^{(1,0)}+F^{(1,1)})(\tau,0)=8$. So only $Bk=0$ contributes, and

$$\begin{aligned}H^S(\rho,\sigma,0)&=\frac{\mathbf{e}(-\rho-\sigma)}{\prod_{k\geq1}(1-p^k)^{16}(1+p^k)^8\prod_{B\geq1}(1-\mathbf{e}(B\rho))^{16}(1+\mathbf{e}(B\rho))^8}\\&=\frac{1}{g(\rho)\,g(\sigma)}=\sum_{k,k'\geq-1}d_1(k)d_1(k')\,\mathbf{e}(k\rho)\,p^{k'}.\end{aligned}$$

The coefficient of $\mathbf{e}(m\rho)p^\nu$ is $d_1(m)d_1(\nu)$. □

For a Farey edge that is not vertical let $G=\left(\begin{smallmatrix}p&q\\r&s\end{smallmatrix}\right)\in SL_2(\mathbb{Z})$, $r,s\geq1$, be the matrix with $G(\infty)=p/r$, $G(0)=q/s$, and define $m_G,n_G,\ell_G$ by (19).

**Corollary 8.7** (Transport at the cusp 0)**.** *Let $C_{\mathrm{out}}$, $C_{\mathrm{in}}$ be the Farey triangles adjacent to the edge of $G$, outside and inside its half-disc. Then $d^S(T;C_{\mathrm{out}})-d^S(T;C_{\mathrm{in}})=\ell_G\,d_1(m_G)\,d_1(n_G)$.*

*Proof.* $G$ maps the edge $(0,\infty)$ onto the edge of $G$, and $\Delta(0)$ onto the triangle with third vertex $G(1)=(p+q)/(r+s)$, which is inside; so $C_{\mathrm{in}}=G(\Delta(0))$ and $C_{\mathrm{out}}=G(\Delta(-1))$. With

$A = \left(\begin{smallmatrix} p & -q \\ -r & s \end{smallmatrix}\right)$ one has $M_A = G$ and $A^*T = (m_G, n_G, \ell_G)$ by (37), and Lemmas 8.5 and 8.6 give the claim. No parity condition on $r, s$ occurs. □

**Lemma 8.8.** (i) *$d^S((m,\nu,\ell);\Delta(0)) = 0$ for $m \le -2$.*

(ii) *Let $\kappa = a'/c'$ in lowest terms with $c' \ge 2$, and let $C$ be the Farey triangle containing $\kappa + i\mathsf{y}$ for all small $\mathsf{y} > 0$ on the line $\mathrm{Re}\,\tau = \kappa$. Then $d^S((m,\nu,\ell);C) = d^S(T';\Delta(0))$ for a charge $T'$ whose first entry is $V(\kappa) = \nu c'^2 - \ell a' c' + m a'^2$; hence it vanishes if $V(\kappa) \le -2$.*

*Proof.* (i) is proved as Lemma 7.12(i). (ii) With $M = \left(\begin{smallmatrix} a' & b' \\ c' & d' \end{smallmatrix}\right) \in SL_2(\mathbb{Z})$, $M^{-1}(\kappa + i\mathsf{y}) = -d'/c' + i/(c'^2\mathsf{y})$ has non-integral real part and, for $\mathsf{y} < 2/c'^2$, imaginary part above $\frac12$, the largest radius of a non-vertical Farey edge; so it lies in $\Delta(n)$, $n = \lfloor -d'/c' \rfloor$, and $C = M_n(\Delta(0))$ with $M_n = MT^n$, whose first column is $(a', c')^T$. Conclude as in Lemma 7.12(ii) with Lemma 8.5. □

8.4. **Identification and the proof of the theorem.**

**Proposition 8.9.** *Theorem 8.1(i) and (iii) hold, and $16\, c^S_m(\nu,\ell) = d^S((m,\nu,\ell);\Delta(0))$ for $m \ge 1$, $\nu \in \mathbb{Z}$, $0 \le \ell \le 2m$.*

*Proof.* By Proposition 8.4(iii), $16\,\psi^S_m = \varphi'_m - d_1(m)\,\mathcal{A}_{2,m}/g(\sigma)$. Each $L'_l$ is holomorphic on $\mathbb{H}\times\mathbb{C}$, even in $v$, with the elliptic property of index $l$, by the argument of Lemma 2.1 applied to the terms of Lemma 8.3; so $\varphi'_m$ is meromorphic, even, of index $m$, with at most double poles along $v \in \mathbb{Z}\sigma + \mathbb{Z}$, the zeros of $A$. At $v = 0$, Lemma 8.3 and $\sum_j C_i(4n - j^2) = 0$ for $n \ge 1$ give $L'_l(\sigma, 0) = \sum_{t|l} \frac{1}{t}\big(16 + 8(-1)^t\big)$, so

$$\sum_{k\ge 0} P_{k-1}(L'(\sigma,0))X^k = \prod_{d\ge1}(1-X^d)^{-16}(1+X^d)^{-8}$$
$$= \prod_{d\ge1}(1-X^d)^{-8}(1-X^{2d})^{-8} = \sum_{k\ge -1} d_1(k)X^{k+1},$$

and $P_m(L'(\sigma,0)) = d_1(m)$. Hence $\varphi'_m = -d_1(m)/(4\pi^2 g(\sigma)v^2) + O(1)$, which is also the principal part of $d_1(m)\mathcal{A}_{2,m}/g(\sigma)$ (Proposition 7.13(e)). Both are even and of index $m$, so $\psi^S_m$ is holomorphic on $\mathbb{H}\times\mathbb{C}$. Both terms are series in $p$ and $y$ with $p$-exponents at least $-1$ (the factor $1/g(\sigma)$) and coefficients in $\mathbb{Z}$, which gives (i). By Proposition 8.4(ii) and Lemma 7.7, $\varphi'_m = \sum_{\nu,\ell} d^S((m,\nu,\ell);\Delta(0))p^\nu y^\ell$ on $\mathcal{S}$, and on $\mathcal{S}$ the expansion of $\mathcal{A}_{2,m}$ has no $y^\ell$ with $0 \le \ell \le 2m$ (Proposition 7.13(f)). Computing the coefficients of the holomorphic function $16\psi^S_m$ on a horizontal torus in $\mathcal{S}$ gives $16c^S_m(\nu,\ell) = d^S((m,\nu,\ell);\Delta(0))$.

For (iii): $\mathcal{A}_{2,m}$ is even, has the elliptic property of index $m$ and the principal part $-1/(4\pi^2v^2)$ at $v = 0$ (Proposition 7.13(e)). The slash $|_{2,m}S$ preserves evenness and the elliptic property of index $m$ (Lemma 9.1(iii)), and maps the principal part at $v = 0$ to $\sigma^{-2}\cdot(-\sigma^2/(4\pi^2v^2))$, the same. So $h_m$ is holomorphic near $v = 0$ and, by the elliptic property, on $\mathbb{H}\times\mathbb{C}$. □

*Proof of Theorem 8.1(ii).* Fix a polar term $T = (m,\nu,\tilde\ell)$, $\nu \in \mathbb{Z}$, $0 \le \tilde\ell < 2m$, and let $\omega_0$, $t$, $\delta$, $\delta' = \delta/2$, $J$ be as in Section 7. Call a non-vertical Farey edge *relevant* if $m_G, n_G \ge -1$; by Corollary 8.7 only relevant edges have a nonzero jump, and Lemma 7.14 applies to them unchanged (its proof uses only Lemma 5.1(ii), valid for every $G$). Choose $\kappa^* = (2j+1)/2^k \in (\omega_0, \beta)$ as in the proof of Theorem 7.1, with $V(\kappa^*) \le -2$, and follow $\gamma(\mathsf{y}) = \kappa^* + i\mathsf{y}$ from $\mathsf{y} > \frac12$, where $\gamma$ lies in $\Delta(0)$, down to 0. It crosses each Farey edge whose interval contains $\kappa^*$ strictly exactly once, from outside to inside; there are finitely many, by (F3), and no vertical one. Below the last one it lies in the triangle $C$ of Lemma 8.8(ii), where $d^S(T;C) = 0$. Summing

Corollary 8.7,

$$d^S(T;\Delta(0)) = \sum_{G:\ q/s<\kappa^*<p/r} \ell_G\, d_1(m_G)\, d_1(n_G),$$

and the argument at the end of the proof of Theorem 7.1 shows that the relevant edges in this sum are exactly the relevant $G \in \mathcal{W}(m,\nu,\tilde\ell)$. Hence $d^S(T;\Delta(0)) = 16\, a_S(\nu,\tilde\ell)$ by (22), and $c_m^S(\nu,\tilde\ell) = a_S(\nu,\tilde\ell)$ by Proposition 8.9. The opposite endpoint convention is Remark 7.15 verbatim: for $\omega_0 = 0$ the path starts in $\Delta(-1)$, and $d^S(T;\Delta(-1)) = d^S(T;\Delta(0))$ by Lemma 8.6 with $\ell = \tilde\ell = 0$. □

**Remark 8.10.** By Theorem 8.1(iii), $h_m = \sum_{\ell \bmod 2m} h_{m,\ell}(\sigma)\vartheta_{m,\ell}(\sigma,v)$ with $\vartheta_{m,\ell} = \sum_{r\equiv\ell\,(2m)} p^{r^2/4m}y^r$ and $h_{m,\ell}$ holomorphic on $\mathbb{H}$, so that

$$\varphi_m^F|_{-6,m}S = \psi_m^S - \frac{d_1(m)}{16\,g(\sigma)} \sum_{\ell \bmod 2m} h_{m,\ell}(\sigma)\,\vartheta_{m,\ell}(\sigma,v).$$

The functions $h_{m,\ell}$ are period integrals: $h_m$ is the function $\mathcal{D} = \mathcal{A}|_{2,m}S - \mathcal{A}$ of the proof of Proposition 9.9, where it is proved that

$$h_{m,\ell}(\sigma) = \frac{\sqrt{2m}}{4\pi}\int_0^\infty \vartheta_{m,\ell}(it,0)\,(t-i\sigma)^{-3/2}\,dt$$

(this is $c_\ell = \frac{\sqrt{2m}}{4\pi}e^{\pi i/4}\tilde P_\ell$ there, with $u = it$ in $\tilde P_\ell$ and $(i(t-i\sigma))^{-3/2} = e^{-3\pi i/4}(t-i\sigma)^{-3/2}$). They are not Fourier series in $\sigma$. The circle method of Section 10 uses at the frame $S$ the theta coefficients $h_{S,\ell}$ of $\psi_m^S = \psi_S - G_S\mathcal{A}$ (Lemma 9.10, with $G_S = d_1(m)/(16g(\sigma))$, Proposition 9.3(iii)), not those of $\varphi_m^F|_{-6,m}S$; the transformation of $\mathcal{A}$ under $S$, including the period integrals $h_{m,\ell}$, is accounted for by the completion $\widehat{\mathcal{A}}$ of Proposition 9.9 in the transformation law (T2b), Theorem 9.11. So the polar data at the frame $S$ are the coefficients $c_m^S$ of Theorem 8.1(ii), with no correction from the Appell–Lerch sum.

## 9. The transformation law of the finite parts

This section proves the theta decomposition of the finite parts in the three frames, the completion of the Appell–Lerch sum and the transformation law of the completed theta coefficients, together with the finiteness of the polar parts in every frame. These are the properties (T1)–(T3) on which the circle method of Section 10 runs; the values of the polar coefficients are those of Theorems 7.1 and 8.1. The modularity of $\varphi_m$ in all three frames is derived from the explicit formula of Proposition 7.13(b) and from Proposition 8.4(iii), without any statement about Siegel modular forms, and the completion is proved directly, by an explicit Mordell integral.

9.1. **Conventions and sources.** Powers are principal, as in Section 6.1. In the Jacobi variables $\tau = \sigma$, $z = v$ we write $q = \mathbf{e}(\tau)$, $y = \mathbf{e}(z)$, $x = \mathbf{e}(\tau/2)$. For an integer matrix $M = \left(\begin{smallmatrix} a & b \\ c & d\end{smallmatrix}\right)$ with $\det M = l \geq 1$, integers $k, m$ (or $k = 2$ for non-holomorphic $\varphi$) and a function $\varphi$ on $\mathbb{H}\times\mathbb{C}$ put

$$(\varphi\|_{k,m}M)(\tau,z) = (c\tau+d)^{-k}\,\mathbf{e}\Big(-\frac{mlcz^2}{c\tau+d}\Big)\,\varphi\Big(M\tau, \frac{lz}{c\tau+d}\Big),$$

and write $\varphi|_{k,m}\gamma = \varphi\|_{k,m}\gamma$ for $\gamma \in SL_2(\mathbb{Z})$; for $f$ a function of $\tau$ alone, $f|_k\gamma = f|_{k,0}\gamma$. For $M = M_{t,u,d} = \left(\begin{smallmatrix} t & u \\ 0 & d\end{smallmatrix}\right)$, $(\varphi\|_{k,m}M)(\tau,z) = d^{-k}\varphi((t\tau+u)/d, tz)$. The *elliptic law of index m* is $\varphi(\tau, z+\lambda\tau+\mu) = \mathbf{e}(-m(\lambda^2\tau+2\lambda z))\varphi(\tau,z)$ for $\lambda,\mu\in\mathbb{Z}$. Let $\Gamma^0(2)$, $R_I, R_T, R_S$ and the frames be as in Section 6.1; $R_T = T = \left(\begin{smallmatrix} 1 & 1 \\ 0 & 1\end{smallmatrix}\right)$, $R_S = S = \left(\begin{smallmatrix} 0 & -1 \\ 1 & 0\end{smallmatrix}\right)$.

**Lemma 9.1.** *Let $M_1, M_2$ be integer matrices with $\det M_i = l_i \geq 1$.*

(i) $\varphi\|_{k,m}(M_1M_2) = (\varphi\|_{k,m}M_1)\|_{k,ml_1}M_2$. *In particular* $|_{k,m}$ *is a right action of* $SL_2(\mathbb{Z})$, $-I$ *acts by* $\varphi(\tau,z)\mapsto(-1)^k\varphi(\tau,-z)$, *and for* $k=0$, $m=1$ *and* $\gamma\in SL_2(\mathbb{Z})$, $\varphi\|(\gamma M) = (\varphi|\gamma)\|M$ *and* $\varphi\|(MS) = (\varphi\|M)|_{0,l}S$.
(ii) $(\varphi_1\varphi_2)\|_{k_1+k_2,m_1+m_2}M = (\varphi_1\|_{k_1,m_1}M)(\varphi_2\|_{k_2,m_2}M)$.
(iii) *If* $\varphi$ *satisfies the elliptic law of index* $m$, *so does* $\varphi|_{k,m}\gamma$ *for every* $\gamma\in SL_2(\mathbb{Z})$; *if* $\varphi$ *is even in* $z$, *so is* $\varphi\|_{k,m}M$.

*Proof.* Write $M_i = \left(\begin{smallmatrix} a_i & b_i \\ c_i & d_i\end{smallmatrix}\right)$, $j_i(\tau) = c_i\tau + d_i$, and $M_1M_2 = \left(\begin{smallmatrix} a & b \\ c & d\end{smallmatrix}\right)$, so $c = c_1a_2 + d_1c_2$, and $j(\tau) = c\tau + d = j_1(M_2\tau)j_2(\tau)$, since $j_1(M_2\tau)j_2(\tau) = c_1(a_2\tau+b_2) + d_1(c_2\tau+d_2)$.

(i) By definition

$$\big((\varphi\|M_1)\|M_2\big)(\tau,z) = j_2^{-k}j_1(M_2\tau)^{-k}\,\mathbf{e}\Big(-\frac{ml_1l_2c_2z^2}{j_2} - \frac{ml_1c_1l_2^2z^2}{j_2^2j_1(M_2\tau)}\Big) \times \varphi\Big(M_1M_2\tau, \frac{l_1l_2z}{j_2j_1(M_2\tau)}\Big).$$

The weight factor is $j^{-k}$, the second argument is $l_1l_2z/j$, and the exponent is $-ml_1l_2z^2X/j$ with

$$X = c_2j_1(M_2\tau) + \frac{c_1l_2}{j_2} = \frac{c_1c_2(a_2\tau+b_2) + c_1(a_2d_2 - b_2c_2)}{j_2} + c_2d_1 = c_1a_2 + c_2d_1 = c.$$

This is $\varphi\|_{k,m}(M_1M_2)$. For $-I$: $c=0$, $d=-1$. The last two identities are the cases $M_1=\gamma$, $l_1=1$, and $M_2=S$, $l_2=1$.

(ii) The weight factors and the exponentials multiply, and the arguments agree.

(iii) Evenness is clear, as $z\mapsto -z$ commutes with $z\mapsto lz/(c\tau+d)$. Let $\gamma = \left(\begin{smallmatrix} a & b \\ c & d\end{smallmatrix}\right)$, $j = c\tau+d$, $\lambda,\mu\in\mathbb{Z}$, and $(\lambda',\mu') = (\lambda,\mu)\gamma^{-1}\in\mathbb{Z}^2$, i.e. $\lambda = \lambda'a+\mu'c$, $\mu = \lambda'b+\mu'd$. Put $w = z+\lambda\tau+\mu$. Then $w/j = z/j + \lambda'\gamma\tau + \mu'$, and expanding both sides (with $ad-bc=1$) gives the identity

$$\frac{cw^2}{j} + \lambda'^2\gamma\tau + \frac{2\lambda'z}{j} = \frac{cz^2}{j} + \lambda^2\tau + 2\lambda z + \big(ab\lambda'^2 + 2bc\lambda'\mu' + cd\mu'^2\big).$$

Hence, by the elliptic law of $\varphi$ at $(\gamma\tau, z/j)$,

$$(\varphi|\gamma)(\tau,w) = j^{-k}\mathbf{e}\Big(-\frac{mcw^2}{j}\Big)\mathbf{e}\Big(-m\Big(\lambda'^2\gamma\tau + \frac{2\lambda'z}{j}\Big)\Big)\varphi\Big(\gamma\tau,\frac{z}{j}\Big) = \mathbf{e}\big(-m(\lambda^2\tau+2\lambda z)\big)(\varphi|\gamma)(\tau,z),$$

the bracket $ab\lambda'^2 + 2bc\lambda'\mu' + cd\mu'^2$ being an integer. □

With $k=0$ and $m=1$, $\|M$ is the operator $F\mapsto F|[M]$ of Section 8, and (i) contains the two identities $F|[\gamma M] = (F|\gamma)|[M]$ and $F|[MS] = (F|[M])|_{0,l}S$ used there in the proof of Lemma 8.3.

Besides [17] and [1], the proofs below rest on three groups of facts.

(P) From Section 2: the genera (2), the products $g$ and (1), the function $\varphi_{-1} = 1/(g(\sigma/2)A)$ of (5), the functions $L_l$ defined by the second line of (6), Lemma 2.1, and the polar part (8). By (2) and [17, Theorem 9.3], each $F^{(r,s)}$ is even in $z$, satisfies the elliptic law of index one, and is a series $\sum_{n\in\frac12\mathbb{Z}_{\geq 0}}\sum_j c(n,j)q^n\zeta^j$ ($\mathfrak{E}(\tau/2)$ and $\mathfrak{E}((\tau+1)/2)$ are series in $q^{1/2}$ without negative powers); $\phi_{-2,1}$ is a weak Jacobi form of weight $-2$ and index one for $SL_2(\mathbb{Z})$.
(Q) From Section 7: Proposition 7.13(b)–(d).

(R) From Section 8: Lemma 8.2 ($F^{(r,s)}|\gamma = F^{(r,s)\gamma}$), and Proposition 8.4(iii) with the function $\varphi'_m$ of Proposition 8.4(ii); its proof uses Lemmas 8.2, 8.3 and 2.1, $\phi_{-2,1}|_{-2,1}S = \phi_{-2,1}$, $g(-1/2\sigma) = 16\sigma^8 g(\sigma)$ and Lemma 9.1.

No statement about Siegel modular forms is used, and the product (3) enters only through (Q), (R).

9.2. **The functions $\psi_m$ and the three frames.** Following Proposition 7.13(b) define, on $\mathbb{H}\times\mathbb{C}$,

(43)
$$\psi_m = \varphi_{-1}\,P_m(L_1,\dots,L_{m+1}) = \frac{B_{m+1}}{g(\sigma/2)\,\phi_{-2,1}}, \qquad B_{m+1} = \sum_{k\ge1}\frac{1}{k!}\sum_{l_1+\cdots+l_k=m+1} L_{l_1}\cdots L_{l_k},$$

where $\exp(\sum_{l\ge1} L_l X^l) = \sum_{k\ge0} P_{k-1}X^k$, so that $P_m = B_{m+1}$. By Proposition 7.13(b), $\psi_m$ is the coefficient of $\mathbf{e}(m\rho)$ in the expansion of $1/\widetilde{\Phi}_6$ in the Fourier–Jacobi region, i.e. the function $\varphi_m$ of Section 2; we write $\psi_m$ to avoid a clash with the frame index. We write $\mathcal{A} = \mathcal{A}_{2,m}$ for the Appell–Lerch sum of (8); note that $\phi_{-2,1}$ (denoted $A$ in Section 2) and $\mathcal{A}$ are different functions. Finally put

$$\varphi'_m = \frac{P_m(L'_1,\dots,L'_{m+1})}{g(\sigma)\,\phi_{-2,1}}, \qquad L'_l = \sum_{td=l}\frac{1}{2dt}\sum_{r,s\in\{0,1\}}(-1)^{rt}\sum_{u=0}^{2d-1}F^{(r,s)}\Big(\frac{t\sigma+u}{d},tv\Big),$$

as in Lemma 8.3 and Proposition 8.4(ii).

**Lemma 9.2.** *$\Gamma^0(2) = \{\left(\begin{smallmatrix}a&b\\c&d\end{smallmatrix}\right) \in SL_2(\mathbb{Z}) : b\equiv 0 \bmod 2\}$ is generated by $T^2$, $U = \left(\begin{smallmatrix}1&0\\1&1\end{smallmatrix}\right)$ and $-I$.*

*Proof.* The three elements lie in $\Gamma^0(2)$. Let $\gamma\in\Gamma^0(2)$ with first row $(a,b)$; $b$ is even, so $a$ is odd. Right multiplication by $T^{2k}$ replaces $b$ by $b+2ka$, and by $U^k$ replaces $a$ by $a+kb$; both preserve $\Gamma^0(2)$. If $b\neq0$ and $|b|>|a|$, choose $k$ with $|b+2ka|\le|a|$; as $b+2ka$ is even and $a$ odd, $|b+2ka|<|a|$. If $0<|b|<|a|$, choose $k$ with $|a+kb|\le|b|/2<|b|$; $a+kb$ stays odd. So $\max(|a|,|b|)$ decreases strictly until $b=0$; then $a=\pm1$, and the matrix is $\pm\left(\begin{smallmatrix}1&0\\c'&1\end{smallmatrix}\right) = \pm U^{c'}$. Hence $\gamma\in\langle T^2,U,-I\rangle$. □

**Proposition 9.3.** (i) *$\psi_m$ is meromorphic on $\mathbb{H}\times\mathbb{C}$, holomorphic off $\mathbb{Z}\sigma+\mathbb{Z}$, with at most double poles on $v\in\mathbb{Z}\sigma+\mathbb{Z}$; it is even in $v$, satisfies the elliptic law of index $m$, and $\psi_m|_{-6,m}\gamma = \psi_m$ for $\gamma\in\Gamma^0(2)$.*

(ii) *Near $v=0$, $\psi_m(\sigma,v) = G_I(\sigma)\,(2\pi iv)^{-2} + O(1)$ with $G_I(\sigma) = \hat f(m)/g(\sigma/2)$.*

(iii) *Put $\psi_F = \psi_m|_{-6,m}R_F$ ($F\in\{I,T,S\}$). Then $\psi_F$ has the properties in* (i) *(with $\Gamma^0(2)$ replaced by $R_F^{-1}\Gamma^0(2)R_F$), $\psi_S = \frac{1}{16}\varphi'_m$, and $\psi_F(\sigma,v) = G_F(\sigma)(2\pi iv)^{-2}+O(1)$ near $v=0$, where $G_F = G_I|_{-8}R_F$:*

$$G_T(\tau) = G_I(\tau+1), \qquad G_S(\tau) = \tau^8 G_I(-1/\tau) = \frac{\hat f(m)}{16\,g(\tau)}.$$

(iv) *For $N = N_{\gamma,\delta}$ and $F = F(\gamma,\delta)$, $\psi_m = \psi_F|_{-6,m}N^{-1}$, and $G_I(N\tau) = j(\tau)^{-8}G_F(\tau)$; this is property* (T2a) *of Section 10.1, with $\lambda_{\gamma,\delta}=1$.*

*Proof.* (i) Meromorphy, the location and order of the poles, evenness and the elliptic law are Proposition 7.13(c): the $L_l$ are holomorphic, even and of index $l$ (Lemma 2.1), $P_m$ is weighted homogeneous of weight $m+1$, $\phi_{-2,1} = -\theta_1^2/\eta^6$ has index one and its zeros are double zeros along $\mathbb{Z}\sigma+\mathbb{Z}$, and $g(\sigma/2)$ has no zeros on $\mathbb{H}$. By Lemma 9.1(i) the set of $\gamma$ with $\psi_m|_{-6,m}\gamma=\psi_m$ is a group, so by Lemma 9.2 it suffices to treat $-I$, $T^2$ and $U^{-1}$.

$-I$: $\psi_m|_{-6,m}(-I)(\sigma,v) = (-1)^{-6}\psi_m(\sigma,-v) = \psi_m(\sigma,v)$ by evenness.

$T^2$: $\psi_m(\sigma+2,v)=\psi_m(\sigma,v)$. Indeed $g(\sigma/2)=x\prod(1-x^n)^8(1-x^{2n})^8$ and $\phi_{-2,1}$ have period 2, and in the term $(t,d,u)$ of (6), with the sign $(-1)^{sd+ru}$, the substitution $\sigma\mapsto\sigma+2$ replaces $u$ by $u+2t$, which permutes the residues modulo $2d$ and multiplies the sign by $(-1)^{2rt}=1$; so each $L_l$ has period 2.

$U^{-1}$: by Proposition 8.4(iii), $\psi_m|_{-6,m}S=\frac{1}{16}\varphi'_m$. The function $\varphi'_m$ has period 1 in $\sigma$: $g(\sigma)$ and $\phi_{-2,1}$ have period 1, and in the term $(t,d)$ of $L'_l$, $\sigma\mapsto\sigma+1$ replaces $u$ by $u+t$, which permutes the residues modulo $2d$, while the sign $(-1)^{rt}$ does not depend on $u$; here $F^{(r,s)}$ has period 2 in $\tau$, by (P). Hence $(\psi_m|S)|T=\psi_m|S$, i.e. $\psi_m|STS^{-1}=\psi_m$, and $STS^{-1}=\left(\begin{smallmatrix}1&0\\-1&1\end{smallmatrix}\right)=U^{-1}$.

(ii) By Proposition 7.13(d), $\psi_m=-d_1(m)/(4\pi^2g(\sigma/2)v^2)+O(1)$, and $d_1(m)=\hat f(m)$ by (21); as $(2\pi iv)^{-2}=-1/(4\pi^2v^2)$, this is (ii).

(iii) Slashing preserves evenness, the elliptic law (Lemma 9.1(iii)) and meromorphy, and maps the lattice $\mathbb{Z}\sigma+\mathbb{Z}$ to itself; since $|_{-6,m}$ is an action, $\psi_F|_{-6,m}R_F^{-1}hR_F=\psi_F$ for $h\in\Gamma^0(2)$. $\psi_S=\frac{1}{16}\varphi'_m$ is Proposition 8.4(iii). Near $v=0$, for $\gamma=\left(\begin{smallmatrix}a&b\\c&d\end{smallmatrix}\right)$, $(\psi_m|_{-6,m}\gamma)(\sigma,v)=(c\sigma+d)^6(1+O(v^2))\big[G_I(\gamma\sigma)(2\pi iv)^{-2}(c\sigma+d)^2+O(1)\big]$, so the coefficient is $(c\sigma+d)^8G_I(\gamma\sigma)=(G_I|_{-8}\gamma)(\sigma)$, and evenness excludes a $v^{-1}$ term. For $R_T$ this is $G_I(\tau+1)$; for $R_S$ it is $\tau^8\hat f(m)/g(-1/2\tau)$, and $\eta(-1/w)=(w/i)^{1/2}\eta(w)$ gives $g(-1/2\tau)=\eta(-1/2\tau)^8\eta(-1/\tau)^8=(2\tau/i)^4\eta(2\tau)^8(\tau/i)^4\eta(\tau)^8=16\tau^8g(\tau)$.

(iv) $NR_F^{-1}\in\Gamma^0(2)$, so $\psi_m|N=\psi_m|R_F=\psi_F$ and $\psi_m=\psi_F|N^{-1}$. By (i) and the computation in (iii), $G_I|_{-8}h=G_I$ for $h\in\Gamma^0(2)$, so $G_I|_{-8}N=G_I|_{-8}R_F=G_F$, i.e. $G_F(\tau)=j(\tau)^8G_I(N\tau)$ ($j(\tau)=-\gamma\tau+\alpha$ is the automorphy factor of $N$). □

**Remark 9.4.** $1/g(\tau/2)=\sum_{j\ge0}\beta(j)x^{j-1}$ by (1), with $\beta(0)=1$. Hence $G_I=\hat f(m)\mathbf{e}(-\tau/2)+\sum_{j\ge1}\hat f(m)\beta(j)\mathbf{e}((j-1)\tau/2)$, $G_T(\tau)=G_I(\tau+1)$ has the coefficients multiplied by $(-1)^{j-1}$, and $G_S=\frac{\hat f(m)}{16}q^{-1}\prod(1-q^n)^{-8}(1-q^{2n})^{-8}$. So $G_F=c_F\mathbf{e}(-\mu_F\tau)+\cdots$ with $c_I=\hat f(m)$, $\mu_I=\frac12$; $c_T=-\hat f(m)$, $\mu_T=\frac12$; $c_S=\hat f(m)/16$, $\mu_S=1$; the remaining exponents lie in $\frac12\mathbb{Z}_{\ge0}$, $\frac12\mathbb{Z}_{\ge0}$, $\mathbb{Z}_{\ge0}$, and the series converge absolutely on $\mathbb{H}$. None of the $G_F$ vanishes identically, and $g(\tau/2)$ has weight 8 and trivial character on $\Gamma^0(2)$ by Proposition 9.3(iii) applied to $h\in\Gamma^0(2)$.

9.3. **The Appell–Lerch sum and its completion.** Recall

$$\mathcal{A}(\tau,z)=\sum_{s\in\mathbb{Z}}q^{ms^2+s}y^{2ms+1}(1-q^sy)^{-2}.$$

This series converges absolutely and locally uniformly off its poles, which are double, at $z\in\mathbb{Z}\tau+\mathbb{Z}$.

**Lemma 9.5.** *$\mathcal{A}$ is even in $z$, $\mathcal{A}(\tau+1,z)=\mathcal{A}(\tau,z)$, it satisfies the elliptic law of index $m$, and $\mathcal{A}(\tau,z)=(2\pi iz)^{-2}+O(1)$ near $z=0$.*

*Proof.* Replacing $z$ by $-z$ in the term $s$ gives $q^{ms^2+s-2s}y^{1-2ms}(1-q^{-s}y)^{-2}$, the term $-s$. The exponents $ms^2+s$ are integers. Under $z\mapsto z+\tau$ the term $s$ becomes $p_s=q^{ms^2+s+2ms+1}y^{2ms+1}(1-q^{s+1}y)^{-2}$, and $ms^2+s+2ms+1=m(s+1)^2+(s+1)-m$, so $\mathcal{A}(\tau,z+\tau)=q^{-m}y^{-2m}\mathcal{A}(\tau,z)$; $z\mapsto z+1$ is clear. Near $z=0$ only the term $s=0$ is singular, and $y(1-y)^{-2}=(4\sinh^2(\pi iz))^{-1}=(2\pi iz)^{-2}+O(1)$. □

**Lemma 9.6** (Fourier expansion in a strip)**.** *For $-\operatorname{Im}\tau<\operatorname{Im}z<0$ $\mathcal{A}(\tau,z)$ is the absolutely convergent series $\sum_{r\in\mathbb{Z}}q^{r^2/4m}a(r)\,y^r$, where $a(0)=0$ and*

$$a(r)=\sum_{\substack{0<n<r\\ n\equiv r\,(2m)}}n\,q^{-n^2/4m}\quad(r>0),\qquad a(r)=-\sum_{\substack{r\le n<0\\ n\equiv r\,(2m)}}n\,q^{-n^2/4m}\quad(r<0).$$

*Proof.* In the strip $|q^s y| < 1$ for $s \geq 1$ and $|q^s y| > 1$ for $s \leq 0$. Expand $(1-w)^{-2} = \sum_{k\geq 0}(k+1)w^k$ for $s \geq 1$ and $(1-w)^{-2} = \sum_{k\geq 0}(k+1)w^{-k-2}$ for $s \leq 0$; the resulting double series converge absolutely. For $s \geq 1$ the coefficient of $y^r$ is $(r-2ms)q^{rs-ms^2}$ when $r \geq 2ms+1$; for $s \leq 0$ it is $(2ms-r)q^{rs-ms^2}$ when $r \leq 2ms-1$. With $n = r-2ms$ one has $rs - ms^2 = (r^2-n^2)/4m$, and the conditions become $0 < n < r$, resp. $r \leq n < 0$, with $n \equiv r \bmod 2m$. □

For $0 \leq \ell < 2m$ put $\vartheta_{m,\ell}(\tau, z) = \sum_{r\equiv \ell\,(2m)} \mathbf{e}(r^2\tau/4m + rz)$ (as in Section 6.1) and $\vartheta_\ell(\tau) = \vartheta_{m,\ell}(\tau, 0)$, and

$$\tilde{P}_\ell(\tau) = \int_0^{i\infty} \vartheta_\ell(u)(u+\tau)^{-3/2} du \qquad (\tau \in \mathbb{H}),$$

along the imaginary axis; it converges absolutely ($|\vartheta_\ell(it)| \leq Ct^{-1/2}$ for $t \leq 1$ by Lemma 9.7 below with $z = 0$, and $\vartheta_\ell(it)$ is bounded for $t \geq 1$) and is holomorphic in $\tau$. The same bound holds in every sector $|\arg u - \frac{\pi}{2}| \leq \frac{\pi}{2} - \epsilon$, so the path may be replaced by any path to 0 inside such a sector.

**Lemma 9.7** (Theta transformation)**.** *Let $M = \left(\begin{smallmatrix}\alpha & \beta\\ \gamma & \delta\end{smallmatrix}\right) \in SL_2(\mathbb{Z})$ with $\gamma \geq 1$ and $\alpha$ as in Section 6.1 (so $M = M_{\gamma,\delta}$), and $J = \gamma\tau + \delta$. Then for $0 \leq \tilde{\ell} < 2m$*

$$\vartheta_{m,\tilde{\ell}}\Big(M\tau, \frac{z}{J}\Big) = J^{1/2}\mathbf{e}\Big(\frac{m\gamma z^2}{J}\Big) \sum_{\ell=0}^{2m-1} W_{\gamma,\delta}(\tilde{\ell}, \ell)\, \vartheta_{m,\ell}(\tau, z),$$

*with $W_{\gamma,\delta}$ as in (24). In particular ($\gamma = 1$, $\delta = 0$)*

$$\vartheta_{m,\ell}(-1/\tau, z/\tau) = (\tau/2mi)^{1/2}\mathbf{e}(mz^2/\tau) \sum_{\ell'} \mathbf{e}(-\ell\ell'/2m)\vartheta_{m,\ell'}(\tau, z).$$

*Proof.* $M\tau = \frac{\alpha}{\gamma} - \frac{1}{\gamma J}$. Write $r \equiv \tilde{\ell} \bmod 2m$ as $r = x + 2m\gamma p$ with $x = \tilde{\ell} - 2mT$, $0 \leq T < \gamma$, $p \in \mathbb{Z}$. Then $\mathbf{e}(\alpha r^2/4m\gamma) = \mathbf{e}(\alpha x^2/4m\gamma)$, and

$$\vartheta_{m,\tilde{\ell}}(M\tau, z/J) = \sum_T \mathbf{e}\Big(\frac{\alpha x^2}{4m\gamma}\Big) \sum_p \Phi(x + 2m\gamma p), \qquad \Phi(s) = \mathbf{e}\Big(-\frac{s^2}{4m\gamma J} + \frac{sz}{J}\Big).$$

Since $\mathrm{Im}(-1/J) > 0$, $\Phi$ is a Schwartz function, with

$$\hat{\Phi}(\xi) = \int \Phi(s)\mathbf{e}(-s\xi)\, ds = (2m\gamma J/i)^{1/2}\, \mathbf{e}\big(m\gamma(z - \xi J)^2/J\big)$$

(the Gaussian integral $\int \exp(\pi i A s^2 + 2\pi i B s) ds = (-iA)^{-1/2}\exp(-\pi i B^2/A)$, $\mathrm{Im}\, A > 0$). Poisson summation,

$$\sum_p \Phi(x + 2m\gamma p) = \frac{1}{2m\gamma}\sum_k \hat{\Phi}\Big(\frac{k}{2m\gamma}\Big)\mathbf{e}\Big(\frac{kx}{2m\gamma}\Big), \qquad \frac{m\gamma}{J}\Big(z - \frac{kJ}{2m\gamma}\Big)^2 = \frac{m\gamma z^2}{J} - kz + \frac{k^2\tau}{4m} + \frac{\delta k^2}{4m\gamma},$$

and $k \mapsto -k$ give

$$\vartheta_{m,\tilde{\ell}}(M\tau, z/J) = \frac{(J/i)^{1/2}}{\sqrt{2m\gamma}}\mathbf{e}\Big(\frac{m\gamma z^2}{J}\Big) \sum_{k\in\mathbb{Z}} \Big[\sum_T \mathcal{S}(x,k)\Big]\mathbf{e}\Big(\frac{k^2\tau}{4m} + kz\Big),$$

where $\mathcal{S}(x,k) = \mathbf{e}\big((\alpha x^2 - 2kx + \delta k^2)/4m\gamma\big)$, all sums converging absolutely. $\mathcal{S}(x,k)$ depends on $x$ only modulo $2m\gamma$, and $\mathcal{S}(x + 2m\delta, k + 2m) = \mathcal{S}(x,k)$ because the difference of the arguments is $4m(\alpha\delta - 1)(x + m\delta)/4m\gamma = \beta(x + m\delta) \in \mathbb{Z}$; as $x \mapsto x + 2m\delta$ permutes the classes $\tilde{\ell} - 2mT$

modulo $2m\gamma$, $\sum_T \mathcal{S}(x,k)$ depends only on $k$ mod $2m$. Grouping $k \equiv \ell \bmod 2m$ gives the claim, since $(J/i)^{1/2}(2m\gamma)^{-1/2} = J^{1/2}(2m\gamma i)^{-1/2}$ for $J \in \mathbb{H}$. □

**Lemma 9.8** (Mordell integral)**.** *Let* $\gamma \geq 1$, $g \in \mathbb{Z}$, *and* $\theta_{\gamma,g}(t) = \sum_{p\in\mathbb{Z}} \exp(-\pi(g + 2m\gamma p)^2 t/2m)$ *for* $t > 0$. *For* $\operatorname{Re} X > 0$,

$$\int_0^\infty \theta_{\gamma,g}(t)\,(t+X)^{-3/2}\,dt = \frac{2\,\mathbf{1}_{2m\gamma|g}}{\sqrt{X}} - \frac{1}{\sqrt{2m}\,\pi\gamma^2 X}\int_{\mathbb{R}} e^{-2\pi m X u^2} f_{\gamma,g}(u)\,du,$$

*all integrals converging absolutely.*

*Proof.* Write $a_p = g + 2m\gamma p$ and $\lambda_p = \pi a_p^2/2m$. If $a_{p_0} = 0$ for some (unique) $p_0$, which happens exactly when $2m\gamma \mid g$, its term contributes $\int_0^\infty (t+X)^{-3/2}dt = 2X^{-1/2}$. For the terms with $a_p \neq 0$ we use, for $\operatorname{Re} s > 0$, $\int_{\mathbb{R}} v^2 e^{-sv^2}dv = \frac{\sqrt{\pi}}{2}s^{-3/2}$ (true for $s > 0$, and both sides are holomorphic), with $s = t + X$:

$$\begin{aligned}\int_0^\infty e^{-\lambda_p t}(t+X)^{-3/2}dt &= \frac{2}{\sqrt{\pi}}\int_{\mathbb{R}} \frac{v^2 e^{-Xv^2}}{\lambda_p + v^2}dv \\ &= 2\sqrt{2m}\int_{\mathbb{R}} e^{-2\pi m X u^2}\frac{4m^2u^2}{a_p^2 + 4m^2u^2}du,\end{aligned}$$

by Fubini (the double integral of $v^2 e^{-t(\lambda_p+v^2)-v^2\operatorname{Re}X}$ is finite) and $v = \sqrt{2\pi m}\,u$. The last integral is at most $\frac{4m^2}{a_p^2}\int u^2 e^{-2\pi m u^2 \operatorname{Re}X}du$, which is summable over $p$; so summation over $p$ and integration over $t$ may be interchanged (also $\sum_p \int_0^\infty e^{-\lambda_p t}|t+X|^{-3/2}dt < \infty$). Now integrate by parts, using $ue^{-2\pi m X u^2} = -\frac{1}{4\pi m X}\frac{d}{du}e^{-2\pi m X u^2}$ and the vanishing at $\pm\infty$ of $\frac{4m^2u}{a_p^2+4m^2u^2}$:

$$\int_{\mathbb{R}} e^{-2\pi m X u^2}\frac{4m^2u^2}{a_p^2+4m^2u^2}du = \frac{1}{4\pi m X}\int_{\mathbb{R}} e^{-2\pi m X u^2}\frac{d}{du}\frac{4m^2u}{a_p^2+4m^2u^2}du,$$

$$\frac{d}{du}\frac{4m^2u}{a_p^2+4m^2u^2} = 2m^2\Big(\frac{1}{(a_p+2miu)^2} + \frac{1}{(a_p-2miu)^2}\Big).$$

The last expression is bounded by $4m^2/a_p^2$, which is summable, so we may sum over $p$ under the integral. With $\pi^2/\sin^2(\pi w) = \sum_{p\in\mathbb{Z}}(w+p)^{-2}$ for $w \notin \mathbb{Z}$ [1, Chapter 5, §2.1], $a_p \pm 2miu = 2m\gamma(p + \frac{g}{2m\gamma} \pm \frac{iu}{\gamma})$, and $\sin(\pi(c \pm \frac{iu}{\gamma})) = \pm i\sinh(\frac{\pi u}{\gamma} \mp \pi i c)$, one finds, for $2m\gamma \nmid g$ and $c = g/2m\gamma$,

$$\sum_p \frac{1}{(a_p \pm 2miu)^2} = -\frac{1}{4m^2\gamma^2}\,f_{\gamma,g}(\pm u),$$

and for $2m\gamma \mid g$, summing over $p \neq p_0$ and subtracting the term $p' = 0$ from $\sum_{p'}(p' \pm iu/\gamma)^{-2}$, the same identity with $f_{\gamma,0}$ (even) for $u \neq 0$, and by continuity at $u = 0$. Hence in both cases

$$\begin{aligned}\sum_{a_p\neq 0}\int_0^\infty e^{-\lambda_p t}(t+X)^{-3/2}dt &= \frac{2\sqrt{2m}}{4\pi m X}\int_{\mathbb{R}} e^{-2\pi m X u^2}\cdot 2m^2\cdot\Big(-\frac{f_{\gamma,g}(u)+f_{\gamma,g}(-u)}{4m^2\gamma^2}\Big)du \\ &= -\frac{1}{\sqrt{2m}\,\pi\gamma^2 X}\int_{\mathbb{R}} e^{-2\pi m X u^2} f_{\gamma,g}(u)\,du,\end{aligned}$$

using that $e^{-2\pi m X u^2}$ is even. Absolute convergence holds because $f_{\gamma,g}$ is bounded, by (33), and $\operatorname{Re} X > 0$. □

For $0 \le \ell < 2m$ and $\tau \in \mathbb{H}$ put

$$\kappa_m = -\sqrt{\frac{m}{4\pi}}\Big(\frac{i}{2\pi}\Big)^{1/2} = -\frac{e^{\pi i/4}\sqrt{m}}{2\sqrt{2}\,\pi}, \qquad E_\ell(\tau) = \int_{-\bar\tau}^{i\infty} \vartheta_\ell(w)\,(w+\tau)^{-3/2}\,dw, \tag{44}$$

the integral along the vertical ray. Integrating term by term along $w = -\bar\tau + it$, $t \ge 0$, and writing $\Gamma(s,x) = \int_x^\infty e^{-t}t^{s-1}\,dt$,

$$E_\ell(\tau) = e^{-\pi i/4}\Big(\frac{2\,\delta_{\ell\equiv 0\,(2m)}}{\sqrt{2\tau_2}} + \sum_{\substack{r\equiv \ell\,(2m)\\ r\neq 0}} \sqrt{\frac{\pi r^2}{2m}}\,\Gamma\Big(-\frac{1}{2}, \frac{\pi r^2\tau_2}{m}\Big)\,\mathbf{e}\Big(-\frac{r^2\tau}{4m}\Big)\Big),$$

so $E_\ell$ is a non-holomorphic Eichler integral of $\vartheta_\ell$; the constant $\kappa_m$ is the one forced by the invariance in the next proposition.

**Proposition 9.9** (Completion of $\mathcal{A}$)**.** *With $\kappa_m$ and $E_\ell$ as in* (44)*, put*

$$\widehat{\mathcal{A}}(\tau,z) = \mathcal{A}(\tau,z) - \kappa_m \sum_{\ell=0}^{2m-1} E_\ell(\tau)\,\vartheta_{m,\ell}(\tau,z).$$

*Then $\widehat{\mathcal{A}}|_{2,m}\gamma = \widehat{\mathcal{A}}$ for every $\gamma \in SL_2(\mathbb{Z})$, and $\widehat{\mathcal{A}}$ satisfies the elliptic law of index $m$.*

*Proof.* The elliptic law holds for $\mathcal{A}$ (Lemma 9.5) and for each $\vartheta_{m,\ell}$. Since $SL_2(\mathbb{Z})$ is generated by $T = \left(\begin{smallmatrix}1&1\\0&1\end{smallmatrix}\right)$ and $S$, it suffices to treat these. Put $\mathcal{R} = \sum_\ell E_\ell\vartheta_{m,\ell}$.

*$T$.* $\mathcal{A}(\tau+1,z) = \mathcal{A}(\tau,z)$. Substituting $w \mapsto w-1$ in $E_\ell(\tau+1) = \int_{-\bar\tau-1}^{i\infty}\vartheta_\ell(w)(w+\tau+1)^{-3/2}dw$ and using $\vartheta_\ell(w-1) = \mathbf{e}(-\ell^2/4m)\vartheta_\ell(w)$ gives $E_\ell(\tau+1) = \mathbf{e}(-\ell^2/4m)E_\ell(\tau)$, while $\vartheta_{m,\ell}(\tau+1,z) = \mathbf{e}(\ell^2/4m)\vartheta_{m,\ell}(\tau,z)$; so $\mathcal{R}(\tau+1,z) = \mathcal{R}(\tau,z)$.

*$S$, the correction term.* We show $\mathcal{R}|_{2,m}S = \mathcal{R} - \sum_\ell \tilde P_\ell\vartheta_{m,\ell}$. By Lemma 9.7,

$$(\mathcal{R}|_{2,m}S)(\tau,z) = \tau^{-3/2}\sum_{\ell'}\Big[\sum_{\ell} W_{1,0}(\ell,\ell')E_\ell(-1/\tau)\Big]\vartheta_{m,\ell'}(\tau,z) \qquad (\tau^{-2}\tau^{1/2} = \tau^{-3/2}).$$

In $E_\ell(-1/\tau) = \int_{1/\bar\tau}^{i\infty}\vartheta_\ell(w)(w-1/\tau)^{-3/2}dw$ substitute $w = -1/u$: $u$ runs from $-\bar\tau$ to $0$ (within a sector as above), $dw = u^{-2}du$, $\vartheta_\ell(-1/u) = (u/2mi)^{1/2}\sum_{\ell''}\mathbf{e}(-\ell\ell''/2m)\vartheta_{\ell''}(u)$, and

$$\Big(-\frac{1}{u}-\frac{1}{\tau}\Big)^{-3/2} = \Big(-\frac{u+\tau}{u\tau}\Big)^{-3/2} = e^{-3\pi i/2}(u+\tau)^{-3/2}u^{3/2}\tau^{3/2} = i\,(u+\tau)^{-3/2}u^{3/2}\tau^{3/2}$$

(the arguments satisfy $\operatorname{Arg}(-\frac{u+\tau}{u\tau}) = \operatorname{Arg}(u+\tau) - \operatorname{Arg}u - \operatorname{Arg}\tau + \pi$: both sides are continuous on $\mathbb{H}\times\mathbb{H}$, congruent modulo $2\pi$, and agree at $u=\tau=i$). With $(u/2mi)^{1/2} = (2m)^{-1/2}e^{-\pi i/4}u^{1/2}$ and $u^{1/2}u^{3/2}u^{-2} = 1$,

$$E_\ell(-1/\tau) = \frac{e^{\pi i/4}\tau^{3/2}}{\sqrt{2m}}\sum_{\ell''}\mathbf{e}\Big(-\frac{\ell\ell''}{2m}\Big)\int_{-\bar\tau}^{0}\vartheta_{\ell''}(u)(u+\tau)^{-3/2}du.$$

Since $W_{1,0}(\ell,\ell') = (2m)^{-1/2}e^{-\pi i/4}\mathbf{e}(-\ell\ell'/2m)$, $\sum_\ell \mathbf{e}(-\ell(\ell'+\ell'')/2m) = 2m\,\mathbf{1}_{\ell''\equiv-\ell'}$ and $\vartheta_{-\ell'} = \vartheta_{\ell'}$,

$$\tau^{-3/2}\sum_\ell W_{1,0}(\ell,\ell')E_\ell(-1/\tau) = \int_{-\bar\tau}^{0}\vartheta_{\ell'}(u)(u+\tau)^{-3/2}du = E_{\ell'}(\tau) - \tilde P_{\ell'}(\tau).$$

*$S$, the sum $\mathcal{A}$.* Let $\mathcal{D} = \mathcal{A}|_{2,m}S - \mathcal{A}$. Both terms are even, satisfy the elliptic law of index $m$, are holomorphic off $\mathbb{Z}\tau+\mathbb{Z}$ and equal $(2\pi iz)^{-2} + O(1)$ at $z=0$ (for $\mathcal{A}|_{2,m}S$: $\tau^{-2}\mathbf{e}(-mz^2/\tau)(2\pi iz/\tau)^{-2}(1+O(z^2))$). So $\mathcal{D}$ is holomorphic at 0, hence on $\mathbb{H}\times\mathbb{C}$ by the

elliptic law, and $\mathcal{D} = \sum_\ell d_\ell(\tau)\vartheta_{m,\ell}(\tau,z)$ with $d_\ell$ holomorphic [17, Theorem 5.1] (the proof there uses only the elliptic law and holomorphy). We show $d_\ell = c_\ell$, $c_\ell := \frac{\sqrt{2m}}{4\pi} e^{\pi i/4}\tilde{P}_\ell$.

*Unfolding.* With $\mathbf{e}(-mz^2/\tau)\, q'^{ms^2+s} y'^{2ms+1} = \mathbf{e}\big((-m(z-s)^2+(z-s))/\tau\big)$ for $q' = \mathbf{e}(-1/\tau)$, $y' = \mathbf{e}(z/\tau)$, and $\mathbf{e}(u)(1-\mathbf{e}(u))^{-2} = -(4\sin^2\pi u)^{-1} = -\frac{1}{4\pi^2}\sum_k (u-k)^{-2}$,

$$(\mathcal{A}|_{2,m}S)(\tau,z) = \sum_{s\in\mathbb{Z}} \Phi(z-s), \qquad \Phi(w) = \frac{1}{(2\pi i)^2}\mathbf{e}\Big(-\frac{mw^2}{\tau}\Big)\sum_{k\in\mathbb{Z}}(w-k\tau)^{-2}.$$

On every horizontal line, $|\mathbf{e}(-mw^2/\tau)| = \exp(-2\pi m(\operatorname{Re} w)^2\tau_2/|\tau|^2 + O(|\operatorname{Re} w|))$, and $\sum_k (w-k\tau)^{-2}$ is bounded on the line away from its poles $w \in \mathbb{Z}\tau$. So for $-\tau_2 < c < 0$ the Fourier coefficient of $\mathcal{A}|_{2,m}S$ on $\operatorname{Im} z = c$ is

$$g_r = \int_{\operatorname{Im} z = c} \Phi(z)\mathbf{e}(-rz)\, dz = \frac{q^{r^2/4m}}{(2\pi i)^2}\sum_k K_{c'}(\nu_k), \qquad \nu_k = k + \frac{r}{2m},\ c' = c + \frac{r\tau_2}{2m},$$

where $K_h(\nu) = \int_{\operatorname{Im} w = h} \mathbf{e}(-mw^2/\tau)(w-\nu\tau)^{-2}dw$ for $h \neq \nu\tau_2$; we completed the square, $-mz^2/\tau - rz = -m(z+r\tau/2m)^2/\tau + r^2\tau/4m$, substituted $w = z + r\tau/2m$, and exchanged $\sum_k$ and $\int$ (uniform convergence on the line, Gaussian decay). The line $\operatorname{Im} w = c'$ meets no pole, as $c/\tau_2 \notin \mathbb{Z}$.

*Moving the line.* Let $h_0 = 0$ if $2m \nmid r$ and $h_0 = -\epsilon$ with $0 < \epsilon < \tau_2$ if $2m \mid r$; no pole $\nu_k\tau$ lies on $\operatorname{Im} w = h_0$. The residue of $\mathbf{e}(-mw^2/\tau)(w-\nu\tau)^{-2}$ at $\nu\tau$ is $-4\pi i m\nu\, q^{-m\nu^2}$, and the Gaussian decay allows moving lines, so $K_{c'}(\nu) = K_{h_0}(\nu) - 8\pi^2 m\nu q^{-m\nu^2}$ if $h_0 < \nu\tau_2 < c'$, $K_{c'}(\nu) = K_{h_0}(\nu) + 8\pi^2 m\nu q^{-m\nu^2}$ if $c' < \nu\tau_2 < h_0$, and $K_{c'} = K_{h_0}$ otherwise. With $n = 2m\nu_k \equiv r \bmod 2m$ and $2mc/\tau_2 \in (-2m, 0)$, the first case is $0 \le n < r$ and the second is $r \le n < 0$ (the term $n = 0$ has residue 0). Each crossed pole contributes $\mp 4\pi^2 n q^{-n^2/4m}$ to $\sum_k K_{c'}(\nu_k)$, i.e. $\pm n q^{-n^2/4m}$ to $g_r q^{-r^2/4m}$; by Lemma 9.6 these contributions add up to exactly $a(r)$. Since $\mathcal{D}$ has Fourier coefficients $g_r - q^{r^2/4m}a(r) = q^{r^2/4m}d_\ell$ $(r \equiv \ell)$,

$$d_\ell(\tau) = -\frac{1}{4\pi^2}\sum_k K_{h_0}\Big(k + \frac{r}{2m}\Big) = -\frac{1}{4\tau^2}\int_{\operatorname{Im} w = h_0} \frac{\mathbf{e}(-mw^2/\tau)}{\sin^2\big(\pi(w/\tau - \frac{r}{2m})\big)}dw,$$

by $\sum_k (w - (k+\frac{r}{2m})\tau)^{-2} = \tau^{-2}\pi^2/\sin^2(\pi(w/\tau - \frac{r}{2m}))$ [1, Chapter 5, §2.1].

*Evaluation on the imaginary axis.* Both sides being holomorphic in $\tau$, take $\tau = it$, $t > 0$, and $w = tu$. Then $\mathbf{e}(-mw^2/\tau) = e^{-2\pi m t u^2}$, and $\sin(\pi(-iu - \frac{r}{2m}))^2 = -\sinh^2(\pi u - \frac{\pi i r}{2m})$, so in the notation of Section 6.1 ($\gamma = 1$), for $2m \nmid r$,

$$d_\ell(it) = -\frac{1}{4\pi^2 t}\int_{\mathbb{R}} e^{-2\pi m t u^2} f_{1,r}(u)\, du.$$

For $2m \mid r$ the integrand is $e^{-2\pi m t u^2}\pi^2/\sinh^2(\pi u) = e^{-2\pi m t u^2}(f_{1,0}(u) + u^{-2})$ on the line $\operatorname{Im} u = -\epsilon/t$; $f_{1,0}$ is holomorphic in $|\operatorname{Im} u| < 1$, so its part may be moved to $\mathbb{R}$, and integration by parts gives $\int_{\operatorname{Im} u = -\epsilon/t} e^{-au^2}u^{-2}du = -2a\int_{\mathbb{R}} e^{-au^2}du = -2\sqrt{\pi a}$, $a = 2\pi m t$. Hence $d_0(it) = -\frac{1}{4\pi^2 t}\big(\int_{\mathbb{R}} e^{-2\pi m t u^2} f_{1,0}du - 2\pi\sqrt{2mt}\big)$. In both cases Lemma 9.8 with $\gamma = 1$, $g = r$, $X = t$, and $\theta_{1,r}(s) = \vartheta_\ell(is)$, gives

$$d_\ell(it) = \frac{\sqrt{2m}}{4\pi}\int_0^\infty \vartheta_\ell(is)(s+t)^{-3/2}ds = \frac{\sqrt{2m}}{4\pi}e^{\pi i/4}\tilde{P}_\ell(it),$$

the last step by $u = is$ in $\tilde{P}_\ell$ and $(i(s+t))^{-3/2} = i^{-3/2}(s+t)^{-3/2}$. So $d_\ell = c_\ell$ on $i\mathbb{R}_{>0}$, hence on $\mathbb{H}$.

*Conclusion.* $\widehat{\mathcal{A}}|_{2,m}S - \widehat{\mathcal{A}} = \mathcal{D} - \kappa_m(\mathcal{R}|_{2,m}S - \mathcal{R}) = \sum_\ell \big(c_\ell + \kappa_m \tilde{P}_\ell\big)\vartheta_{m,\ell} = 0$, because $\kappa_m = -\frac{\sqrt{2m}}{4\pi}e^{\pi i/4}$. □

9.4. **Theta decomposition and the transformation law.** For each frame put $\psi_F^F = \psi_F - G_F\mathcal{A}$.

**Lemma 9.10.** *$\psi_F^F$ is holomorphic on $\mathbb{H}\times\mathbb{C}$, even, satisfies the elliptic law of index $m$, and $\psi_F^F = \sum_{\ell=0}^{2m-1} h_{F,\ell}(\tau)\vartheta_{m,\ell}(\tau,z)$ with $h_{F,\ell}$ holomorphic on $\mathbb{H}$. For $F = I$, $\psi_I^F$ is the finite part $\varphi_m^F$ of (8) and*

$$h_{I,\ell}(\tau) = \sum_{n\in\mathbb{Z}} c_m^F(n,\ell)\,\mathbf{e}\Big(\frac{(2mn-\ell^2)\tau}{4m}\Big); \tag{45}$$

*$\psi_T^F(\tau,z) = \psi_I^F(\tau+1,z)$, so $h_{T,\ell}(\tau) = \mathbf{e}(\ell^2/4m)h_{I,\ell}(\tau+1)$; and $h_{F,\ell}(\tau)\mathbf{e}(\ell^2\tau/4m)$ has period 2 for $F = I, T$ and period 1 for $F = S$.*

*Proof.* By Proposition 9.3 and Lemma 9.5, $\psi_F$ and $G_F\mathcal{A}$ are even, of index $m$, holomorphic off the lattice, and have the same principal part $G_F(2\pi iz)^{-2}$ at $z = 0$; so the difference is holomorphic at 0 and, by the elliptic law, everywhere. Expand $\psi_F^F(\tau,z) = \sum_r A_r(\tau)\mathbf{e}(rz)$ (period 1 in $z$); the elliptic law with $\lambda = 1$ gives $A_{r+2m}(\tau) = \mathbf{e}(((r+2m)^2 - r^2)\tau/4m)A_r(\tau)$, so $h_{F,\ell} := \mathbf{e}(-r^2\tau/4m)A_r$ depends only on $r \equiv \ell \bmod 2m$, which is the theta decomposition. For $F = I$, $G_I\mathcal{A}$ is the polar part (8) (Proposition 9.3(ii)), and inserting $\varphi_m^F = \sum c_m^F(n,\ell)x^n y^\ell$ gives (45). For $F = T$ use $\mathcal{A}(\tau+1,z) = \mathcal{A}(\tau,z)$ and $\vartheta_{m,\ell}(\tau+1,z) = \mathbf{e}(\ell^2/4m)\vartheta_{m,\ell}(\tau,z)$. The periods: $\psi_I$ is invariant under $\tau\mapsto\tau+2$ ($\left(\begin{smallmatrix}1&2\\0&1\end{smallmatrix}\right)\in\Gamma^0(2)$), and so is $G_I\mathcal{A}$; for $S$, $STS^{-1} = \left(\begin{smallmatrix}1&0\\-1&1\end{smallmatrix}\right)\in\Gamma^0(2)$ gives $\psi_S(\tau+1,z) = \psi_S(\tau,z)$, and $G_S$, $\mathcal{A}$ have period 1. □

Put

$$\widehat{h}_{F,\ell}(\tau) = h_{F,\ell}(\tau) + h^-_{F,\ell}(\tau), \qquad h^-_{F,\ell}(\tau) = \kappa_m\, G_F(\tau)\, E_\ell(\tau), \tag{46}$$

with $\kappa_m$ and $E_\ell$ as in (44); these are the theta coefficients of the completion $\widehat{\psi}_F = \psi_F - G_F\widehat{\mathcal{A}}$.

**Theorem 9.11** (Transformation law)**.** *Put $\widehat{\psi}_F = \psi_F - G_F\widehat{\mathcal{A}} = \sum_\ell \widehat{h}_{F,\ell}\vartheta_{m,\ell}$. Then $\widehat{\psi}_F|_{-6,m}N^{-1} = \widehat{\psi}_I$ for $N = N_{\gamma,\delta}$, $F = F(\gamma,\delta)$, and for $0 \le \ell < 2m$, $\tau\in\mathbb{H}$,*

$$\widehat{h}_{I,\ell}(N\tau) = j(\tau)^{-13/2}\sum_{\ell'=0}^{2m-1} W_{\gamma,\delta}(\ell',\ell)\,\widehat{h}_{F,\ell'}(\tau).$$

*Together with Proposition 9.3(iv) this is property* (T2) *of Section 10.1, with $\lambda_{\gamma,\delta} = 1$.*

*Proof.* By Proposition 9.3(iv), Proposition 9.9 and Lemma 9.1(ii),

$$\widehat{\psi}_F|_{-6,m}N^{-1} = \psi_F|N^{-1} - (G_F|_{-8}N^{-1})(\widehat{\mathcal{A}}|_{2,m}N^{-1}) = \psi_m - G_I\widehat{\mathcal{A}} = \widehat{\psi}_I.$$

Now $N^{-1} = M_{\gamma,\delta} = M$; with $J = \gamma\tau+\delta$ and Lemma 9.7,

$$\begin{aligned}\widehat{\psi}_I(\tau,z) &= J^6\mathbf{e}\Big(-\frac{m\gamma z^2}{J}\Big)\sum_{\ell'}\widehat{h}_{F,\ell'}(M\tau)\vartheta_{m,\ell'}\Big(M\tau,\frac{z}{J}\Big)\\ &= J^{13/2}\sum_\ell\Big[\sum_{\ell'}W_{\gamma,\delta}(\ell',\ell)\widehat{h}_{F,\ell'}(M\tau)\Big]\vartheta_{m,\ell}(\tau,z)\end{aligned}$$

($J^6J^{1/2} = J^{13/2}$, $J\in\mathbb{H}$). The $\vartheta_{m,\ell}(\tau,\cdot)$, $0\le\ell<2m$, have disjoint Fourier supports in $z$, so $\widehat{h}_{I,\ell}(\tau) = J^{13/2}\sum_{\ell'}W(\ell',\ell)\widehat{h}_{F,\ell'}(M\tau)$. Replace $\tau$ by $N\tau$: $MN\tau = \tau$ and $J(N\tau)\,j(\tau) = 1$

(cocycle of $MN=1$). As $j(\tau)$ lies in the lower half-plane, $\operatorname{Arg}(1/j) = -\operatorname{Arg} j$ and $(1/j)^{13/2} = j^{-13/2}$. □

**Remark 9.12.** The completion $\widehat{\psi}_F$ is not holomorphic: $\partial_{\bar\tau}\widehat{h}_{F,\ell} = \kappa_m G_F \overline{\vartheta_\ell}(2i\tau_2)^{-3/2}$ (see the proof of Lemma 10.9), so the shadow of $\psi_F^F$ is $\kappa_m G_F$ times the vector $(\overline{\vartheta_\ell})$.

9.5. **Growth and finiteness of the polar parts.** With the operators $\|M$ of Section 9.1 (weight 0, index one), so (6) reads

$$L_l = \sum_{td=l} \frac{1}{2dt} \sum_{r,s} (-1)^{sd} \sum_{u=0}^{2d-1} (-1)^{ru} F^{(r,s)} \| M_{t,u,d}. \tag{47}$$

**Lemma 9.13.** *For $\gamma \in SL_2(\mathbb{Z})$ and $l \geq 1$ there is $C$ such that $|(L_l|_{0,l}\gamma)(\sigma,v)| \leq Ce^{\pi l \operatorname{Im}\sigma}$ on $\operatorname{Im} v = -\frac{1}{2}\operatorname{Im}\sigma$, $\operatorname{Im}\sigma \geq l$.*

*Proof.* For each term of (47), $M_{t,u,d}\gamma = \gamma' M'$ with $\gamma' \in SL_2(\mathbb{Z})$ and $M' = \left(\begin{smallmatrix} t' & u' \\ 0 & d' \end{smallmatrix}\right)$, $d' \geq 1$, $t'd' = l$ (Hermite normal form; replace $\gamma'$ by $-\gamma'$ if necessary, $F^{(r,s)}$ being even). So $F^{(r,s)}\|M_{t,u,d}|_{0,l}\gamma = F^{(r,s)}\|(\gamma' M') = (F^{(r,s)}|_{0,1}\gamma')\|M' = F^{(r',s')}((t'\sigma+u')/d', t'v)$ by Lemma 9.1(i) and Lemma 8.2. Let $F$ be a genus, $F = h_0\vartheta_{1,0} + h_1\vartheta_{1,1}$ its theta decomposition; by (P) $h_\mu = \sum_{n\geq 0} c(n,\mu) q^{n-\mu^2/4}$, so $|h_\mu(\tau')| \leq Ce^{\pi \operatorname{Im}\tau'/2}$ for $\operatorname{Im}\tau' \geq 1$. With $T = \operatorname{Im}\tau'$, $Y = \operatorname{Im} z'$, $|\vartheta_{1,\mu}(\tau',z')| \leq e^{2\pi Y^2/T} \sum_r e^{-\frac{\pi T}{2}(r+2Y/T)^2} \leq (1+\sqrt{2/T}) e^{2\pi Y^2/T}$. At $\tau' = (t'\sigma+u')/d'$, $z' = t'v$: $T = t'\tau_2/d' \in [\tau_2/l, l\tau_2]$ and $Y^2/T = l\tau_2/4$, so each term is at most $Ce^{\pi l\tau_2/2 + \pi l\tau_2/2}$. □

**Theorem 9.14** (Finiteness of the polar parts)**.** *For each frame $F$ and $0 \leq \ell < 2m$, $h_{F,\ell}(\tau)\mathbf{e}(\ell^2\tau/4m)$ is a Laurent series in $\mathbf{e}(\tau/w_F)$ ($w_I = w_T = 2$, $w_S = 1$) with finitely many negative powers, converging absolutely and locally uniformly on $\mathbb{H}$. Consequently property* (T1) *of Section 10.1 holds, and $h_{F,\ell}(\tau) = \sum_\lambda b(\lambda)\mathbf{e}(\lambda\tau)$ with $\lambda \in \frac{1}{4m}\mathbb{Z}$, only finitely many $\lambda < 0$, absolutely convergent on $\mathbb{H}$; in particular at $\operatorname{Im}\tau = \frac{1}{2}$. With Remark 9.4 this is property* (T3) *with finite sets $P_F$ ($\tilde\nu \in \frac{1}{2}\mathbb{Z}$ for $F = I, T$, $\tilde\nu \in \mathbb{Z}$ for $F = S$).*

*Proof.* By Lemma 9.10, $H(\tau) = h_{F,\ell}(\tau)\mathbf{e}(\ell^2\tau/4m)$ is holomorphic with period $w_F$, so it is an absolutely and locally uniformly convergent Laurent series $\sum_n b_n \mathbf{e}(n\tau/w_F)$ on $\mathbb{H}$ (Laurent expansion on $0 < |\mathbf{e}(\tau/w_F)| < 1$). It suffices to show $|H(\tau)| \leq Ce^{C'\tau_2}$ for $\tau_2 \geq \tau_0$, uniformly in $\tau_1$: then $|b_{-n}| = |\frac{1}{w_F}\int_0^{w_F} H(\tau)\mathbf{e}(n\tau/w_F)d\tau_1| \leq Ce^{(C'-2\pi n/w_F)\tau_2} \to 0$ for $n > C'w_F/2\pi$.

Shift the coefficient integral to the line $\operatorname{Im} z = -\tau_2/2$ ($\psi_F^F$ is entire and 1-periodic in $z$): $h_{F,\ell}(\tau)\mathbf{e}(\ell^2\tau/4m) = \int_0^1 \psi_F^F(\tau, x_0 - i\tau_2/2)\mathbf{e}(-\ell(x_0 - i\tau_2/2))dx_0$, and $|\mathbf{e}(-\ell z)| = e^{-\pi\ell\tau_2}$ there. On this line:

(a) $|\mathcal{A}| \leq C$ for $\tau_2 \geq 1$: $|y| = e^{\pi\tau_2}$, the term $s$ has modulus $e^{-2\pi\tau_2(ms(s-1)+s-\frac{1}{2})}|1-q^s y|^{-2}$ and $|q^s y| = e^{\pi\tau_2(1-2s)}$; the terms $s = 0, 1$ are $O(e^{-\pi\tau_2})$ and the others are smaller.

(b) $|G_F(\tau)| \leq Ce^{2\pi\tau_2}$ by Remark 9.4.

(c) $|\psi_F| \leq Ce^{C'\tau_2}$. By (43), $\psi_m = B_{m+1}/(g(\sigma/2)\phi_{-2,1})$, with $B_{m+1}$ of weight 0 and index $m+1$. Hence, by Lemma 9.1(ii),

$$\psi_F = \frac{B_{m+1}|_{0,m+1}R_F}{(g(\sigma/2)|_8 R_F)\,(\phi_{-2,1}|_{-2,1}R_F)},$$

with $\phi_{-2,1}|_{-2,1}R_F = \phi_{-2,1}$, $g(\sigma/2)|_8 R_T = g((\sigma+1)/2)$ and $g(\sigma/2)|_8 R_S = 16g(\sigma)$ (proof of Proposition 9.3(iii)). $B_{m+1}|R_F$ is a polynomial in the $L_l|_{0,l}R_F$, bounded by Lemma 9.13; $1/|g| \leq Ce^{2\pi\tau_2}$; and from the product $\theta_1(\tau,z) = -iq^{1/8}y^{1/2}\prod_{n\geq 1}(1-q^n)(1-q^n y)(1-q^{n-1}y^{-1})$,

$|q^n y|, |q^{n-1}y^{-1}| \le e^{-\pi(2n-1)\tau_2}$ on the line, so $|\theta_1| \ge ce^{\pi\tau_2/4}$, $|\eta|^6 \le Ce^{-\pi\tau_2/2}$ and $|\phi_{-2,1}| = |\theta_1|^2/|\eta|^6 \ge ce^{\pi\tau_2}$ for $\tau_2 \ge 1$.

So $|\psi_F^F| \le Ce^{C'\tau_2}$ on the line, for $\tau_2 \ge m+1$, uniformly in $\tau_1$, and the claim follows. The exponents of $h_{F,\ell}$ are $(2mn-\ell^2)/4m$ ($F = I, T$, $n \in \mathbb{Z}$) and $(4mn-\ell^2)/4m$ ($F = S$), in $\frac{1}{4m}\mathbb{Z}$; finitely many are negative, and these form $P_F$ with $\tilde\nu = n/2$, resp. $n$. □

## 10. The circle method for the level-two expansion

This section proves Theorem 6.2 by the circle method on Farey arcs, in the form used by Rademacher and Zuckerman [25, 34] and, for mixed mock modular forms, by Bringmann and Manschot [6]. Because the weight $-\frac{13}{2}$ is very negative, every error term is estimated absolutely; no cancellation in Kloosterman-type sums is used. The conventions are those of Section 6.1.

10.1. **The three properties.** The coefficients $c_m^F(n,\ell)$ are those of the finite part $\varphi_m^F(\sigma,v) = \sum c_m^F(n,\ell)\, x^n y^\ell$, $x = \mathbf{e}(\sigma/2)$, $y = \mathbf{e}(v)$, and we write $\tau$ for $\sigma$. The circle method uses only the following three properties of $\varphi_m^F$; Proposition 10.1 below records where each is proved.

**(T1) Theta decomposition.** For each $\ell \in \{0,\dots,2m-1\}$ the series $h_{I,\ell}$ of (45) has only finitely many nonzero terms with $2mn-\ell^2 < 0$ and converges absolutely and locally uniformly on $\mathbb{H}$, and $\varphi_m^F(\tau,v) = \sum_{\ell=0}^{2m-1} h_{I,\ell}(\tau)\,\vartheta_{m,\ell}(\tau,v)$.

**(T2) Completion and modularity.** There are holomorphic functions $h_{F,\ell}$ on $\mathbb{H}$ ($F \in \{I,T,S\}$, $0 \le \ell < 2m$; for $F = I$ they are (45)) and holomorphic functions $G_I, G_T, G_S$ on $\mathbb{H}$, none identically zero, with the following properties. Let $\widehat{h}_{F,\ell} = h_{F,\ell} + \kappa_m G_F E_\ell$ be as in (46), with $\kappa_m$ and $E_\ell$ as in (44). Then for every $\gamma \ge 1$ and $(\gamma,\delta) \in \mathcal{C}_\gamma$, with $N = N_{\gamma,\delta}$, $F = F(\gamma,\delta)$ and $j = j_{\gamma,\delta}$:

(T2a) $G_I(N\tau) = \lambda_{\gamma,\delta}\, j(\tau)^{-8}\, G_F(\tau)$ for a constant $\lambda_{\gamma,\delta} \ne 0$;
(T2b) for $0 \le \ell < 2m$ and $\tau \in \mathbb{H}$,

$$\widehat{h}_{I,\ell}(N\tau) = j(\tau)^{-13/2} \sum_{\ell'=0}^{2m-1} W_{\gamma,\delta}(\ell',\ell)\, \widehat{h}_{F,\ell'}(\tau). \tag{48}$$

**(T3) Expansions in the frames.** For each $F$ there are a finite set $P_F$ of pairs $(\tilde\nu,\tilde\ell)$ with $0 \le \tilde\ell < 2m$ and $\tilde D := 4m\tilde\nu - \tilde\ell^2 < 0$, numbers $a_F(\tilde\nu,\tilde\ell)$, numbers $\mu_F > 0$ and $c_F \in \mathbb{C}$, such that for $0 \le \ell' < 2m$

$$h_{F,\ell'}(\tau) = \sum_{\tilde\nu:\ (\tilde\nu,\ell')\in P_F} a_F(\tilde\nu,\ell')\, \mathbf{e}\Big(\frac{\tilde D\tau}{4m}\Big) + \sum_{\lambda\in\Lambda_{F,\ell'}} b_{F,\ell'}(\lambda)\, \mathbf{e}(\lambda\tau), \tag{49}$$

$$G_F(\tau) = c_F\, \mathbf{e}(-\mu_F\tau) + \sum_{\lambda\in\Lambda_F^G} g_F(\lambda)\, \mathbf{e}(\lambda\tau), \tag{50}$$

where $\Lambda_{F,\ell'}, \Lambda_F^G$ are countable subsets of $[0,\infty)$ and the two series over $\lambda$ converge absolutely on $\mathbb{H}$.

We use the abbreviations

$$C_h=\max_{F,\ell'}\sum_{\lambda\in\Lambda_{F,\ell'}}|b_{F,\ell'}(\lambda)|e^{-\pi\lambda},\qquad C_G=\max_F\sum_{\lambda\in\Lambda_F^G}|g_F(\lambda)|e^{-\pi\lambda},$$

$$a_*=\max_F\sum_{(\tilde\nu,\tilde\ell)\in P_F}|a_F(\tilde\nu,\tilde\ell)|,\qquad A_*=\max_F\max_{P_F}\frac{|\tilde D|}{4m},\tag{51}$$

$$c_*=\max_F|c_F|,\qquad \mu_*=\max_F\mu_F,$$

($A_*=0$ if every $P_F$ is empty). $C_h$ and $C_G$ are finite by the absolute convergence in (T3) at $\operatorname{Im}\tau=\frac12$, and by (T3), for $\operatorname{Im}\tau\ge\frac12$,

$$\Big|h_{F,\ell'}(\tau)-\sum_{(\tilde\nu,\ell')\in P_F}a_F\,\mathbf{e}\Big(\frac{\tilde D\tau}{4m}\Big)\Big|\le C_h,\qquad |G_F(\tau)-c_F\mathbf{e}(-\mu_F\tau)|\le C_G.\tag{52}$$

**Proposition 10.1.** *The finite part $\varphi_m^F$ has the properties* (T1)–(T3)*, with $\lambda_{\gamma,\delta}=1$, with $P_F$ and $a_F$ the polar data* (27)*, with the constants $c_F,\mu_F$ of* (28)*, and with $\Lambda_{F,\ell'}\subset\frac{1}{4m}\mathbb{Z}_{\ge0}$, $\Lambda_I^G=\Lambda_T^G=\frac12\mathbb{Z}_{\ge0}$, $\Lambda_S^G=\mathbb{Z}_{\ge0}$.*

*Proof.* (T1) is Lemma 9.10 together with Theorem 9.14. In (T2) the functions are the $h_{F,\ell}$ of Lemma 9.10 and $G_I=\hat f(m)/g(\tau/2)$, $G_T(\tau)=G_I(\tau+1)$, $G_S(\tau)=\tau^8G_I(-1/\tau)=\hat f(m)/(16\,g(\tau))$ of Proposition 9.3(iii); (T2a) with $\lambda_{\gamma,\delta}=1$ is Proposition 9.3(iv), and (T2b) is Theorem 9.11. In (T3) the finiteness of the polar parts and the absolute convergence are Theorem 9.14, and the expansions of $G_F$, with the leading terms (28), are Remark 9.4. The polar coefficients are $a_I$ and $a_T=(-1)^{2\tilde\nu}a_I$ in the frames $I$ and $T$ by Theorem 7.1, and $a_S$ in the frame $S$ by Theorem 8.1(ii); the latter applies because the $h_{S,\ell}$ are the theta coefficients of $\psi_S-G_S\mathcal{A}=\psi_m^S$ (Proposition 9.3(iii) and (41); see Remark 8.10). Terms with exponent exactly 0 belong to the remainders. □

10.2. **The statement.** For $D>0$ and $(\gamma,\delta)\in\mathcal{C}_\gamma$, with $F=F(\gamma,\delta)$, $\alpha$ attached to $(\gamma,\delta)$, $\tilde D=4m\tilde\nu-\tilde\ell^2$ and $u_F=\sqrt{\mu_F/m}$, put

$$\mathcal{M}_{\gamma,\delta}=\sum_{(\tilde\nu,\tilde\ell)\in P_F}a_F(\tilde\nu,\tilde\ell)\,\mathbf{e}\Big(\frac{\alpha\tilde D+\delta D}{4m\gamma}\Big)W_{\gamma,\delta}(\tilde\ell,\ell)\,\frac{2\pi}{\gamma}\Big(\frac{|\tilde D|}{D}\Big)^{\frac{15}{4}}I_{\frac{15}{2}}\Big(\frac{\pi\sqrt{|\tilde D|D}}{m\gamma}\Big),\tag{53}$$

$$\mathcal{K}_{2;\gamma,\delta}=-\sqrt{2m}\,\gamma^{-\frac12}c_F\,\mathbf{e}\Big(\frac{-4m\mu_F\alpha+\delta D}{4m\gamma}\Big)W_{\gamma,\delta}(0,\ell)\Big(\frac{4m\mu_F}{D}\Big)^4I_8\Big(\frac{2\pi}{\gamma}\sqrt{\frac{\mu_FD}{m}}\Big),\tag{54}$$

$$\mathcal{K}_{3;\gamma,\delta}=\frac{\gamma^{-2}}{2\pi}\sum_{g=-m\gamma}^{m\gamma-1}c_F\,\mathbf{e}\Big(\frac{-\alpha(4m\mu_F+g^2)+\delta D}{4m\gamma}\Big)W_{\gamma,\delta}(g,\ell)\Big(\frac{4m\mu_F}{D}\Big)^{\frac{17}{4}}\tag{55}$$

$$\times\int_{-u_F}^{u_F}f_{\gamma,g}(u)\Big(1-\frac{mu^2}{\mu_F}\Big)^{\frac{17}{4}}I_{\frac{17}{2}}\Big(\frac{2\pi}{\gamma}\sqrt{\frac{(\mu_F-mu^2)D}{m}}\Big)\,du,$$

and $\mathcal{M}=\sum_{\gamma\ge1}\sum_{\mathcal{C}_\gamma}\mathcal{M}_{\gamma,\delta}$, $\mathcal{K}_2$, $\mathcal{K}_3$ likewise; by Proposition 10.1 these are the series (29)–(31). Let $\mathcal{M}^{(N)},\mathcal{K}_2^{(N)},\mathcal{K}_3^{(N)}$ be the partial sums over $\gamma\le N$.

**Theorem 10.2.** *Let $0\le\ell<2m$, $n\in\mathbb{Z}$ and $D=2mn-\ell^2>0$, and put $B=D/4m$. Then the series $\mathcal{M}$, $\mathcal{K}_2$, $\mathcal{K}_3$ converge absolutely, and*

$$c_m^F(n,\ell)=\tfrac12\,e^{-3\pi i/4}\big(\mathcal{M}+\mathcal{K}_2+\mathcal{K}_3\big),\tag{56}$$

*equivalently* $(-1)^{\ell+1}c_m^F(n,\ell) = \frac{1}{2}(-1)^\ell\sqrt{i}\,(\mathcal{M}+\mathcal{K}_2+\mathcal{K}_3)$. *More precisely, for every integer* $N \geq 1$,

$$\left| c_m^F(n,\ell) - \tfrac{1}{2}\, e^{-3\pi i/4}\big(\mathcal{M}^{(N)} + \mathcal{K}_2^{(N)} + \mathcal{K}_3^{(N)}\big)\right| \leq C_*\, N^{-6}, \tag{57}$$

*where, with the constants* (51),

$$C_* = \frac{2^{17/2}a_*}{\sqrt{2m}}e^{8\pi(A_*+B)} + 2^{13/4}\sqrt{2m}\,C_h e^{2\pi B} + \frac{2^{5/2}}{\pi}C_G e^{2\pi B} + 2^{1/4}\pi^2 m\,(C_G + c_*)e^{2\pi B} + \frac{2^8}{\pi}c_* e^{8\pi(\mu_*+B)} + 2^7\pi^2 m\sqrt{\mu_*}\,c_* e^{8\pi(\mu_*+B)}.$$

The two forms of (56) agree because $-e^{-3\pi i/4} = e^{\pi i/4} = \sqrt{i}$, and for $0 \leq \ell \leq m$ the second form is (32). So Theorem 10.2 contains Theorem 6.2. The constant $\frac{1}{2}$ is the reciprocal of the width 2 of the cusp $\infty$, and $e^{-3\pi i/4}$ is the value of $j_{\gamma,\delta}^{-13/2}/z^{13/2}$ on every arc (see (62)); for $\gamma = 1$, $\delta = 0$ it combines with $(2mi)^{-1/2}$ in $W_{1,0}$ to the real negative number $-(2m)^{-1/2}$, the factor $i^{-6} = -1$ familiar from weight $-6$.

10.3. **Auxiliary results.**

**Lemma 10.3.** $W_{\gamma,\delta}(\tilde{\ell}+2m,\ell) = W_{\gamma,\delta}(\tilde{\ell},\ell)$, *and* $|W_{\gamma,\delta}(\tilde{\ell},\ell)| \leq \sqrt{\gamma/2m}$.

*Proof.* Replacing $\tilde{\ell}$ by $\tilde{\ell}+2m$ replaces $T$ by $T-1$ in $x$. The summand is periodic in $T$ with period $\gamma$: replacing $x$ by $x - 2m\gamma$ changes the argument of $\mathbf{e}$ by $-\alpha x + \alpha m\gamma + \ell \in \mathbb{Z}$. So the sum over $T$ is unchanged. The bound is the triangle inequality: $\gamma$ terms of modulus one, divided by $\sqrt{2m\gamma}$. □

(The sharper bound of Lemma 6.1 is not needed here.)

**Lemma 10.4.** *Let* $c > 0$, $\nu > 0$ *and* $x > 0$. *Then*

$$\frac{1}{2\pi i}\int_{c-i\infty}^{c+i\infty} t^{-\nu-1}\exp\Big(t + \frac{x^2}{4t}\Big)\,dt = \Big(\frac{x}{2}\Big)^{-\nu} I_\nu(x), \tag{58}$$

*the integral converging absolutely. Consequently, for* $A, B, k > 0$, $r > 0$ *and* $c > 0$,

$$\frac{1}{2\pi i}\int_{c-i\infty}^{c+i\infty} t^{-r-2}\exp\Big(\frac{2\pi}{k}\Big(At + \frac{B}{t}\Big)\Big)\,dt = \Big(\frac{A}{B}\Big)^{\frac{r+1}{2}} I_{r+1}\Big(\frac{4\pi\sqrt{AB}}{k}\Big). \tag{59}$$

*Moreover* $I_\nu(x)/x^\nu$ *is increasing on* $(0,\infty)$, *so* $I_\nu(x/\gamma) \leq \gamma^{-\nu} I_\nu(x)$ *for* $\gamma \geq 1$.

*Proof.* First, for $\operatorname{Re} s > 1$,

$$\frac{1}{\Gamma(s)} = \frac{1}{2\pi i}\int_{c-i\infty}^{c+i\infty} e^t t^{-s}\,dt. \tag{60}$$

Indeed, let $\phi(x) = x^{s-1}e^{-cx}$ for $x > 0$ and $\phi(x) = 0$ for $x \leq 0$. Its Fourier transform is $\hat{\phi}(y) = \int_0^\infty x^{s-1}e^{-(c+iy)x}dx = \Gamma(s)(c+iy)^{-s}$ (principal branch; both sides are holomorphic in $c+iy$ on the right half-plane and agree for $y = 0$), which is integrable since $\operatorname{Re} s > 1$. By Fourier inversion, $\phi(1) = \frac{1}{2\pi}\int_{\mathbb{R}} \hat{\phi}(y)e^{iy}dy$, the equality holding at $x = 1$ because $\phi$ is continuous on $(0,\infty)$ and the inverse transform is continuous. With $t = c+iy$ this is (60). (It is Hankel's formula [33, §12.22] with the loop opened to a line.)

On the line $\operatorname{Re} t = c$ one has $|t| \geq c$, $|e^t| = e^c$ and $|e^{x^2/4t}| \leq e^{x^2/4c}$, and $|t^{-\nu-1}| = |t|^{-\nu-1}$ is integrable since $\nu > 0$; so (58) converges absolutely. Expanding $e^{x^2/4t} = \sum_{p\geq 0}(x^2/4)^p t^{-p}/p!$,

the series of absolute values is dominated by $e^c e^{x^2/4c}|t|^{-\nu-1}$, so we may integrate termwise and use (60) with $s=\nu+1+p>1$:

$$\frac{1}{2\pi i}\int_{c-i\infty}^{c+i\infty} t^{-\nu-1}e^{t+x^2/4t}dt=\sum_{p\ge 0}\frac{(x^2/4)^p}{p!\,\Gamma(\nu+p+1)}=\Big(\frac{x}{2}\Big)^{-\nu}I_\nu(x).$$

For (59) substitute $t=\frac{k}{2\pi A}s$; the line $\operatorname{Re}t=c$ goes to the line $\operatorname{Re}s=2\pi Ac/k>0$, $t^{-r-2}=(\frac{2\pi A}{k})^{r+2}s^{-r-2}$ (a positive real factor), and $\frac{2\pi B}{kt}=\frac{x^2}{4s}$ with $x=4\pi\sqrt{AB}/k$. So the left side of (59) is $(\frac{2\pi A}{k})^{r+1}(\frac{x}{2})^{-r-1}I_{r+1}(x)=(\frac{A}{B})^{(r+1)/2}I_{r+1}(x)$. Finally $I_\nu(x)/x^\nu=\sum_p 2^{-\nu-2p}x^{2p}/(p!\Gamma(\nu+p+1))$ has positive coefficients. □

**Lemma 10.5** (Farey arcs)**.** *Let $N\ge 1$. To each $(\gamma,\delta)\in\mathcal{C}_\gamma$ with $\gamma\le N$ attach the fraction $-\delta/\gamma\in[0,2)$, and let $\gamma_1,\gamma_2$ be the denominators of its left and right neighbours in the sequence of reduced fractions in $[0,2]$ with denominator at most $N$, where the left neighbour of $0/1$ is taken to be $-1/N$. Put $\omega'_{\gamma,\delta}=1/(\gamma(\gamma+\gamma_1))$, $\omega''_{\gamma,\delta}=1/(\gamma(\gamma+\gamma_2))$ and $J_{\gamma,\delta}=[-\delta/\gamma-\omega'_{\gamma,\delta},\,-\delta/\gamma+\omega''_{\gamma,\delta}]$. Then the intervals $J_{\gamma,\delta}$ have disjoint interiors, their union is $[-\frac{1}{N+1},2-\frac{1}{N+1}]$, and*

$$\frac{1}{2\gamma N}\le\omega'_{\gamma,\delta},\ \omega''_{\gamma,\delta}<\frac{1}{\gamma N}.$$

*Proof.* The fractions in $[0,1]$ form the Farey sequence $\mathfrak{F}_N$, those in $[1,2]$ form $1+\mathfrak{F}_N$, and the extension by $-1/N$ is the translate by $-1$ of the last step $(N-1)/N<1/1$ of $\mathfrak{F}_N$. For consecutive fractions $h_1/k_1<h_2/k_2$ in $\mathfrak{F}_N$ one has $h_2k_1-h_1k_2=1$ and $k_1+k_2>N$ [21, Chapter III]; translation by an integer preserves both properties. Hence the mediant $\frac{h_1+h_2}{k_1+k_2}$ satisfies $\frac{h_1+h_2}{k_1+k_2}-\frac{h_1}{k_1}=\frac{1}{k_1(k_1+k_2)}$ and $\frac{h_2}{k_2}-\frac{h_1+h_2}{k_1+k_2}=\frac{1}{k_2(k_1+k_2)}$, so $J_{\gamma,\delta}$ runs from the mediant with the left neighbour to the mediant with the right neighbour. Consecutive intervals therefore share an endpoint, the first starts at $\frac{0-1}{1+N}$ and the last, belonging to $\frac{2N-1}{N}$ with right neighbour $\frac{2}{1}$, ends at $\frac{2N-1+2}{N+1}=2-\frac{1}{N+1}$. The bounds follow from $N+1\le\gamma+\gamma_j\le 2N$. □

For $(\gamma,\delta)\in\mathcal{C}_\gamma$ and $\varphi\in\mathbb{R}$ put

$$z=\gamma\Big(\frac{1}{N^2}-i\varphi\Big),\qquad \tau=\frac{-\delta+iz}{\gamma}=-\frac{\delta}{\gamma}+\varphi+\frac{i}{N^2},\qquad \tau'=M_{\gamma,\delta}\tau=\frac{\alpha}{\gamma}+\frac{i}{\gamma z}. \tag{61}$$

The last equality holds because $\gamma\tau+\delta=iz$ and $\alpha\tau+\beta=(-1+i\alpha z)/\gamma$. Then $\tau=N_{\gamma,\delta}\tau'$, and $j_{\gamma,\delta}(\tau')=-\gamma\tau'+\alpha=-i/z$. Since $\operatorname{Re}z>0$, $\operatorname{Arg}(-i/z)=-\frac{\pi}{2}-\operatorname{Arg}z\in(-\pi,0)$, so $\operatorname{Log}j_{\gamma,\delta}(\tau')=-\operatorname{Log}z-\frac{\pi i}{2}$ and

$$j_{\gamma,\delta}(\tau')^{-13/2}=e^{13\pi i/4}z^{13/2}=e^{-3\pi i/4}z^{13/2}. \tag{62}$$

**Lemma 10.6.** *Let $1\le\gamma\le N$ and $|\varphi|\le\frac{1}{\gamma N}$. Then $\operatorname{Re}z=\gamma/N^2$, $|z|\le\sqrt{2}/N$, $\gamma|z|\le\sqrt{2}$ and $\operatorname{Im}\tau'=\operatorname{Re}\frac{1}{\gamma z}\ge\frac12$. If moreover $|\varphi|\ge\frac{1}{2\gamma N}$, then $\operatorname{Re}\frac{1}{\gamma z}\le 4$.*

*Proof.* $|z|^2=\gamma^2(N^{-4}+\varphi^2)\le\gamma^2N^{-4}+N^{-2}\le 2N^{-2}$, using $\gamma\le N$. Next $\operatorname{Re}\frac{1}{\gamma z}=\frac{N^{-2}}{\gamma^2(N^{-4}+\varphi^2)}$, which is at least $\frac{N^{-2}}{\gamma^2N^{-4}+N^{-2}}=\frac{1}{1+\gamma^2N^{-2}}\ge\frac12$, and, if $\varphi^2\ge\frac{1}{4\gamma^2N^2}$, at most $N^{-2}\cdot 4N^2=4$. □

**Lemma 10.7** (The arc integral)**.** *Let $1\le\gamma\le N$, $(\gamma,\delta)\in\mathcal{C}_\gamma$, $r\ge1$, $A>0$, $B>0$, and $z$ as in (61). Then*

$$\int_{-\omega'_{\gamma,\delta}}^{\omega''_{\gamma,\delta}} z^r\exp\Big(\frac{2\pi}{\gamma}\Big(\frac{A}{z}+Bz\Big)\Big)d\varphi=\frac{2\pi}{\gamma}\Big(\frac{A}{B}\Big)^{\frac{r+1}{2}}I_{r+1}\Big(\frac{4\pi\sqrt{AB}}{\gamma}\Big)+\varepsilon,$$

*where* $|\varepsilon| \le 2^{r+2} e^{8\pi(A+B)} \gamma^{-1} N^{-r-1}$.

*Proof.* Write $\Psi(z) = z^r e^{\frac{2\pi}{\gamma}(A/z + Bz)}$ and split the interval as $[-\frac{1}{\gamma N}, \frac{1}{\gamma N}]$ minus $[-\frac{1}{\gamma N}, -\omega')$ and $(\omega'', \frac{1}{\gamma N}]$ (both nonempty or empty half-open intervals, by Lemma 10.5). On the two removed pieces $\frac{1}{2\gamma N} \le |\varphi| \le \frac{1}{\gamma N}$, so by Lemma 10.6 $|\Psi(z)| \le (\sqrt{2}/N)^r e^{8\pi A} e^{2\pi B/N^2}$; each piece has length at most $\frac{1}{2\gamma N}$, so together they contribute at most $2^{r/2} e^{8\pi A + 2\pi B}/(\gamma N^{r+1})$.

On the symmetric interval, $d\varphi = \frac{i}{\gamma} dz$ and $z$ runs down the segment from $z_1 = \frac{\gamma}{N^2} + \frac{i}{N}$ to $z_2 = \bar{z}_1$. Put $t = 1/z$; then $z^r = t^{-r}$ (as $\operatorname{Arg} t = -\operatorname{Arg} z$), $dz = -dt/t^2$, and $t$ runs from $t_1 = 1/z_1 = c - iT$ to $t_2 = c + iT$ along the arc of the circle $|t - \frac{N^2}{2\gamma}| = \frac{N^2}{2\gamma}$ (the image of the line $\operatorname{Re} z = \gamma/N^2$) through $t = N^2/\gamma$, where

$$c = \frac{\gamma N^2}{\gamma^2 + N^2}, \qquad T = \frac{N^3}{\gamma^2 + N^2} \ge \frac{N}{2}.$$

This arc is the part of the circle in $\{\operatorname{Re} t \ge c\}$, and $t^{-r-2} e^{\frac{2\pi}{\gamma}(At + B/t)}$ is holomorphic on $\operatorname{Re} t > 0$; by Cauchy's theorem the arc may be replaced by the segment $[t_1, t_2]$. Hence

$$\begin{aligned} \int_{-1/\gamma N}^{1/\gamma N} \Psi \, d\varphi &= -\frac{i}{\gamma} \int_{c-iT}^{c+iT} t^{-r-2} e^{\frac{2\pi}{\gamma}(At + B/t)} dt \\ &= \frac{2\pi}{\gamma} \cdot \frac{1}{2\pi i} \Big( \int_{c-i\infty}^{c+i\infty} - \int_{|\operatorname{Im} t| > T} \Big) t^{-r-2} e^{\frac{2\pi}{\gamma}(At + B/t)} dt. \end{aligned}$$

The full line gives the main term by (59). On $t = c + is$, $|s| > T$: $\frac{2\pi}{\gamma} Ac = \frac{2\pi A N^2}{\gamma^2 + N^2} \le 2\pi A$ and $\frac{2\pi B}{\gamma} \operatorname{Re} \frac{1}{t} = \frac{2\pi B c}{\gamma(c^2 + s^2)} \le \frac{2\pi B}{T^2} \le 8\pi B$; so the tail is at most $\frac{1}{\gamma} \cdot 2 e^{2\pi A + 8\pi B} \int_T^\infty s^{-r-2} ds \le \frac{2^{r+2}}{(r+1)} \frac{e^{2\pi A + 8\pi B}}{\gamma N^{r+1}}$. Adding the two errors and using $2^{r/2} + 2^{r+2}/(r+1) \le 2^{r+2}$ for $r \ge 1$ gives the bound. □

**Lemma 10.8.** *For* $\gamma \ge 1$, $\displaystyle\sum_{g=-m\gamma}^{m\gamma-1} \sup_{u \in \mathbb{R}} |f_{\gamma,g}(u)| \le \frac{\pi^4 m^2 \gamma^2}{2}$.

*Proof.* For $g = 0$, $f_{\gamma,0}(u) = \pi^2(\sinh^{-2} y - y^{-2})$ with $y = \pi u/\gamma$, and $\sinh^{-2} y - y^{-2} \in [-\frac{1}{3}, 0)$, so $|f_{\gamma,0}| \le \pi^2/3$. For $0 < |g| \le m\gamma$, $|\sinh(a - ib)|^2 = \sinh^2 a + \sin^2 b \ge \sin^2 b$ with $b = \pi g/2m\gamma$, $|b| \le \frac{\pi}{2}$, and $|\sin b| \ge \frac{2}{\pi}|b| = |g|/m\gamma$; so $|f_{\gamma,g}| \le \pi^2 m^2 \gamma^2/g^2$. Summing, $\frac{\pi^2}{3} + 2\pi^2 m^2 \gamma^2 \zeta(2) \le \frac{\pi^4 m^2 \gamma^2}{2}$. □

The next lemma extracts from (T2) the transformation of the shadow. With $\lambda_{\gamma,\delta} = 1$ it is the classical transformation law of the vector $(\vartheta_\ell)$, proved directly in Lemma 9.7; the argument below shows that it is forced by (T2).

**Lemma 10.9.** *For* $(\gamma, \delta) \in \mathcal{C}_\gamma$, $0 \le \ell < 2m$ *and* $u \in \mathbb{H}$,

$$\vartheta_\ell(N^*_{\gamma,\delta} u) = \lambda^{-1}_{\gamma,\delta} (\gamma u + \alpha)^{1/2} \sum_{\ell'=0}^{2m-1} W_{\gamma,\delta}(\ell', \ell) \, \vartheta_{\ell'}(u).$$

*Proof.* Write $N, j, \lambda, W$ without indices. The function $(\tau, s) \mapsto \int_s^{i\infty} \vartheta_\ell(w)(w + \tau)^{-3/2} dw$ is holomorphic on $\mathbb{H} \times \mathbb{H}$ (the integrand is holomorphic in $(w, \tau)$, as $w + \tau \in \mathbb{H}$, and decays like $(\operatorname{Im} w)^{-3/2}$ uniformly on compacta), and $E_\ell(\tau)$ is its value at $s = -\bar{\tau}$. Hence, by (25),

$$\partial_{\bar\tau} E_\ell(\tau) = \vartheta_\ell(-\bar\tau)(\tau - \bar\tau)^{-3/2} = \overline{\vartheta_\ell(\tau)} \, (2i\tau_2)^{-3/2} \qquad (\tau_2 = \operatorname{Im} \tau),$$

and $\partial_{\bar\tau}\widehat{h}_{F,\ell} = \kappa_m G_F(\tau)\overline{\vartheta_\ell(\tau)}(2i\tau_2)^{-3/2}$. Apply $\partial_{\bar\tau}$ to (48). On the left, by the chain rule and $\frac{d}{d\tau}N\tau = j(\tau)^{-2}$, we get $(\partial_{\bar w}\widehat{h}_{I,\ell})(N\tau)\,\overline{j(\tau)}^{-2}$, and $\operatorname{Im} N\tau = \tau_2/|j|^2$, so $(2i\operatorname{Im} N\tau)^{-3/2} = |j|^3(2i\tau_2)^{-3/2}$. Cancelling $\kappa_m(2i\tau_2)^{-3/2} \neq 0$ and using (T2a),

$$G_F(\tau)\Big[\lambda\, j^{-8}|j|^3\bar j^{-2}\,\overline{\vartheta_\ell(N\tau)} - j^{-13/2}\sum_{\ell'} W(\ell',\ell)\overline{\vartheta_{\ell'}(\tau)}\Big] = 0.$$

The zeros of $G_F \not\equiv 0$ are isolated and the bracket is continuous, so the bracket vanishes on $\mathbb{H}$. Since $j(\tau)$ lies in the lower half-plane, $\operatorname{Arg}\bar j = -\operatorname{Arg} j$ and $|j|^3 = j^{3/2}\bar j^{3/2}$; thus $j^{-8}|j|^3\bar j^{-2} = j^{-13/2}\bar j^{-1/2}$ and $\overline{\vartheta_\ell(N\tau)} = \lambda^{-1}\bar j^{1/2}\sum_{\ell'} W(\ell',\ell)\overline{\vartheta_{\ell'}(\tau)}$. Take $\tau = -\bar u$. By (26) and (25), $\overline{\vartheta_\ell(N(-\bar u))} = \overline{\vartheta_\ell(-\overline{N^*u})} = \vartheta_\ell(N^*u)$, $\overline{\vartheta_{\ell'}(-\bar u)} = \vartheta_{\ell'}(u)$, and $\overline{j(-\bar u)} = \gamma u + \alpha$. □

**Proposition 10.10** (The transformation on an arc)**.** *Let $(\gamma,\delta) \in \mathcal{C}_\gamma$, $F = F(\gamma,\delta)$, $\operatorname{Re} z > 0$, and let $\tau,\tau'$ be as in* (61)*. Then for $0 \le \ell < 2m$*

$$\begin{aligned} h_{I,\ell}(\tau) &= e^{-3\pi i/4} z^{13/2} \sum_{\ell'=0}^{2m-1} W_{\gamma,\delta}(\ell',\ell)\, h_{F,\ell'}(\tau') \\ &\quad - \frac{\sqrt{m}}{2\sqrt{2}\,\pi} e^{-3\pi i/4} z^{13/2} G_F(\tau') \sum_{g=-m\gamma}^{m\gamma-1} W_{\gamma,\delta}(g,\ell)\,\mathbf{e}\Big(-\frac{\alpha g^2}{4m\gamma}\Big) \\ &\qquad \times \Big[2\cdot\mathbf{1}_{g=0}(\gamma z)^{1/2} - \frac{z}{\sqrt{2m}\,\pi\gamma}\int_{\mathbb{R}} e^{-\frac{2\pi m u^2}{\gamma z}} f_{\gamma,g}(u)\,du\Big]. \end{aligned} \tag{63}$$

*Proof.* Write $N, N^*, j, \lambda, W$ without indices, and

$$\Phi_\ell(\tau') = h_{I,\ell}(N\tau') - j(\tau')^{-13/2}\sum_{\ell'} W(\ell',\ell) h_{F,\ell'}(\tau').$$

By (48) and (46),

$$\Phi_\ell(\tau') = \kappa_m\Big[j(\tau')^{-13/2} G_F(\tau')\sum_{\ell'} W(\ell',\ell)E_{\ell'}(\tau') - G_I(N\tau')E_\ell(N\tau')\Big].$$

*Path independence.* For $a \in \mathbb{H}$ and $\tau \in \mathbb{H}$ the integral of $\vartheta_\ell(w)(w+\tau)^{-3/2}$ from $a$ to $i\infty$ is the same along every piecewise smooth path in $\mathbb{H}$ that is vertical from some point on: for $\operatorname{Im} w \ge 1$ one has $|\vartheta_\ell(w)| \le \sum_r e^{-\pi r^2/2m}$ and $|(w+\tau)^{-3/2}| \le (\operatorname{Im} w)^{-3/2}$, so horizontal connecting segments of bounded length at height $Y$ contribute $O(Y^{-3/2})$, and Cauchy's theorem applies. Also $|\vartheta_\ell(x+it)| \le 2 + (2mt)^{-1/2}$ for $t > 0$ (compare the sum over the progression $r \equiv \ell \pmod{2m}$ with an integral).

*Change of variables.* Put $\tau = N\tau'$; by (26), $-\bar\tau = N^*(-\bar\tau')$. Write $E_\ell(\tau)$ as the integral along the segment from $-\bar\tau$ to $w_1 = \frac{\delta}{\gamma} + i(\operatorname{Im}\tau + 1)$ followed by the vertical ray from $w_1$ upwards. Substitute $w = N^*u$: the segment becomes a path $\Sigma$ in $\mathbb{H}$ from $-\bar\tau'$ to $u_1 = (N^*)^{-1}w_1$, and, since $N^*(-\frac{\alpha}{\gamma} + it) = \frac{\delta}{\gamma} + \frac{i}{\gamma^2 t}$, the ray becomes the vertical segment from $u_1$ down to $-\alpha/\gamma$, which we call $\Sigma'$. By Lemma 10.9, (26) and $dw = (\gamma u+\alpha)^{-2}du$,

$$E_\ell(\tau) = \lambda^{-1}\sum_{\ell'} W(\ell',\ell)\int_{\Sigma\cup\Sigma'} \vartheta_{\ell'}(u)\,(\gamma u+\alpha)^{1/2}\Big(\frac{u+\tau'}{(\gamma u+\alpha)j(\tau')}\Big)^{-3/2}(\gamma u+\alpha)^{-2}du.$$

For $u,\tau' \in \mathbb{H}$ the four numbers $u+\tau'$, $\gamma u+\alpha$, $\overline{j(\tau')}$ and $w+\tau = \frac{u+\tau'}{(\gamma u+\alpha)j(\tau')}$ lie in $\mathbb{H}$, so $\operatorname{Arg}(u+\tau') - \operatorname{Arg}(\gamma u+\alpha) - \operatorname{Arg} j(\tau') - \operatorname{Arg}(w+\tau) \in 2\pi\mathbb{Z}$ is a continuous function on the connected set $\mathbb{H}\times\mathbb{H}$, hence constant; at $u = \tau' = it$ it is $\frac{\pi}{2} - \vartheta_0 + \vartheta_0 - \frac{\pi}{2} = 0$ with $\vartheta_0 =$

$\operatorname{Arg}(\alpha+i\gamma t)$. Therefore $(w+\tau)^{-3/2}=(u+\tau')^{-3/2}(\gamma u+\alpha)^{3/2}j(\tau')^{3/2}$, the powers of $\gamma u+\alpha$ cancel, and

$$E_\ell(N\tau')=\lambda^{-1}j(\tau')^{3/2}\sum_{\ell'}W(\ell',\ell)\int_{\Sigma\cup\Sigma'}\vartheta_{\ell'}(u)(u+\tau')^{-3/2}du.$$

With (T2a), $G_I(N\tau')E_\ell(N\tau')=j(\tau')^{-13/2}G_F(\tau')\sum_{\ell'}W(\ell',\ell)\int_{\Sigma\cup\Sigma'}$. On the other hand $E_{\ell'}(\tau')$ is, by path independence, the integral along $\Sigma$ followed by the vertical ray from $u_1$ upwards. Subtracting,

$$\Phi_\ell(\tau')=\kappa_m\,j(\tau')^{-13/2}G_F(\tau')\sum_{\ell'}W(\ell',\ell)\int_{-\alpha/\gamma}^{i\infty}\vartheta_{\ell'}(u)\,(u+\tau')^{-3/2}\,du, \tag{64}$$

the integral along the vertical line $\operatorname{Re}u=-\alpha/\gamma$; it converges absolutely by the bound for $|\vartheta_{\ell'}(x+it)|$.

*Evaluation.* Let $u=-\frac{\alpha}{\gamma}+it$ and $X=\frac{1}{\gamma z}$, so $\operatorname{Re}X>0$ and $u+\tau'=i(t+X)$ by (61). As $\operatorname{Arg}i+\operatorname{Arg}(t+X)\in(0,\pi)$, $(u+\tau')^{-3/2}=i^{-3/2}(t+X)^{-3/2}$, and $du=i\,dt$. Splitting $r=g+2m\gamma p$ with $g\in[-m\gamma,m\gamma)$ and using $\mathbf{e}(-\alpha r^2/4m\gamma)=\mathbf{e}(-\alpha g^2/4m\gamma)$,

$$\vartheta_{\ell'}\Big(-\frac{\alpha}{\gamma}+it\Big)=\sum_{\substack{-m\gamma\le g<m\gamma\\ g\equiv\ell'\,(2m)}}\mathbf{e}\Big(-\frac{\alpha g^2}{4m\gamma}\Big)\theta_{\gamma,g}(t).$$

By Lemma 10.3, $\sum_{\ell'}W(\ell',\ell)\sum_{g\equiv\ell'}=\sum_{g=-m\gamma}^{m\gamma-1}W(g,\ell)$. Lemma 9.8 with $X^{-1/2}=(\gamma z)^{1/2}$, $X^{-1}=\gamma z$, together with $\kappa_m i^{-1/2}=-\frac{\sqrt{m}}{2\sqrt{2\pi}}$ and (62), turns (64) into the second line of (63); the first line is $j(\tau')^{-13/2}\sum W h_{F,\ell'}(\tau')$ by (62). □

10.4. **Proof of Theorem 10.2.** Fix $\ell$, $n$ with $D=2mn-\ell^2>0$, $B=D/4m$, and an integer $N\ge1$. Write $W=W_{\gamma,\delta}$ when $(\gamma,\delta)$ is clear.

*Step 1: the coefficient as an integral.* By (T1) and Proposition 10.1, $h_{I,\ell}(\tau)\mathbf{e}(-D\tau/4m)=\sum_{n'}c_m^F(n',\ell)\mathbf{e}((n'-n)\tau/2)$, an absolutely and uniformly convergent series on the line $\operatorname{Im}\tau=N^{-2}$ (a Laurent series in $\mathbf{e}(\tau/2)$ with finitely many negative powers). Integrating termwise over a period of length 2,

$$c_m^F(n,\ell)=\frac12\int_{-1/(N+1)}^{2-1/(N+1)}h_{I,\ell}\Big(x+\frac{i}{N^2}\Big)\,\mathbf{e}\Big(-\frac{D}{4m}\Big(x+\frac{i}{N^2}\Big)\Big)dx. \tag{65}$$

By Lemma 10.5 the interval is the union of the arcs $J_{\gamma,\delta}$, $\gamma\le N$. On $J_{\gamma,\delta}$ write $x=-\delta/\gamma+\varphi$, $\varphi\in[-\omega',\omega'']$, so that $x+i/N^2=\tau$ with $z,\tau,\tau'$ as in (61), and

$$\mathbf{e}\Big(-\frac{D\tau}{4m}\Big)=\mathbf{e}\Big(\frac{\delta D}{4m\gamma}\Big)\exp\Big(\frac{2\pi}{\gamma}Bz\Big),\qquad\Big|\exp\Big(\frac{2\pi}{\gamma}Bz\Big)\Big|=e^{2\pi B/N^2}\le e^{2\pi B}. \tag{66}$$

By Lemma 10.6, $\operatorname{Im}\tau'\ge\frac12$ on every arc, so (52) applies at $\tau'$.

*Step 2: decomposition of the integrand.* Insert (63), and in it the expansions (T3) at $\tau'$. Since $\mathbf{e}(\tilde D\tau'/4m)=\mathbf{e}(\alpha\tilde D/4m\gamma)\exp(\frac{2\pi}{\gamma}\frac{|\tilde D|}{4mz})$ and $\mathbf{e}(-\mu_F\tau')=\mathbf{e}(-\alpha\mu_F/\gamma)\exp(\frac{2\pi}{\gamma}\frac{\mu_F}{z})$, the integrand of (65) on $J_{\gamma,\delta}$ equals $\mathbf{e}(\delta D/4m\gamma)$ times the sum of the following terms, with $F=F(\gamma,\delta)$ and

$\kappa' = \sqrt{m}/(2\sqrt{2}\pi)$:

$$\Pi_1 = e^{-3\pi i/4} \sum_{(\tilde\nu,\tilde\ell)\in P_F} a_F(\tilde\nu,\tilde\ell)\,\mathbf{e}\Big(\frac{\alpha\tilde D}{4m\gamma}\Big) W(\tilde\ell,\ell)\, z^{13/2} e^{\frac{2\pi}{\gamma}(\frac{|\tilde D|}{4mz}+Bz)},$$

$$\Pi_2 = -e^{-3\pi i/4}\kappa'\cdot 2\gamma^{1/2} c_F\,\mathbf{e}\Big(-\frac{\alpha\mu_F}{\gamma}\Big) W(0,\ell)\, z^{7} e^{\frac{2\pi}{\gamma}(\frac{\mu_F}{z}+Bz)},$$

$$\Pi_3 = e^{-3\pi i/4}\frac{\kappa'}{\sqrt{2m}\,\pi\gamma} c_F\,\mathbf{e}\Big(-\frac{\alpha\mu_F}{\gamma}\Big)\sum_g W(g,\ell)\mathbf{e}\Big(-\frac{\alpha g^2}{4m\gamma}\Big) \times \int_{|u|<u_F} f_{\gamma,g}(u)\, z^{15/2} e^{\frac{2\pi}{\gamma}(\frac{\mu_F - mu^2}{z}+Bz)}du,$$

and the error terms

$$\Xi_1 = e^{-3\pi i/4} z^{13/2}\sum_{\ell'} W(\ell',\ell)\Big(h_{F,\ell'}(\tau') - \sum_{P_F} a_F \mathbf{e}\Big(\frac{\tilde D\tau'}{4m}\Big)\Big) e^{\frac{2\pi}{\gamma}Bz},$$

$$\Xi_2 = -e^{-3\pi i/4}\kappa'\, z^{13/2}\big(G_F(\tau') - c_F\mathbf{e}(-\mu_F\tau')\big) W(0,\ell)\cdot 2(\gamma z)^{1/2} e^{\frac{2\pi}{\gamma}Bz},$$

$$\Xi_3 = e^{-3\pi i/4}\frac{\kappa' z^{15/2}}{\sqrt{2m}\,\pi\gamma}\sum_g W(g,\ell)\mathbf{e}\Big(-\frac{\alpha g^2}{4m\gamma}\Big)\Big[\big(G_F(\tau') - c_F\mathbf{e}(-\mu_F\tau')\big)\int_{\mathbb{R}} e^{-\frac{2\pi m u^2}{\gamma z}} f_{\gamma,g}\,du + c_F\mathbf{e}(-\mu_F\tau')\int_{|u|\ge u_F} e^{-\frac{2\pi m u^2}{\gamma z}} f_{\gamma,g}\,du\Big] e^{\frac{2\pi}{\gamma}Bz}.$$

Here $\sum_g$ runs over $-m\gamma \le g < m\gamma$, and we used $z^{13/2}(\gamma z)^{1/2} = \gamma^{1/2} z^7$ and $z^{13/2} z = z^{15/2}$ (principal powers of $z$, $\operatorname{Re} z > 0$), and exchanged the order of $\sum_g$ and $\int du$ (finite sum). Note $\kappa'/(\sqrt{2m}\pi) = 1/(4\pi^2)$ and $4\pi\kappa' = \sqrt{2m}$.

*Step 3: the error terms.* We use Lemma 10.3 ($|W| \le \sqrt{\gamma/2m} \le \sqrt{N/2m}$), Lemma 10.6, (52), (66), and the fact that the arcs have total length 2; the factor $\frac12$ of (65) then cancels the length.

- $|\Xi_1| \le (\sqrt{2}/N)^{13/2}\cdot 2m\sqrt{N/2m}\cdot C_h e^{2\pi B} = 2^{13/4}\sqrt{2m}\,C_h e^{2\pi B} N^{-6}$.
- $|\Xi_2| \le \kappa'(\sqrt{2}/N)^{13/2} C_G\sqrt{N/2m}\cdot 2\cdot 2^{1/4} e^{2\pi B} = \frac{2^{5/2}}{\pi} C_G e^{2\pi B} N^{-6}$, using $\gamma|z| \le \sqrt{2}$.
- In $\Xi_3$, $|e^{-2\pi m u^2/\gamma z}| = e^{-2\pi m u^2 \operatorname{Re}(1/\gamma z)} \le e^{-\pi m u^2}$, and for $|u| \ge u_F$,
  $|\mathbf{e}(-\mu_F\tau') e^{-2\pi m u^2/\gamma z}| = e^{-2\pi(mu^2-\mu_F)\operatorname{Re}(1/\gamma z)} \le e^{-\pi m(u^2-u_F^2)} \le e^{-\pi m(|u|-u_F)^2}$.
  Both Gaussian integrals are at most $m^{-1/2}$. With Lemma 10.8,
  $|\Xi_3| \le \dfrac{(\sqrt{2}/N)^{15/2}}{4\pi^2\gamma}\sqrt{\dfrac{\gamma}{2m}}\,\dfrac{C_G + c_*}{\sqrt{m}}\,\dfrac{\pi^4 m^2\gamma^2}{2} e^{2\pi B} \le 2^{1/4}\pi^2 m\,(C_G + c_*) e^{2\pi B} N^{-6}$,
  using $\gamma^{3/2} \le N^{3/2}$.

So the error terms contribute at most $\big(2^{13/4}\sqrt{2m}C_h + \frac{2^{5/2}}{\pi}C_G + 2^{1/4}\pi^2 m(C_G + c_*)\big) e^{2\pi B} N^{-6}$ to (65).

*Step 4: the principal terms.* Apply Lemma 10.7 on $J_{\gamma,\delta}$: to $\Pi_1$ termwise with $r = \frac{13}{2}$, $A = |\tilde D|/4m \le A_*$; to $\Pi_2$ with $r = 7$, $A = \mu_F$; and to $\Pi_3$, after exchanging $\int d\varphi$ and $\int du$ (continuous integrand on a compact set), for each $u$ with $|u| < u_F$, with $r = \frac{15}{2}$ and

$A = \mu_F - mu^2 \in (0, \mu_*]$. Since

$$\frac{4\pi}{\gamma}\sqrt{\frac{|\tilde D|}{4m}\cdot\frac{D}{4m}} = \frac{\pi\sqrt{|\tilde D| D}}{m\gamma}, \qquad \frac{4\pi}{\gamma}\sqrt{\frac{\mu_F D}{4m}} = \frac{2\pi}{\gamma}\sqrt{\frac{\mu_F D}{m}},$$
$$\Big(\frac{4m(\mu_F - mu^2)}{D}\Big)^{\frac{17}{4}} = \Big(\frac{4m\mu_F}{D}\Big)^{\frac{17}{4}}\Big(1 - \frac{mu^2}{\mu_F}\Big)^{\frac{17}{4}},$$

and $2\pi\cdot 2\kappa' = \sqrt{2m}$, $\frac{\kappa'}{\sqrt{2m\pi}}\cdot 2\pi = \frac{1}{2\pi}$, the main terms of Lemma 10.7 give exactly

$$\frac12\,\mathbf{e}\Big(\frac{\delta D}{4m\gamma}\Big)\int_{J_{\gamma,\delta}} (\Pi_1+\Pi_2+\Pi_3)\,d\varphi = \frac12 e^{-3\pi i/4}\big(\mathcal{M}_{\gamma,\delta} + \mathcal{K}_{2;\gamma,\delta} + \mathcal{K}_{3;\gamma,\delta}\big) + \frac12\,\mathbf{e}\Big(\frac{\delta D}{4m\gamma}\Big)(\varepsilon_1+\varepsilon_2+\varepsilon_3),$$

where, by the error bound of Lemma 10.7, Lemma 10.3 and Lemma 10.8,

$$|\varepsilon_1| \le a_*\sqrt{\frac{\gamma}{2m}}\,\frac{2^{17/2}e^{8\pi(A_*+B)}}{\gamma N^{15/2}},$$
$$|\varepsilon_2| \le 2\kappa'\gamma^{1/2}c_*\sqrt{\frac{\gamma}{2m}}\,\frac{2^9 e^{8\pi(\mu_*+B)}}{\gamma N^8},$$
$$|\varepsilon_3| \le \frac{c_*}{4\pi^2\gamma}\sqrt{\frac{\gamma}{2m}}\,2u_*\frac{\pi^4m^2\gamma^2}{2}\,\frac{2^{19/2}e^{8\pi(\mu_*+B)}}{\gamma N^{17/2}},$$

with $u_* = \sqrt{\mu_*/m}$. There are at most $2\gamma$ pairs in $\mathcal{C}_\gamma$. Summing $\frac12(|\varepsilon_1|+|\varepsilon_2|+|\varepsilon_3|)$ over them and over $\gamma \le N$, with $\sum_{\gamma\le N}\gamma^{1/2} \le N^{3/2}$, $\sum_{\gamma\le N}\gamma \le N^2$, $\sum_{\gamma\le N}\gamma^{3/2} \le N^{5/2}$ and $2\kappa'(2m)^{-1/2} = \frac{1}{2\pi}$, gives at most

$$\frac{2^{17/2}a_*}{\sqrt{2m}}e^{8\pi(A_*+B)}N^{-6} + \frac{2^8}{\pi}c_*e^{8\pi(\mu_*+B)}N^{-6} + 2^7\pi^2 m\sqrt{\mu_*}\,c_*e^{8\pi(\mu_*+B)}N^{-6}.$$

*Step 5: conclusion.* Steps 1–4 give (57). The absolute convergence and the limit $N\to\infty$ follow from the next lemma, which shows that $\mathcal{M}^{(N)} + \mathcal{K}_2^{(N)} + \mathcal{K}_3^{(N)} \to \mathcal{M} + \mathcal{K}_2 + \mathcal{K}_3$; letting $N\to\infty$ in (57) gives (56). □

**Lemma 10.11.** *There is a constant $C$, depending on $m, D$ and the data of* (T3)*, such that $|\mathcal{M}_{\gamma,\delta}| + |\mathcal{K}_{2;\gamma,\delta}| + |\mathcal{K}_{3;\gamma,\delta}| \le C\gamma^{-8}$ for all $(\gamma,\delta)\in\mathcal{C}_\gamma$. In particular the three series converge absolutely, and their tails over $\gamma > N$ are at most $2C\sum_{\gamma>N}\gamma^{-7} \le \frac{C}{3}N^{-6}$.*

*Proof.* Use $|W| \le \sqrt{\gamma/2m}$ and $I_\nu(x/\gamma) \le \gamma^{-\nu}I_\nu(x)$ for $\gamma \ge 1$ (Lemma 10.4). In $\mathcal{M}_{\gamma,\delta}$ the factor

$$(|\tilde D|/D)^{15/4} I_{15/2}\big(\pi\sqrt{|\tilde D| D}/m\gamma\big)$$

is increasing in $|\tilde D| \le 4mA_*$, so

$$|\mathcal{M}_{\gamma,\delta}| \le a_*\sqrt{\gamma/2m}\,\frac{2\pi}{\gamma}\,\gamma^{-15/2}(A_*/B)^{15/4}I_{15/2}(4\pi\sqrt{A_*B}).$$

Similarly

$$|\mathcal{K}_{2;\gamma,\delta}| \le \sqrt{2m}\gamma^{-1/2}c_*\sqrt{\gamma/2m}\,\gamma^{-8}(\mu_*/B)^4 I_8(4\pi\sqrt{\mu_*B}).$$

In $\mathcal{K}_3$, $(1-mu^2/\mu_F)^{17/4}\le 1$ and $I_{17/2}(\frac{2\pi}{\gamma}\sqrt{(\mu_F-mu^2)D/m})\le \gamma^{-17/2}I_{17/2}(4\pi\sqrt{\mu_* B})$, so by Lemma 10.8

$$|\mathcal{K}_{3;\gamma,\delta}|\le \frac{\gamma^{-2}}{2\pi}c_*\sqrt{\gamma/2m}\,(\mu_*/B)^{17/4}\,2u_*\frac{\pi^4m^2\gamma^2}{2}\gamma^{-17/2}I_{17/2}(4\pi\sqrt{\mu_* B}).$$

Each bound is $O(\gamma^{-8})$, and $|\mathcal{C}_\gamma|\le 2\gamma$. □

10.5. **Remarks.**

**Remark 10.12** (Signs and normalisations)**.** The derivation fixes every constant of (29)–(31): the factor $\frac12$ (width of $\infty$), the phase $e^{-3\pi i/4}=j^{-13/2}z^{-13/2}$ of (62), the factor $2\pi/\gamma$ and the Bessel orders $r+1=\frac{15}{2},8,\frac{17}{2}$ of Lemma 10.7 for $r=\frac{13}{2},7,\frac{15}{2}$, the coefficient $-\sqrt{2m}\gamma^{-1/2}$ of $\mathcal{K}_2$ (from the term $2X^{-1/2}$ of Lemma 9.8) and $\frac{1}{2\pi}\gamma^{-2}$ of $\mathcal{K}_3$, both through $\kappa_m$, i.e. through the normalisation of the completion term forced by the invariance of $\widehat{\mathcal{A}}$ (Proposition 9.9). With $W_{\gamma,\delta}$ as in (24) and the frame data (28), the result is the right side of (32). Three features of the formula are forced by the derivation, and each would change under a wrong convention: $W$ enters with the polar index in the first slot and the charge index $\ell$ in the second; the phase of $\mathcal{K}_3$ carries $-\alpha g^2$ (from $\vartheta_{\ell'}(-\alpha/\gamma+it)$, Lemma 10.9 being an identity for $\vartheta_\ell\circ N^*$, not $\vartheta_\ell\circ N$); and $\mathcal{K}_2$ and $\mathcal{K}_3$ have opposite signs relative to the $g=0$ term of $\mathcal{K}_3$.

**Remark 10.13** (What is not used)**.** No Kloosterman cancellation is used: all sums over $\delta$ are estimated by $|\mathcal{C}_\gamma|\le 2\gamma$ and the trivial bound for $W$. The only properties of the remainder series in (T3) that are used are (52), which follows from absolute convergence at height $\frac12$; terms with exponent 0 (constant terms) are harmless because the weight is negative. Lemma 10.9 shows that (T2b) forces the shadow to transform by the matrix $W$ under $N^*_{\gamma,\delta}$, in agreement with the classical transformation of Lemma 9.7.

*Proof of Theorem* 6.2. By Proposition 10.1 the finite part has the properties (T1)–(T3) with the data of Section 6, so Theorem 10.2 applies; for $0\le\ell\le m$ its second form is (32), and the absolute convergence is part of it. □

## 11. The leading family and an eta quotient

Fix $m\ge 1$ and a charge $(m,n,\ell)$ with $0\le\ell\le m$ and $D>0$; write $x=\sqrt{D}/m$, $\theta=\pi\ell/m$ and $t_0=m^2+4m$. The *extreme term* is

$$\mathcal{E}(D,m)=\frac{\pi}{\sqrt{2m}}\cdot\frac{m+2}{16}\Big(\frac{t_0}{D}\Big)^{\frac{15}{4}}I_{\frac{15}{2}}\big(\pi x\sqrt{t_0}\big)>0. \tag{67}$$

It is the contribution to $\frac12(-1)^\ell\sqrt{i}\,\mathcal{M}$ of the frame $S$, $\gamma=1$, and the most polar term $\tilde\nu=-1$, $\tilde\ell=m$, whose only wall is $G_0=\left(\begin{smallmatrix}1&0\\1&1\end{smallmatrix}\right)$ with $m_{G_0}=n_{G_0}=-1$ and $\ell_{G_0}=m+2$.

11.1. **The terms with $\gamma=1$.** For $\gamma=1$ the set $\mathcal{C}_1$ consists of $\delta=0$ and $\delta=-1$, both with $\alpha=0$ and $\beta=-1$. The matrix $\left(\begin{smallmatrix}0&1\\-1&0\end{smallmatrix}\right)$ has frame $S$, and $\left(\begin{smallmatrix}-1&1\\-1&0\end{smallmatrix}\right)$ has frame $T$. So the terms with $\gamma=1$ are one term in frame $S$ and one in frame $T$.

In frame $S$, $W_{1,0}(\tilde\ell,\ell)=(2mi)^{-1/2}\mathbf{e}(-\ell\tilde\ell/2m)$ and the phase in (29) is 1. Since $\sqrt{i}/\sqrt{2mi}=1/\sqrt{2m}$,

$$\tfrac12(-1)^\ell\sqrt{i}\,\mathcal{M}_{\gamma=1,S}=\frac{\pi}{\sqrt{2m}}\sum_{(\tilde\nu,\tilde\ell)}a_S(\tilde\nu,\tilde\ell)\,(-1)^\ell\mathbf{e}\Big(-\frac{\ell\tilde\ell}{2m}\Big)\Big(\frac{t}{D}\Big)^{\frac{15}{4}}I_{\frac{15}{2}}(\pi x\sqrt{t}),\qquad t=-\tilde D. \tag{68}$$

Split $a_S = a_S^{\mathrm{lead}} + a_S^{\mathrm{rest}}$, where $a_S^{\mathrm{lead}}$ is the term of $G_0$ in (22) and $a_S^{\mathrm{rest}}$ the sum over the walls with $rs \geq 2$. Lemma 5.1(vii) identifies the polar terms carrying $G_0$ with the set

$$\mathcal{L}_m = \big\{(a,b) \in \mathbb{Z}^2 :\ a,b \geq -1,\ k = m-a-b \geq 1,\ t_{a,b} = k^2 - 4ab > 0,\ -m \leq b-a < m\big\},$$

by $\tilde\nu = b$, $\tilde\ell = m+b-a$, with $a_S^{\mathrm{lead}} = \frac{1}{16} k\, d_1(a) d_1(b)$ and $t = t_{a,b}$. For $(a,b) \in \mathcal{L}_m$ put $S = a+b+2$ and

$$\rho_{a,b}(x) = \frac{k}{m+2} \Big(\frac{t_{a,b}}{t_0}\Big)^{\frac{15}{4}} \frac{I_{15/2}(\pi x \sqrt{t_{a,b}})}{I_{15/2}(\pi x \sqrt{t_0})}. \tag{69}$$

**Theorem 11.1.** *The leading part of* (68) *is*

$$\begin{aligned} &\frac{\pi}{\sqrt{2m}} \sum_{(\tilde\nu,\tilde\ell)} a_S^{\mathrm{lead}}(\tilde\nu,\tilde\ell)\, (-1)^{\ell} \mathbf{e}\Big(-\frac{\ell\tilde\ell}{2m}\Big) \Big(\frac{t}{D}\Big)^{\frac{15}{4}} I_{\frac{15}{2}}(\pi x\sqrt{t}) \\ &\qquad = \mathcal{E}(D,m) \sum_{(a,b)\in\mathcal{L}_m} d_1(a) d_1(b)\, \rho_{a,b}(x) \cos\big((a-b)\theta\big). \end{aligned} \tag{70}$$

*Moreover* $\rho_{-1,-1} = 1$, *and with* $z = e^{-\pi x + i\theta}$,

$$\sum_{a,b\geq -1} d_1(a) d_1(b)\, e^{-\pi x(a+b+2)} \cos\big((a-b)\theta\big) = |P(z)|^2, \qquad P(z) = \prod_{n\geq 1} (1-z^n)^{-8} (1-z^{2n})^{-8}. \tag{71}$$

*Proof.* With $\tilde\ell = m+b-a$ one has $(-1)^{\ell}\mathbf{e}(-\ell\tilde\ell/2m) = (-1)^{\ell}(-1)^{\ell}\mathbf{e}(\ell(a-b)/2m) = e^{i(a-b)\theta}$. Dividing the term of $(a,b)$ by $\mathcal{E}(D,m)$ leaves $d_1(a)d_1(b)\rho_{a,b}(x) e^{i(a-b)\theta}$. The conditions defining $\mathcal{L}_m$ other than $-m \leq b-a < m$ are symmetric in $(a,b)$, and the weight $d_1(a)d_1(b)\rho_{a,b}$ is symmetric; so the terms of $(a,b)$ and $(b,a)$ are complex conjugate when both lie in $\mathcal{L}_m$, and the only unpaired terms have $|a-b| = m$, where $e^{i(a-b)\theta} = e^{\pm i\pi\ell}$ is real. Hence the sum equals its real part, which is (70). For $a = b = -1$ one has $k = m+2$ and $t = (m+2)^2 - 4 = t_0$. Finally $P(z) = \sum_{a\geq -1} d_1(a) z^{a+1}$ by (21), so $|P(z)|^2 = \sum_{a,b} d_1(a)d_1(b) z^{a+1}\bar z^{b+1}$; since $z^{a+1}\bar z^{b+1} = e^{-\pi x S} e^{i(a-b)\theta}$ and the sum is symmetric, it is real and equals (71). □

The comparison between (70) and (71) rests on the following estimates for the exponents. Write $\Delta_s = \sqrt{t_0} - \sqrt{t_{a,b}}$.

**Lemma 11.2.** *Let* $(a,b) \in \mathcal{L}_m$ *and* $S = a+b+2 \geq 1$. *Then*

(i) $t_0 - t_{a,b} = S(2m-S) + 4(a+1)(b+1) \leq 2mS$, *and* $t_{a,b} \geq m(m+4-2S) = t_0 - 2mS$;

(ii) $S - \Delta_s \leq \eta_- := \dfrac{S(S+4)}{2m+4}$;

(iii) $\Delta_s - S \leq \eta_+$, *where* $\eta_+ = \max\big(0, \sqrt{m(m+4)} - \sqrt{m(m+4-2S)} - S\big)$ *if* $2S \leq m+4$ *and* $\eta_+ = m+2-S$ *otherwise;*

(iv) $\Delta_s \geq \dfrac{S(2m-S)}{2(m+2)}$, *and* $S \leq m+1$.

*Proof.* (i) is Lemma 5.1(iv) with $E_{G_0} = 0$, together with $4(a+1)(b+1) \leq S^2$; and $4ab \leq (a+b)^2 = (S-2)^2$ gives $t_{a,b} = (m+2-S)^2 - 4ab \geq (m+2-S)^2 - (S-2)^2 = m(m+4-2S)$. (iv) $S \leq m+1$ because $k \geq 1$; then $\Delta_s = (t_0 - t)/(\sqrt{t_0} + \sqrt{t}) \geq S(2m-S)/(2\sqrt{t_0})$ and $\sqrt{t_0} \leq m+2$. (ii) follows from (iv): $S - \Delta_s \leq S - \frac{S(2m-S)}{2m+4} = \frac{S(S+4)}{2m+4}$. (iii) If $2S \leq m+4$, then $\sqrt{t} \geq \sqrt{m(m+4-2S)}$ by (i); otherwise $\Delta_s \leq \sqrt{t_0} \leq m+2$. □

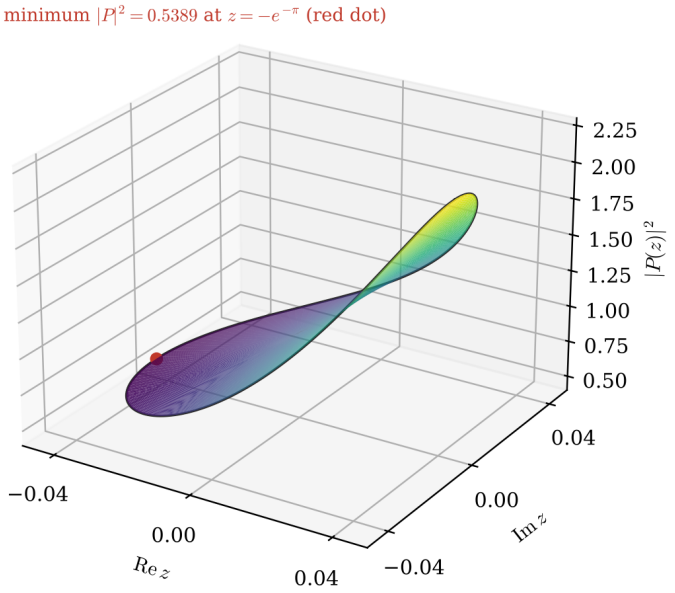



FIGURE 3. The surface $|P(z)|^2$ over the disc $|z| \le e^{-\pi}$, where $z = e^{-\pi x + i\theta}$ with $x \ge 1$. The rim is the image of $x = 1$; the lowest point is $z = -e^{-\pi}$, which corresponds to $x = 1$ and $\ell = m$ odd.

**Proposition 11.3.** *On the closed disc* $|z| \le e^{-\pi}$,

$$|P(z)|^2 \;\ge\; 0.53816,$$

*and the minimum is attained at* $z = -e^{-\pi}$, *where* $|P|^2 = 0.53887\ldots$.

*Proof.* $P$ is holomorphic and zero-free on $|z| < 1$, so by the minimum modulus principle the minimum of $|P|$ on the disc is attained on the circle $|z| = r_0 = e^{-\pi}$. Since the coefficients of $P$ and $zP'$ are nonnegative, $|P(z)| \le P(r_0)$ and $|zP'(z)| \le r_0 P'(r_0)$ on the circle, so $\theta \mapsto |P(r_0 e^{i\theta})|^2$ is Lipschitz with constant $L = 2P(r_0)\, r_0 P'(r_0)$. Since $|P(\bar z)| = |P(z)|$ it suffices to take $\theta \in [0, \pi]$. We evaluate $P$ at $\theta_k = \pi k/4000$, $0 \le k \le 4000$, in ball arithmetic [22], with the product truncated at $n = 60$ and the tail enclosed by $|\log \prod_{n>60}(1 - z^n)^{-8}(1 - z^{2n})^{-8}| \le 16 r_0^{61}/((1 - r_0)(1 - r_0^{61}))$. The least lower bound obtained, minus $L\pi/8000$, is 0.53816. □

**Remark 11.4.** The function $1/(\eta(\tau)^8 \eta(2\tau)^8)$, whose coefficients are $d_1$, is the generating function of the electric half-BPS degeneracies of the model [28]. Theorem 11.1 therefore says that, to leading order, the single-centred index of a quarter-BPS black hole is the extreme Bessel term times the squared modulus of the half-BPS generating function, evaluated at the point $z = e^{-\pi\sqrt{D}/Q^2 + i\pi Q\cdot P/Q^2}$. The positivity of $|P|^2$ is automatic; what has to be proved is that the corrections cannot overturn it, and Proposition 11.3 gives the room.

## 12. Facts on Bessel functions

All Bessel functions below have real order $\nu \ge \frac{1}{2}$ and positive argument. We use

$$g(y) = \sqrt{2\pi y}\, e^{-y} I_{15/2}(y), \qquad \phi(y) = \frac{y\, g'(y)}{g(y)}. \tag{72}$$

**Lemma 12.1.** (i) *$I_\nu(z)/z^\nu$ is increasing; hence $I_\nu(z/\gamma)\le\gamma^{-\nu}I_\nu(z)$ for $\gamma\ge1$.*
(ii) *$z\,I_\nu'(z)/I_\nu(z)$ is increasing in $z$. Consequently, for $0<c\le1$ the function $y\mapsto I_\nu(cy)/I_\nu(y)$ is nonincreasing.*
(iii) *For $\nu'\ge\nu$, $z\,I_{\nu'}'(z)/I_{\nu'}(z)\le(\nu'-\nu)+z\,I_\nu'(z)/I_\nu(z)$.*
(iv) *For fixed $z>0$, $I_\nu(z)$ is nonincreasing in $\nu\ge0$.*
(v) $I_\nu(z)\le\dfrac{(z/2)^\nu}{\Gamma(\nu+1)}e^{z^2/4}$.

*Proof.* (i) $I_\nu(z)/z^\nu=2^{-\nu}\sum_k(z^2/4)^k/(k!\,\Gamma(\nu+k+1))$ has positive coefficients. (v) follows from the same series and $\Gamma(\nu+k+1)\ge\Gamma(\nu+1)$. For (ii) and (iii) we use the Weierstrass product $I_\nu(z)=\frac{(z/2)^\nu}{\Gamma(\nu+1)}\prod_{k\ge1}\big(1+z^2/j_{\nu,k}^2\big)$, where $0<j_{\nu,1}<j_{\nu,2}<\dots$ are the positive zeros of $J_\nu$ [32, §15.41]. It gives

$$\frac{zI_\nu'(z)}{I_\nu(z)}=\nu+\sum_{k\ge1}\frac{2z^2}{j_{\nu,k}^2+z^2},$$

each summand increasing in $z$, which is the first part of (ii). For the second, $\frac{d}{dy}\log\frac{I_\nu(cy)}{I_\nu(y)}=\frac1y\Big(\frac{cyI_\nu'(cy)}{I_\nu(cy)}-\frac{yI_\nu'(y)}{I_\nu(y)}\Big)\le0$. For (iii), the zeros $j_{\nu,k}$ increase with $\nu$ [32, §15.6], so each summand for $\nu'$ is at most the one for $\nu$. (iv) is a theorem of Cochran [11]. □

**Lemma 12.2.** (i) *$0<g(y)<1$ for all $y>0$, and $g(y)\to1$ as $y\to\infty$.*
(ii) *$\phi$ is positive and decreasing on $[3,\infty)$; in particular $g$ is increasing on $[3,\infty)$, and for $0<c\le1$ the ratio $g(cy)/g(y)$ is nondecreasing on $\{y:\ cy\ge3\}$.*
(iii) *$y\,\phi(y)\le29$ for $y\ge30$; hence $g'(y)\le29/y^2$ there.*
(iv) *$\sqrt{2\pi y}\,e^{-y}I_\nu(y)\le1$ for every $\nu\ge\frac{15}{2}$ and $y>0$.*

*Proof.* For half-integral order there is the closed form [32, §3.71]

$$g(y)=p\Big(-\frac{1}{2y}\Big)+e^{-2y}p\Big(\frac{1}{2y}\Big),\qquad p(w)=\sum_{k=0}^{7}\frac{(7+k)!}{k!\,(7-k)!}w^k,\tag{73}$$

which gives $g(y)\to1$. Write $r=I_{17/2}/I_{15/2}$; the recurrences give $\phi=8-y+yr$ and $\phi'=y-1-15r-yr^2$ (we use $I_\nu'=I_{\nu+1}+\frac{\nu}{y}I_\nu$ and $r'=1-\frac{16}{y}r-r^2$). On $[3,60]$ we enclose $\phi'$ on a cover of the interval by boxes, in ball arithmetic, by the mean value form $\phi'(c)+[-\frac{h}{2},\frac{h}{2}]\,\phi''(\text{box})$; every enclosure is negative. On $[60,\infty)$ we use (73): with $w=1/2y$, the main part $w\,p'(-w)/p(-w)$ of $\phi$ has positive derivative in $w$ on $(0,\frac{1}{120}]$ (checked on 600 subintervals), and the remaining part is $O(y^2e^{-2y})$ together with its derivative, which is far smaller. This proves that $\phi$ is decreasing on $[3,\infty)$, and $\phi(60)>0$ together with $\phi\to0^+$ gives positivity. Since $g$ increases on $[3,\infty)$ to the limit 1, $g<1$ there; on $(0,3]$, (v) of Lemma 12.1 gives $g(y)\le\sqrt{6\pi}\,(3/2)^{15/2}e^{9/4-3}/\Gamma(17/2)<0.01$. The monotonicity of $g(cy)/g(y)$ is (ii) of Lemma 12.1 with $\phi$ in place of $zI'/I$. (iii) on $[30,60]$ is checked on a grid of step 0.05 using that $\phi$ is decreasing, and on $[60,\infty)$ from the closed form, where $y\phi(y)\le\frac12p'(-w)/p(-w)+O(y^3e^{-2y})<28$. Finally (iv) follows from (i) and Lemma 12.1(iv). □

The following monotonicity in $x$ is what makes every bound of Section 13, proved at the smallest admissible $x$, valid for all larger $x$.

**Lemma 12.3.** *Let $x=\sqrt{D}/m$ and let $\mathcal{E}(D,m)$ be as in (67). For a constant $0<c\le\pi\sqrt{t_0}$, each of the following ratios is a nonincreasing function of $x>0$:*

$$\frac{D^{-15/4}I_{15/2}(cx)}{\mathcal{E}(D,m)},\qquad\frac{D^{-4}I_8(cx)}{\mathcal{E}(D,m)},\qquad\frac{D^{-17/4}I_{17/2}(cx)}{\mathcal{E}(D,m)}.$$

*Proof.* Up to constants, $\mathcal{E} = D^{-15/4} I_{15/2}(c_0 x)$ with $c_0 = \pi\sqrt{t_0} \geq c$, and $D = m^2 x^2$. The first ratio is $I_{15/2}(cx)/I_{15/2}(c_0 x)$, nonincreasing by Lemma 12.1(ii). The logarithmic derivative in $x$ of the second is

$$\frac{1}{x}\Big( -\frac{1}{2} + \frac{cxI_8'(cx)}{I_8(cx)} - \frac{c_0 x I_{15/2}'(c_0 x)}{I_{15/2}(c_0 x)}\Big) \leq \frac{1}{x}\Big( -\frac{1}{2} + \frac{1}{2} + \frac{cxI_{15/2}'(cx)}{I_{15/2}(cx)} - \frac{c_0 x I_{15/2}'(c_0 x)}{I_{15/2}(c_0 x)}\Big) \leq 0$$

by Lemma 12.1(iii) and (ii). The third is the same with $-1$ and $+1$. □

**Remark 12.4.** In Lemma 12.3 the extra power $D^{-1/4}$ of the shadow terms exactly compensates the higher order of the Bessel function. This is the level-two counterpart of the fact, used by Bringmann and Manschot [6] and by Rosselló at level one, that the shadow terms of a mixed mock form grow at most like its principal-part terms.

**Lemma 12.5.** *For $n \geq -1$, $0 < d_1(n) \leq e^{2\pi\sqrt{2(n+1)}}$. More generally $d_1(n) \leq P(q)\, q^{-(n+1)}$ for every $0 < q < 1$.*

*Proof.* Positivity is clear from the product. The second bound holds since $P(q) = \sum d_1(a) q^{a+1}$ has nonnegative coefficients. With $q = e^{-\tau}$, comparison of the sum with an integral gives $\sum_{k\geq 1} -\log(1-e^{-k\tau}) \leq \int_0^\infty -\log(1-e^{-u\tau})\, du = \pi^2/6\tau$, so $\log P(e^{-\tau}) \leq 8\cdot\frac{\pi^2}{6\tau} + 8\cdot\frac{\pi^2}{12\tau} = \frac{2\pi^2}{\tau}$. Taking $\tau = \pi\sqrt{2/(n+1)}$ gives the first bound. □

## 13. Bounds for the remaining terms

Throughout this section $(m,n,\ell)$ is a reduced charge, $x = \sqrt{D}/m \geq x_0 \geq 1$, and all bounds are uniform in $\ell$ and in $x \geq x_0$. We write $B_\gamma(t;x) = (t/t_0)^{15/4} I_{15/2}(\pi x\sqrt{t}/\gamma)/I_{15/2}(\pi x\sqrt{t_0})$ and $u = e^{-\pi}$. By Lemma 12.1(i),(ii), $B_\gamma(t;x)$ is increasing in $t$ and nonincreasing in $x$, and $B_\gamma \leq \gamma^{-15/2} B_1$.

13.1. **Decomposition.** By Theorem 6.2, $d^*(m,n,\ell) = \frac{1}{2}(-1)^\ell\sqrt{i}(\mathcal{M} + \mathcal{K}_2 + \mathcal{K}_3)$, and Theorem 11.1 splits this as

$$\frac{d^*(m,n,\ell)}{\mathcal{E}(D,m)} = |P(z)|^2 - \mathcal{X}_{\text{miss}} + \mathcal{X}_{\text{main}} + \mathcal{X}_{rs} + \mathcal{X}_T + \mathcal{X}_{\gamma\geq 2} + \mathcal{X}_{K_2} + \mathcal{X}_{K_3}, \tag{74}$$

where

- $\mathcal{X}_{\text{miss}} = \sum d_1(a) d_1(b) e^{-\pi x S}\cos((a-b)\theta)$ over the pairs $a, b \geq -1$ not in $\mathcal{L}_m$;
- $\mathcal{X}_{\text{main}} = \sum_{(a,b)\in\mathcal{L}_m} d_1(a) d_1(b)\big(\rho_{a,b}(x) - e^{-\pi x S}\big)\cos((a-b)\theta)$;
- $\mathcal{X}_{rs}$ is the contribution of the walls with $rs \geq 2$ to the frame $S$ term with $\gamma = 1$, divided by $\mathcal{E}$;
- $\mathcal{X}_T$ is the frame $T$ term with $\gamma = 1$, divided by $\mathcal{E}$;
- $\mathcal{X}_{\gamma\geq 2}$ collects the terms of $\mathcal{M}$ with $\gamma \geq 2$, and $\mathcal{X}_{K_j}$ the term $\frac{1}{2}(-1)^\ell\sqrt{i}\,\mathcal{K}_j$, divided by $\mathcal{E}$.

We bound each piece in absolute value.

13.2. **Parametrising the walls.**

**Lemma 13.1.** *The map sending a polar term $(\tilde\nu, \tilde\ell)$ with $0 \leq \tilde\ell < 2m$ and a wall $G \in \mathcal{W}(m, \tilde\nu, \tilde\ell)$ to $(r, s, m_G, n_G)$ is injective. Given $(r,s,a,b)$ in its image, $\delta$ is the unique solution of*

$$\sqrt{\delta^2 + \frac{a}{mr^2}} + \sqrt{\delta^2 + \frac{b}{ms^2}} = \frac{1}{rs}, \tag{75}$$

*so $t = 4m^2\delta^2$ is a function $t(r,s,a,b)$, and $\ell_G \leq m - a - b$.*

*Proof.* By Lemma 5.1(ii), $U=\sqrt{\delta^2+a/mr^2}$ and $W=\sqrt{\delta^2+b/ms^2}$, and $U+W=1/rs$; the left side of (75) is increasing in $\delta^2$, which gives uniqueness, and $\ell_G=2mrs(UW+\delta^2)$ is then determined. The form $\phi\circ G$ has coefficients $(n_G,-\ell_G,m_G)$ up to the order of the variables, so it is determined; given $(r,s)$, the matrices $G$ are $\left(\begin{smallmatrix}p_0+kr & q_0+ks\\ r & s\end{smallmatrix}\right)$, $k\in\mathbb{Z}$; for a fixed form $\phi\circ G$ the choice of $k$ shifts $\omega_0=\tilde\ell/2m$ by the integer $k$, that is, $\tilde\ell$ by $2mk$; the condition $0\le\tilde\ell<2m$ fixes $k$, and then $\phi=(\phi\circ G)\circ G^{-1}$ is determined. Finally $\ell_G=m-a-b-E_G\le m-a-b$ by Lemma 5.1(iii). □

13.3. **The leading family.**

**Lemma 13.2.** *Let $(a,b)\in\mathcal{L}_m$ with $S\ge1$, and put $c_0=\frac{k}{m+2}(t/t_0)^{7/2}$ and $\hat g_0=g(\pi x_0\sqrt{t})/g(\pi x_0\sqrt{t_0})$. Then*

$$\sup_{x\ge x_0}\left|\rho_{a,b}(x)-e^{-\pi xS}\right|\le\max_{C\in\{c_0,\,c_0\hat g_0\}}\ \sup_{x\ge x_0}\left|Ce^{-\pi\Delta_s x}-e^{-\pi Sx}\right|.$$

*Proof.* From $I_{15/2}(y)=g(y)e^y/\sqrt{2\pi y}$ one gets $\rho_{a,b}(x)=c_0\,\hat g(x)\,e^{-\pi\Delta_s x}$ with $\hat g(x)=g(\pi x\sqrt{t})/g(\pi x\sqrt{t_0})$. Since $t\ge1$, $\pi x\sqrt{t}\ge3$, and Lemma 12.2(ii) shows $\hat g_0\le\hat g(x)\le1$. The expression is affine in $\hat g$, so its modulus is at most the larger of the values at the two ends. □

The supremum of $|Ce^{-\alpha x}-e^{-\beta x}|$ on $[x_0,\infty)$ is attained at $x_0$ or at the unique critical point $x_*=\log(\beta/C\alpha)/(\beta-\alpha)$ when $x_*>x_0$, and is computed exactly. Summing,

$$|\mathcal{X}_{\rm main}|\le\sum_{(a,b)\in\mathcal{L}_m,\ S\ge1}d_1(a)d_1(b)\max_{C}\sup_{x\ge x_0}\left|Ce^{-\pi\Delta_s x}-e^{-\pi Sx}\right|=:\epsilon_{\rm main}(m,x_0),\tag{76}$$

$$|\mathcal{X}_{\rm miss}|\le\sum_{(a,b)\notin\mathcal{L}_m}d_1(a)d_1(b)u^{x_0S}=:\epsilon_{\rm miss}(m,x_0).\tag{77}$$

The pairs not in $\mathcal{L}_m$ have $S\ge m/2+2$ (if $t\le0$ then $(m+2-S)^2\le4ab\le(S-2)^2$) or $|a-b|\ge m$ (so $S\ge m$) or $k\le0$; we sum them exactly for $S\le4m+10$, and for larger $S$ use $\sum_{a+b=S-2}d_1(a)d_1(b)\le P(q_1)^2q_1^{-S}$ with $q_1=u^{x_0/2}$, by Lemma 12.5.

For large $m$ a coarser bound is convenient.

**Lemma 13.3.** *Let $m\ge3$, $x_0=1$, $1\le S\le m+1$, and let $\eta_\pm$ be as in Lemma 11.2. Assume $S-\eta_-\ge1/\pi$. Then every $(a,b)\in\mathcal{L}_m$ with $a+b+2=S$ satisfies*

$$\sup_{x\ge1}\left|\rho_{a,b}(x)-e^{-\pi xS}\right|\le u^S\,\varphi(S,m),\qquad\varphi(S,m)=\epsilon_1+e^{\pi\eta_-}-1+\min(1,\pi\eta_+),$$

*where $\epsilon_1=\min\left(1,\ \frac{8S}{m}+\frac{29\pi(S+\eta_+)}{y_t^2g(y_t)}\right)$ with $y_t=\pi\sqrt{t_0-2mS}$ if $y_t\ge30$, and $\epsilon_1=1$ otherwise.*

*Proof.* Write $\rho=c_0\hat g\,e^{-\pi x\Delta_s}=c\,e^{\pi xy}e^{-\pi xS}$ with $c=c_0\hat g\le1$ and $y=S-\Delta_s\in[-\eta_+,\eta_-]$. Then $|ce^{\pi xy}-1|\le c|e^{\pi xy}-1|+(1-c)\le(e^{\pi x\eta_-}-1)+\min(1,\pi x\eta_+)+(1-c)$. First, $1-c\le(1-\frac{k}{m+2})+(1-(t/t_0)^{7/2})+(1-\hat g)$; here $1-\frac{k}{m+2}=\frac{S}{m+2}$, $1-(t/t_0)^{7/2}\le\frac72\cdot\frac{2mS}{t_0}\le\frac{7S}{m}$ by Lemma 11.2(i), and, with $z=\pi x\sqrt{t}\ge y_t$ and $z_0=\pi x\sqrt{t_0}$, Lemma 12.2(ii),(iii) give

$$1-\hat g\le\frac{g(z_0)-g(z)}{g(z)}\le\frac{29(z_0-z)}{z^2g(z)}\le\frac{29\pi x(S+\eta_+)}{\pi^2x^2t\,g(y_t)}\le\frac{29\pi(S+\eta_+)}{y_t^2g(y_t)}$$

when $y_t\ge30$; trivially $1-c\le1$. Multiply by $e^{-\pi xS}$. The function $e^{-\pi xS}(e^{\pi x\eta_-}-1)=e^{-\pi x(S-\eta_-)}-e^{-\pi xS}$ is nonincreasing on $[1,\infty)$ as soon as $S/(S-\eta_-)\le e^{\pi\eta_-}$, which follows from $e^{\pi\eta_-}\ge1+\pi\eta_-$ and $S-\eta_-\ge1/\pi$; the function $xe^{-\pi xS}$ is decreasing for $x\ge1/\pi S$; and the bound for $1-c$ is nonincreasing in $x$. So the supremum is at most the value at $x=1$. □

13.4. **The other walls at $\gamma = 1$.** Dividing (68) restricted to $a_S^{\mathrm{rest}}$, and the frame $T$ term, by $\mathcal{E}$, and using $|W_{1,\delta}| = (2m)^{-1/2}$, Lemma 13.1 and the monotonicity of $B_1$,

$$|\mathcal{X}_{rs}| \le \frac{1}{m+2} \sum_{\substack{r,s\le 2m,\ rs\ge 2\\ a,b\ge -1}} (m-a-b)\, d_1(a) d_1(b)\, B_1\big(t(r,s,a,b);x_0\big) =: \epsilon_{rs}, \tag{78}$$

$$|\mathcal{X}_T| \le \frac{16}{m+2} \sum_{\substack{r,s\le 2m\\ r \text{ or } s \text{ even}}} \sum_{a,b\in\frac12\mathbb{Z}} (m-a-b)\, w(r,s,a,b)\, B_1\big(t(r,s,a,b);x_0\big) =: \epsilon_T, \tag{79}$$

where $w = d_1(a)d_1(2b)$ if $r$ is even and $w = d_1(2a)d_1(b)$ if $s$ is even. The ranges are finite by Lemma 5.1(v): $a \le m/s^2$, $b \le m/r^2$. Only quadruples for which (75) has a solution occur; in the certificate an upper bound for $\delta^2$ is found by bisection and then certified in ball arithmetic, and a quadruple is discarded only when the left side of (75) exceeds $1/rs$ already at the least admissible $\delta^2$, again certified.

For $m \ge 101$ we use instead the following bound.

**Lemma 13.4.** *For $m \ge 101$, with $c = \frac{m-1}{4(m+2)}$, $q = e^{-\pi c}$, $A_1 = \frac{m}{2} - 3 - \frac{2}{m}$ and $g_0 = g(\pi\sqrt{t_0})$,*

$$|\mathcal{X}_{rs}| + |\mathcal{X}_T| \le \frac{(2m+1)^2\big(P(q)^2 + 64\sqrt{q}\,P(q)P(\sqrt{q})\big)e^{-\pi A_1/2}}{g_0}.$$

*Proof.* For a wall with $rs \ge 2$, Lemma 5.1(vi) gives $\sqrt{t} \le \frac{m}{2} + 5$, and $\sqrt{t_0} \ge m + 2 - \frac{2}{m}$, so $\Delta_s = \sqrt{t_0} - \sqrt{t} \ge A_1$. Also $\Delta_s \ge S(2m-S)/(2(m+2))$ with $S = a+b+2 \le m+1$ (by Lemma 5.1(ii), $a+b \le m(r^2U^2 + s^2W^2)$, and on the segment $U+W = 1/rs$ this convex function is at most $m\max(r^{-2}, s^{-2}) \le m$; $a+b = m$ is excluded since $\ell_G \le m-a-b$ must be positive), hence $\Delta_s \ge 2cS$. So $e^{-\pi x\Delta_s} \le e^{-\pi A_1/2} q^S$. As in Lemma 13.2, $B_1(t;x) \le (t/t_0)^{7/2} e^{-\pi x\Delta_s} g(\pi x\sqrt{t})/g(\pi x\sqrt{t_0}) \le e^{-\pi x\Delta_s}/g_0$. With $m-a-b \le m+2$, the sum over $(a,b)$ of $d_1(a)d_1(b)q^S$ is at most $P(q)^2$, the sum of $d_1(a)d_1(2b)q^{a+b+2}$ over $a \in \mathbb{Z}$, $b \in \frac12\mathbb{Z}$ is at most $\sqrt{q}\,P(q)P(\sqrt{q})$, and there are at most $(2m)^2$ pairs $(r,s)$. The factor 64 accounts for the factor 16, the two parity types and a factor 2 of room. □

13.5. **The terms with $\gamma \ge 2$.** Let $N_F(\gamma)$ be the number of $\delta$ with $(\gamma,\delta) \in \mathcal{C}_\gamma$ and frame $F$; then $N_I + N_T + N_S \le 2\gamma$. Each term of $\mathcal{M}$ with $\gamma \ge 2$, divided by $\mathcal{E}$, is at most $\frac{16}{m+2}|a_F| \cdot \sqrt{2m}\,|W| \cdot \gamma^{-1} B_\gamma(t;x)$, and $\sqrt{2m}|W| \le \sqrt{2\gcd(m,\gamma)}$ by Lemma 6.1. Bounding $|a_S|$ and $|a_I|$ by the sums of the absolute values of their walls,

$$\begin{aligned} |\mathcal{X}_{\gamma\ge 2}| \le \sum_{\gamma=2}^{G} \frac{\sqrt{2\gcd(m,\gamma)}}{\gamma}\big(N_S(\gamma)\Sigma_S(\gamma) + (N_I+N_T)(\gamma)\,\Sigma_T(\gamma)\big) \\ + \frac{2\sqrt{2}\,(\Sigma_S(1)+\Sigma_T(1))}{6G^6} =: \epsilon_{\gamma\ge 2}, \end{aligned} \tag{80}$$

where $\Sigma_S(\gamma)$ and $\Sigma_T(\gamma)$ are the sums (78) (including the leading family) and (79) with $B_1$ replaced by $B_\gamma$. The tail uses $B_\gamma \le \gamma^{-15/2} B_1$, $N \le 2\gamma$, $\gcd(m,\gamma) \le \gamma$ and $\sum_{\gamma>G} \gamma^{-7} \le G^{-6}/6$. We take $G = 12$.

**Lemma 13.5.** *For $m \ge 101$, with $q, g_0$ as in Lemma 13.4,*

$$|\mathcal{X}_{\gamma\ge 2}| \le 2^9\big(\zeta(7)-1\big)\sqrt{2}\,e^{-\pi\sqrt{t_0}/2}\,\frac{(2m+1)^2\big(P(q)^2 + 64\sqrt{q}\,P(q)P(\sqrt{q})\big)}{g_0}.$$

*Proof.* By Lemma 12.1(i), $I(z/\gamma) \le (2/\gamma)^{15/2} I(z/2)$ for $\gamma \ge 2$. By Lemma 12.2(i), $I(z/2) \le e^{z/2}/\sqrt{\pi z}$, and $I(z_0) = g(z_0)e^{z_0}/\sqrt{2\pi z_0}$, so $(t/t_0)^{15/4} I(z/2)/I(z_0) \le \sqrt{2} e^{-z_0/2} e^{-(z_0-z)/2}/g_0 \le \sqrt{2} e^{-\pi\sqrt{t_0}/2} q^S/g_0$, using $(t/t_0)^{15/4}(t_0/t)^{1/4} \le 1$ and $\Delta_s \ge 2cS$ as before. Finally $\sum_{\gamma\ge 2} 2\gamma \cdot \sqrt{2\gamma}\,\gamma^{-1}(2/\gamma)^{15/2} = 2^9 \sum_{\gamma\ge 2} \gamma^{-7}$. □

13.6. **The shadow terms.** By (33) and Lemma 6.1, with $\psi_1 = (2m)^{-1/2}$ and $\psi_\gamma = \sqrt{\gcd(m,\gamma)/m}$, and $D_0 = m^2 x_0^2$,

$$|\mathcal{X}_{K_2}| \le \sum_{\gamma\ge 1} \sum_F \frac{N_F(\gamma)}{2} \cdot \frac{\sqrt{2m}\,|c_F|\,\psi_\gamma}{\sqrt{\gamma}} \Big(\frac{4m\mu_F}{D_0}\Big)^4 \frac{I_{15/2}\big(2\pi x_0 \sqrt{\mu_F m}/\gamma\big)}{\mathcal{E}(D_0, m)}, \tag{81}$$

$$|\mathcal{X}_{K_3}| \le \sum_{\gamma\ge 1} \sum_F \frac{N_F(\gamma)}{2} \cdot \frac{|c_F|\,\psi_\gamma}{2\pi\gamma^2} \Big(\frac{4m\mu_F}{D_0}\Big)^{\frac{17}{4}} \cdot 2\sqrt{\frac{\mu_F}{m}} \cdot \frac{4\pi^2 m^2\gamma^2}{3} \cdot \frac{I_{15/2}\big(2\pi x_0 \sqrt{\mu_F m}/\gamma\big)}{\mathcal{E}(D_0, m)}. \tag{82}$$

Indeed, the arguments of $I_8$ and $I_{17/2}$ are $cx$ with $c \le 2\pi\sqrt{m} \le \pi\sqrt{t_0}$, so by Lemma 12.3 each term divided by $\mathcal{E}$ is largest at $x = x_0$; there $I_8, I_{17/2} \le I_{15/2}$ by Lemma 12.1(iv), the factor $(1 - mu^2/\mu_F)^{17/4}$ is at most 1, $I_{17/2}$ is increasing in its argument, and the interval of integration has length $2\sqrt{\mu_F/m}$. The certificate sums $\gamma \le G = 12$ exactly and bounds the rest using $I_{15/2}(z/\gamma) \le \gamma^{-15/2} I_{15/2}(z)$, $N \le 2\gamma$ and $\psi_\gamma \le 1$, with $\sum_{\gamma>G} \gamma^{-7} \le G^{-6}/6$ for (81) and $\sum_{\gamma>G} \gamma^{-13/2} \le \frac{2}{11} G^{-11/2}$ for (82).

**Lemma 13.6.** *For $m \ge 101$, with $z_0 = \pi\sqrt{t_0}$, $z_1 = 2\pi\sqrt{m}$ and $J = \sqrt{z_0/z_1}\, e^{z_1 - z_0}/g(z_0)$,*

$$|\mathcal{X}_{K_2}| \le \frac{\sqrt{2m}\,\frac{17}{16} e^{2\pi\sqrt{2m+2}} (4/m)^4 \zeta(7) J}{\frac{\pi}{\sqrt{2m}}\frac{m+2}{16}}, \qquad |\mathcal{X}_{K_3}| \le \frac{\frac{17}{16} e^{2\pi\sqrt{2m+2}} \frac{1}{2\pi} (4/m)^{17/4} \frac{2}{\sqrt{m}} \frac{4\pi^2 m^2}{3} \zeta(\frac{13}{2}) J}{\frac{\pi}{\sqrt{2m}}\frac{m+2}{16}}.$$

*Proof.* Take $x_0 = 1$ in (81), (82). Use $|c_F| \le \hat{f}(m) \le e^{2\pi\sqrt{2m+2}}$ (Lemma 12.5), $\mu_F \le 1$, $\psi_\gamma \le 1$, $N \le 2\gamma$, $I_{15/2}(z_1/\gamma) \le \gamma^{-15/2} I_{15/2}(z_1)$, and $\mathcal{E}(m^2, m) \ge \frac{\pi}{\sqrt{2m}}\frac{m+2}{16} I_{15/2}(z_0)$. Finally $I_{15/2}(z_1)/I_{15/2}(z_0) \le J$ by Lemma 12.2(i). □

## 14. Proof of Theorem A

Let $\epsilon(m, x_0) = \epsilon_{\text{main}} + \epsilon_{\text{miss}} + \epsilon_{rs} + \epsilon_T + \epsilon_{\gamma\ge 2} + \epsilon_{K_2} + \epsilon_{K_3}$ be the sum of the bounds of Section 13, where $\epsilon_{K_j}$ denote the right sides of (81) and (82) with the tails included.

**Theorem 14.1.** *Put $x_0(1) = 4$, $x_0(2) = 3$, $x_0(m) = \sqrt{500}/m$ for $3 \le m \le 7$ and $x_0(m) = 1$ for $m \ge 8$. Then*

$$\epsilon\big(m, x_0(m)\big) \le 0.347 \qquad (1 \le m \le 100),$$

*with the largest value at $m = 8$. Moreover $\epsilon(1,4) \le 0.304$, $\epsilon(2,3) \le 0.105$, and $\epsilon \le 0.0025$ for $3 \le m \le 7$.*

*Proof.* Every term of $\epsilon(m, x_0)$ is a finite sum of explicit quantities, which we evaluate in ball arithmetic with 96 bits [22]. Bessel functions are evaluated by Arb's certified routines, the suprema in (76) by the exact critical points, and the solutions of (75) are enclosed as described after (79). The values are listed in Appendix C and plotted in Figure 4; the computation takes about twelve minutes.[12] □

[12]At $m = 1$ and $m = 2$ the largest pieces are the leading family and the two shadow terms: $\epsilon_{\text{main}} = 0.126$, $\epsilon_{K_2} = 0.058$, $\epsilon_{K_3} = 0.121$ at $m = 1$, and 0.070, 0.007, 0.028 at $m = 2$. The thresholds $x_0 = 4, 3$ are far below the values $x \ge \sqrt{800}/m$ needed in the proof of Theorem A.

**Theorem 14.2.** *For every $m \geq 101$, $\epsilon(m,1) \leq 0.3382$.*

*Proof. Leading family and missing pairs.* Group the pairs by $S$. For $1 \leq S \leq m+1$, Lemma 11.2 gives $S - \eta_- = S(2m-S)/(2m+4) \geq (2m-1)/(2m+4) > 1/\pi$, so Lemma 13.3 applies, and the pairs with $a+b+2 = S$ contribute at most $p_2(S)u^S\varphi(S,m)$, where $p_2(S) = \sum_{a+b=S-2} d_1(a)d_1(b)$. Missing pairs have $S \geq m/2+2$ and contribute at most $p_2(S)u^S$. For $S > 300$ we use, for the pairs in $\mathcal{L}_m$, $|\rho_{a,b} - e^{-\pi x S}| \leq e^{-\pi\Delta_s} + e^{-\pi S} \leq 2e^{-\pi S(m-1)/(2m+4)} \leq 2e^{-0.485\pi S}$ (Lemma 11.2(iv)), and $p_2(S) \leq P(q)^2 q^{-S}$ with $q = e^{-0.4\pi}$, so these terms add up to at most $2P(q)^2\varrho^{301}/(1-\varrho)$ with $\varrho = e^{-0.485\pi}/q$. Evaluated in ball arithmetic, the total is at most 0.33815 for every $101 \leq m \leq 300$, the largest value being at $m = 101$.

For $m > 300$ we compare with $m = 300$. The bound $\eta_-$ decreases in $m$; for $S \leq 3$ one checks $\eta_+ = 0$ (for $S = 3$ this is $\sqrt{1+4/m} + \sqrt{1-2/m} \geq 2$, true for $m \geq 3$), and for $S \geq 4$ the function $\sqrt{m(m+4)} - \sqrt{m(m+4-2S)} = 2S/\big(\sqrt{1+4/m} + \sqrt{1+(4-2S)/m}\big)$ decreases in $m$; $y_t = \pi\sqrt{m(m+4-2S)}$ increases, and $8S/m$ decreases. Hence $\varphi(S,m) \leq \varphi(S,300)$ whenever $2S \leq 304$. For $152 < S \leq 300$ the term $\varphi(S,300)$ has $\epsilon_1 = 1$ and $\min(1, \pi\eta_+) = 1$, so again $\varphi(S,m) \leq \varphi(S,300)$. The missing pairs for $m > 300$ have $S \geq m/2+2 > 152$, and their terms $p_2(S)u^S$ with $S \leq 300$ are counted at $m = 300$; those with $S > 300$ are in the tail. So the bound at $m = 300$, which is 0.1090, holds for all $m > 300$.

*The other pieces.* By Lemmas 13.4, 13.5 and 13.6, with $q = q(101)$ (since $c$ increases with $m$, the bound with $q(101)$ holds for all $m \geq 101$),

$$|\mathcal{X}_{rs}| + |\mathcal{X}_T| \leq 3.02 \cdot 10^{-8}, \quad |\mathcal{X}_{\gamma\geq 2}| \leq 3 \cdot 10^{-45}, \quad |\mathcal{X}_{K_2}| \leq 6 \cdot 10^{-79}, \quad |\mathcal{X}_{K_3}| \leq 7 \cdot 10^{-77}$$

at $m = 101$. Each bound decreases in $m$: in Lemma 13.4 the factor $(2m+1)^2 e^{-\pi m/4}$ decreases for $m \geq 101$ and $g_0$ increases; in Lemma 13.5 the factor $e^{-\pi\sqrt{t_0}/2}$ dominates; in Lemma 13.6 the exponent $2\pi\sqrt{2m+2} + z_1 - z_0 \leq 2\pi\sqrt{2m+2} + 2\pi\sqrt{m} - \pi m$ decreases. □

*Proof of Theorem* A. The published input is [8, Theorem 5.4(a)]: for a charge in the set $\mathcal{A}$ of (10) the single-centred index is the finite-part coefficient, $d^* = (-1)^{\ell+1} c_m^F(n,\ell)$, as recorded in (11). By Proposition 4.6, which rests on it and on the $\Gamma_1(2)$ invariance of $d^*$, it suffices to prove $d^*(m,n,\ell) > 0$ for reduced charges. If $D \leq 800$, this is Theorem 4.7. Let $D > 800$. Then $x = \sqrt{D}/m \geq x_0(m)$: for $m \leq 7$ because $\sqrt{800} \geq 4$, $\sqrt{800}/2 \geq 3$ and $800 > 500$, and for $m \geq 8$ because a reduced charge has $D = 2mn - \ell^2 \geq 2m^2 - m^2 = m^2$. Theorem 6.2 expresses $c_m^F(n,\ell)$ by the convergent series (32), and by (74) and the bounds of Section 13, which are uniform in $x \geq x_0$ and in $\ell$,

$$\frac{d^*(m,n,\ell)}{\mathcal{E}(D,m)} \;\geq\; |P(z)|^2 - \epsilon\big(m, x_0(m)\big) \;\geq\; 0.53816 - 0.347 > 0$$

by Proposition 11.3 (note $|z| = e^{-\pi x} \leq e^{-\pi}$) and Theorems 14.1 and 14.2. Since $\mathcal{E}(D,m) > 0$, $d^*(m,n,\ell) > 0$. □

**Corollary 14.3.** *For every reduced charge with $x \geq x_0(m)$,*

$$0.19\,\mathcal{E}(D,m) \;\leq\; d^*(m,n,\ell) \;\leq\; 2.51\,\mathcal{E}(D,m).$$

*In particular* $\log d^*(Q,P) = \pi\sqrt{1+4/Q^2}\,\sqrt{D} - 4\log D + O_{Q^2}(1)$ *as $D \to \infty$ with $Q^2$ fixed, and* $\log d^* = \pi\sqrt{Q^2P^2 - (Q\cdot P)^2} + O(\log D)$ *when $Q^2 \to \infty$ with $\sqrt{D}/Q^2$ bounded, the Bekenstein–Hawking entropy of the black hole.*

*Proof.* The lower bound is the proof above. For the upper bound, $|P(z)|^2 \leq P(e^{-\pi})^2 \leq 2.156$ and $\epsilon \leq 0.347$. The asymptotics follow from (67), $I_\nu(y) \sim e^y/\sqrt{2\pi y}$, and $\pi\sqrt{D}\,(\sqrt{1+4/m} - 1) \leq 2\pi\sqrt{D}/m$. □

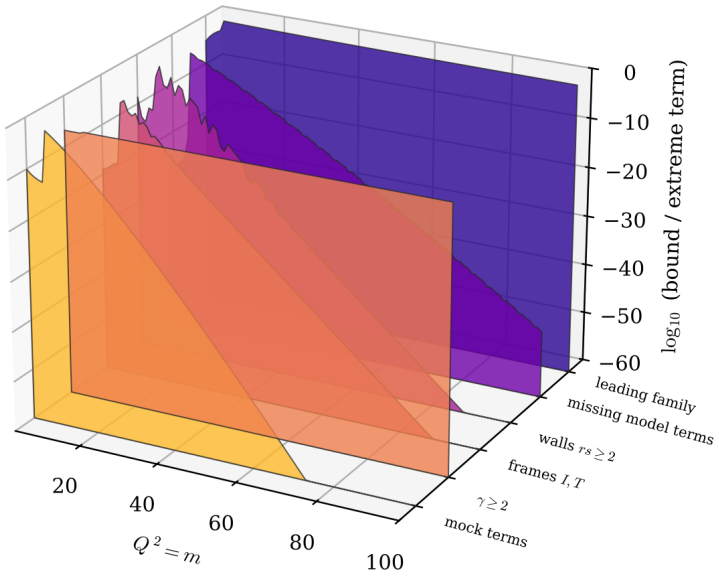



FIGURE 4. The pieces of the certificate for $3 \le Q^2 \le 100$, on a logarithmic scale and divided by the extreme term $\mathcal{E}$: shadow terms, terms with $\gamma \ge 2$, the frames $I, T$, the walls with $rs \ge 2$, the missing model terms and the error of the leading family. Only the last one matters numerically; the others are below $10^{-2}$ at $Q^2 = 8$ and fall off quickly.

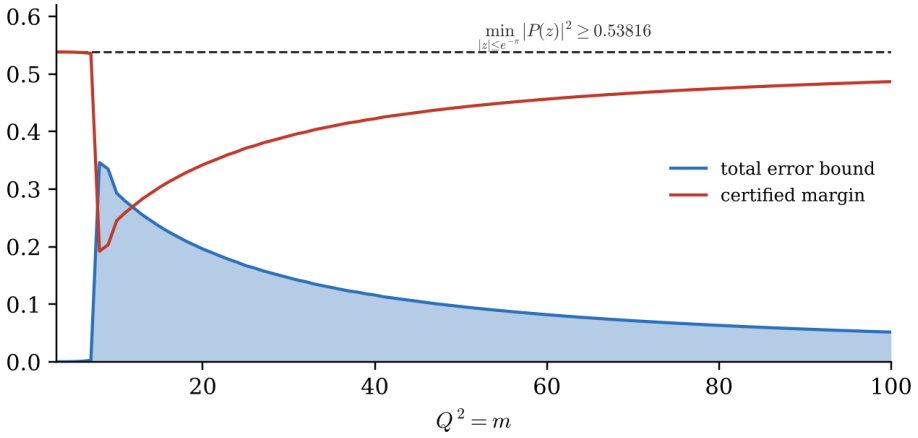



FIGURE 5. The total error bound $\epsilon(m, x_0(m))$ and the margin $0.53816 - \epsilon$ for $3 \le Q^2 \le 100$. The margin is smallest at $Q^2 = 8$, where it is 0.19.

**Remark 14.4.** The minimum 0.5389 of Proposition 11.3 is reached in the limit: as $m \to \infty$ along charges with $x = 1$ and $\ell = m$ odd, $z = -e^{-\pi}$, and the bounds in the proof of Theorem 14.2 tend to 0, since $\varphi(S, m) \to 0$ for each fixed $S$. So $d^*/\mathcal{E}$ comes arbitrarily

| $Q^2=1$ | | $Q^2=2$ | | | | | |
|---|---|---|---|---|---|---|---|
| $D$ | $d^*$ | $D$ | $d^*$ ($\ell$ odd) | $D$ | $d^*$ ($\ell\equiv 0$) | $D$ | $d^*$ ($\ell\equiv 2$) |
| 1 | 360 | 3 | 4352 | 4 | 18944 | 4 | 6912 |
| 2 | 2164 | 7 | 67008 | 8 | 198144 | 8 | 101376 |
| 3 | 4352 | 11 | 671744 | 12 | 1580672 | 12 | 992256 |
| 4 | 18944 | 15 | 5189696 | 16 | 10586880 | 16 | 7550400 |
| 5 | 36024 | 19 | 33581312 | 20 | 62035968 | 20 | 48253952 |
| 6 | 125860 | 23 | 190095232 | 24 | 327011328 | 24 | 270195712 |
| 7 | 233080 | 27 | 968248576 | 28 | 1578794128 | 28 | 1362534400 |
| 8 | 705536 | 31 | 4522209472 | 32 | 7075522048 | 32 | 6304638848 |

TABLE 1. The first single-centred indices on the rows $Q^2=1$ and $Q^2=2$, as functions of $D=Q^2P^2-(Q\cdot P)^2$ and, at $Q^2=2$, of $\ell=Q\cdot P$ modulo 4.

close to 0.5389, and the factor of the leading family cannot be replaced by a single term: an argument with one dominant term, which suffices at level one, does not reach level two without the resummation of Theorem 11.1. For moderate charges the picture is already accurate: at $Q^2=3$, $P^2/2=10$, $Q\cdot P=3$ the exact index gives $d^*/\mathcal{E}=0.9485$ while $|P(z)|^2=0.9910$, and at $Q^2=4$, $P^2/2=58$, $Q\cdot P=4$ the two numbers are 0.99990 and 0.99999.

## 15. Values and comparison with the product formula

Table 1 lists the first single-centred indices on the rows $Q^2=1$ and $Q^2=2$, computed exactly from the product (3) by the program of Appendix A; at $Q^2=1$ they are the values of (13). We compared the expansion of the product in exact rational arithmetic with (12) on every coefficient $x^e y^\ell$ with $e\le 49$ and $|\ell|\le 18$; the two sides agree. Every entry of Table 1 that lies in Sen's Table 2 [29] coincides with it: at $Q^2=1$ the entries 360, 2164, 4352, 18944, 36024, 125860 and the value 2164 at $(P^2,Q\cdot P)=(6,2)$, and at $Q^2=2$ the entries 198144, 67008, 6912 of the row $P^2=4$ and 1580672, 671744, 101376, 4352 of the row $P^2=6$.[13] Figure 6 shows $\log_{10} d^*$ on the two rows.

[13] In these entries the charge lies in the cone (3.9) of [29], and the index of the full partition function equals the single-centred index. The entries of Sen's table with $D\le 0$, such as $-2$ at $(Q^2,P^2,Q\cdot P)=(1,2,2)$ or $-16$ at $(2,6,4)$, are outside the scope of the theorem, as they should be.

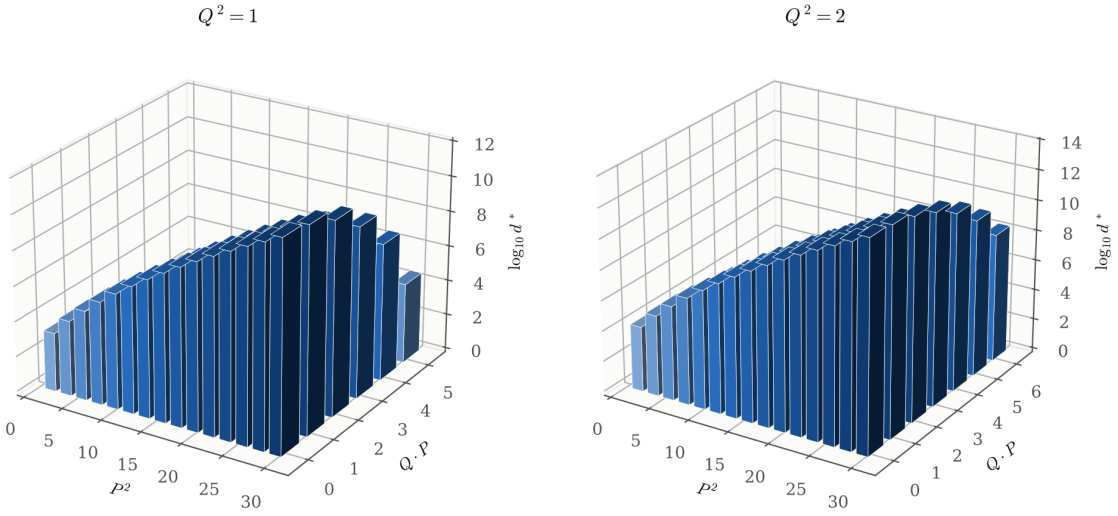



FIGURE 6. $\log_{10} d^*(Q, P)$ for $Q^2 = 1$ (left) and $Q^2 = 2$ (right), $2 \le P^2 \le 30$ and $0 \le Q \cdot P \le \min(6, P^2)$ with $Q^2P^2 > (Q \cdot P)^2$. Every bar is positive, as Theorems 3.3 and A assert.

## 16. Other CHL models

The method is not special to $N = 2$. For the $\mathbb{Z}_N$ CHL models with $N = 3, 5, 7$ the leading family of walls again resums, now to $|1/\eta(\tau)^{k+2}\eta(N\tau)^{k+2}|^2$ evaluated at $e^{-2\pi x/N + i\theta}$ with $k + 2 = 24/(N + 1)$. Positivity then reduces in the same way to a lower bound for this modulus on the disc $|z| \le e^{-2\pi/N}$, to the analogue of Theorem 6.2 for $\Gamma^0(N)$, whose proof requires the polar data at every cusp of $\Gamma^0(N)$ and the transformation law of the finite part under the corresponding frames, and to an exact computation below the discriminant at which the error bounds take over.

## Declarations

**Data availability.** The programs and data that support the findings of this study are openly available in the archive [4], at `https://doi.org/10.5281/zenodo.24140567`. They comprise the exact computation of Appendix A, the interval certificates of Appendix B, the independent checks of the Rademacher expansion listed there, the programs that verify Proposition 3.1 and the values of Table 1, and the scripts that draw the figures.

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**Conflict of interest.** The author has no conflicts of interest to declare that are relevant to the content of this article.

**Ethics statement.** This study involves no human participants, animals or personal data; no ethical approval was required.

## Appendix A. The exact computation

The program `chl_coefficients.py` of the archive [4] proves Theorem 4.7 in exact integer arithmetic, in five steps.

(B1) *Orbits.* A charge $(m, n, \ell)$ corresponds to the binary quadratic form $f(s,t) = ms^2 + 2\ell st + 2nt^2$ of discriminant $-8D$, and $\Gamma_1(2)$ acts by $f \mapsto f \circ g$. The orbits with given $D$ are enumerated through the Gauss-reduced forms of discriminant $-8D$, each with a class of bottom rows modulo 2, taken modulo the automorphs of the reduced form.

(B2) *A representative in $\mathcal{A}$.* In every orbit the program finds a charge with $0 \le \ell < 2m$ satisfying (10), which it tests in the form $2(jm+\ell m_1)^2+m_1^2\cdot 2D \ge 2m^2$ for $m_1 \in 2\mathbb{Z}\backslash\{0\}$ and odd $j$ with $m_1 \mid (j^2-1)/4$, the range of $(m_1, j)$ being finite because the left side grows.

(B3) *Coefficients.* The coefficients $c_{\mathcal{R}}(m, n, \ell)$ are computed from (3) by the recursion for the exponential of (6), with polynomial products in FLINT [20]. By Lemma 4.3(ii) each charge is read in the cheaper of its two orientations.

(B4) *Signs.* For every orbit, $d^* = (-1)^{\ell+1} c_{\mathcal{R}}(m, n, \ell)$ is computed for every representative found in the box, the values are compared, and the sign is checked. At $m = 1$ the values are compared with Proposition 3.1.

(B5) *Least representatives.* On the whole range, every charge of least $Q^2$ in its orbit is checked to lie in $\mathcal{A}$, as Lemma 4.4 predicts.

For $D \le 800$ there are 22972 orbits, with $Q^2 \le 28$ for the least representative; all have $d^* > 0$. The smallest value relative to the extreme term occurs at small $D$, where $x$ is close to 1; this is the region in which the leading-family error of Lemma 13.2 is largest, and Theorem 4.7 covers it for every $Q^2$.

## Appendix B. Interval computations

All the numerical statements used in the proofs are certified in ball arithmetic with Arb [22] through python-flint; each program prints its enclosures.

(I1) `pmin.py`: Proposition 11.3. Product truncated at $n = 60$ with an explicit tail enclosure, 4001 points on the half circle, Lipschitz constant $L = 2P(r_0)r_0P'(r_0)$.

(I2) `gmono2.py`: Lemma 12.2. Monotonicity of $\phi$ on $[3, 60]$ by mean-value enclosures on an adaptive cover; the range $[60, \infty)$ through the closed form (73).

(I3) `cert.py`: Theorem 14.1. For each $m$, $1 \le m \le 100$, the program lists the leading family $\mathcal{L}_m$, the walls with $rs \ge 2$ and the parity walls through their parameters $(r, s, a, b)$, encloses $t(r, s, a, b)$, and evaluates the seven bounds of Section 13. In (79) it uses 32 in place of 16.

(I4) `analytic.py`, `analytic2.py`: Theorem 14.2.

(I5) `polar.py`: the wall sums (22), (23), which supply the polar data in (I3).

The programs in the folder `proof_checks` of the archive check the proof of Theorem 6.2 independently, in exact or high-precision arithmetic; none of them is used in the proof. `polar_check.py` tests the walls, the chambers and the wall sum (23) of Section 7; `polar0_check.py` tests the transformation of the product, the second product and the wall sum (22) of Section 8; `transform_check.py`, with `tc_series.py`, tests the lemmas and the transformation law of Section 9; `chk_theta.py` tests Lemma 10.9, and `chk_mordell.py`, `chk_mordell2.py` and `hp.py` test Lemma 9.8; `expansion_check.py`, with `expansion_series.py`, evaluates the truncated right side of (32) (Remark 6.4). Their outputs are stored next to them.

## Appendix C. The certified margins

Table 2 lists, for $3 \le Q^2 \le 100$, the total bound $\epsilon(m, x_0(m))$ of Theorem 14.1 and the margin $0.53816 - \epsilon$. At $Q^2 = 1$ and $Q^2 = 2$, with $x_0 = 4$ and $x_0 = 3$, the bounds are 0.304 and 0.105, and the margins 0.234 and 0.433.

TABLE 2. The certified total error $\epsilon(m, x_0(m))$ of Theorem 14.1 and the margin $0.53816 - \epsilon$, for $m = Q^2$.

| $m$ | $\epsilon$ | margin | $m$ | $\epsilon$ | margin | $m$ | $\epsilon$ | margin | $m$ | $\epsilon$ | margin |
|---|---|---|---|---|---|---|---|---|---|---|---|
| 3 | $6.04 \cdot 10^{-6}$ | 0.5380 | 28 | 0.1540 | 0.3842 | 53 | 0.0913 | 0.4467 | 78 | 0.0648 | 0.4732 |
| 4 | $5.47 \cdot 10^{-5}$ | 0.5379 | 29 | 0.1500 | 0.3883 | 54 | 0.0898 | 0.4482 | 79 | 0.0640 | 0.4740 |
| 5 | $2.87 \cdot 10^{-4}$ | 0.5377 | 30 | 0.1460 | 0.3921 | 55 | 0.0884 | 0.4496 | 80 | 0.0633 | 0.4747 |
| 6 | $9.79 \cdot 10^{-4}$ | 0.5370 | 31 | 0.1420 | 0.3958 | 56 | 0.0870 | 0.4510 | 81 | 0.0626 | 0.4754 |
| 7 | 0.0024 | 0.5356 | 32 | 0.1390 | 0.3993 | 57 | 0.0857 | 0.4523 | 82 | 0.0619 | 0.4761 |
| 8 | 0.3460 | 0.1917 | 33 | 0.1350 | 0.4026 | 58 | 0.0844 | 0.4536 | 83 | 0.0612 | 0.4768 |
| 9 | 0.3350 | 0.2030 | 34 | 0.1320 | 0.4058 | 59 | 0.0831 | 0.4549 | 84 | 0.0605 | 0.4775 |
| 10 | 0.2930 | 0.2450 | 35 | 0.1290 | 0.4089 | 60 | 0.0819 | 0.4561 | 85 | 0.0599 | 0.4781 |
| 11 | 0.2790 | 0.2594 | 36 | 0.1260 | 0.4118 | 61 | 0.0807 | 0.4573 | 86 | 0.0592 | 0.4788 |
| 12 | 0.2670 | 0.2714 | 37 | 0.1230 | 0.4145 | 62 | 0.0795 | 0.4585 | 87 | 0.0586 | 0.4794 |
| 13 | 0.2550 | 0.2828 | 38 | 0.1210 | 0.4172 | 63 | 0.0784 | 0.4596 | 88 | 0.0580 | 0.4800 |
| 14 | 0.2450 | 0.2930 | 39 | 0.1180 | 0.4197 | 64 | 0.0773 | 0.4607 | 89 | 0.0574 | 0.4806 |
| 15 | 0.2350 | 0.3026 | 40 | 0.1160 | 0.4222 | 65 | 0.0763 | 0.4617 | 90 | 0.0568 | 0.4812 |
| 16 | 0.2260 | 0.3115 | 41 | 0.1130 | 0.4245 | 66 | 0.0753 | 0.4627 | 91 | 0.0563 | 0.4817 |
| 17 | 0.2180 | 0.3199 | 42 | 0.1110 | 0.4268 | 67 | 0.0742 | 0.4638 | 92 | 0.0557 | 0.4823 |
| 18 | 0.2100 | 0.3277 | 43 | 0.1090 | 0.4289 | 68 | 0.0733 | 0.4647 | 93 | 0.0551 | 0.4829 |
| 19 | 0.2030 | 0.3350 | 44 | 0.1070 | 0.4310 | 69 | 0.0723 | 0.4657 | 94 | 0.0546 | 0.4834 |
| 20 | 0.1960 | 0.3419 | 45 | 0.1050 | 0.4330 | 70 | 0.0714 | 0.4666 | 95 | 0.0541 | 0.4839 |
| 21 | 0.1900 | 0.3483 | 46 | 0.1030 | 0.4350 | 71 | 0.0705 | 0.4675 | 96 | 0.0536 | 0.4844 |
| 22 | 0.1840 | 0.3544 | 47 | 0.1010 | 0.4368 | 72 | 0.0696 | 0.4684 | 97 | 0.0530 | 0.4850 |
| 23 | 0.1780 | 0.3601 | 48 | 0.0994 | 0.4386 | 73 | 0.0688 | 0.4692 | 98 | 0.0525 | 0.4855 |
| 24 | 0.1730 | 0.3655 | 49 | 0.0976 | 0.4404 | 74 | 0.0679 | 0.4701 | 99 | 0.0520 | 0.4860 |
| 25 | 0.1670 | 0.3706 | 50 | 0.0960 | 0.4420 | 75 | 0.0671 | 0.4709 | 100 | 0.0516 | 0.4864 |
| 26 | 0.1630 | 0.3754 | 51 | 0.0943 | 0.4437 | 76 | 0.0663 | 0.4717 | | | |
| 27 | 0.1580 | 0.3799 | 52 | 0.0928 | 0.4452 | 77 | 0.0655 | 0.4725 | | | |

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